A Reduction of the Slice–Ribbon Conjecture

Certified Quenching and a Single Completion Condition

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Abstract

We introduce a certified quenching mechanism for stable projections of smooth slice disks. Its state space records double-decker data, boundary collars, height orderings, and finite geometric certificates; every nonconstant certified transition strictly decreases a well-founded energy $\mathrm{E} : \mathfrak{S} \to \mathbb{N}^7$. Hence every selected certified history terminates, while boundary fixation and liftability preserve the boundary knot and slice realization.

We then package the remaining geometric requirements into one quenching completion condition H_{Q}. It states that non-clean terminal states admit projected-disk recovery packages and that clean blocked cores satisfy the reduction and Milnor-applicability properties required for the terminal contradiction. We prove

$$\mathsf{H}_{\mathrm{Q}} \Longrightarrow \forall K \subset S^3 \left(\mathrm{Slice}(K) \Rightarrow \mathrm{Ribbon}(K)\right).$$

Thus our two principal contributions are the general certified quenching framework and a reduction of the slice–ribbon conjecture to the single explicit condition H_{Q}. No unconditional proof of H_{Q} or effective certificate-search algorithm is claimed.

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General Introduction

Aim, scope, and chapter architecture

We study the implication Slice(K) $\Rightarrow$ Ribbon(K) for smooth knots $K \subset S^3$. The paper has two principal contributions. First, we define a general certified quenching mechanism whose states, transitions, energy, selectors, and terminal alternatives are independent of any claimed solution of the conjecture. Second, we isolate one explicit quenching completion condition H_{Q} and prove that H_{Q} implies the slice–ribbon conjecture.

The condition H_{Q} is a single theorem interface with three technical clauses: projected-disk recovery for the remaining non-clean branch, reduction of Milnor-trivial clean cores, and Milnor applicability for the projected terminal core. These clauses are denoted locally by H_{nc}, $\mathsf{H}_{\mathrm{red}}$, and $\mathsf{H}_{\mathrm{app}}$, but they are used globally only through their conjunction H_{Q}. Relative branch cancellation, Roseman/movie

decomposition, and framed cusp merge are established unconditionally in Theorems 10.2 to 10.4.

We divide the argument into ten numbered chapters. Chapters 1–4 define the geometric objects and the reductio state. Chapters 5–6 construct the quenching mechanism and prove soundness and finite termination. Chapters 7–9 analyze terminal states and the Milnor obstruction. Chapter 10 establishes the available relative local theorems, states the remaining completion interface, and proves the reduction $\mathsf{H}_{\mathrm{Q}} \Rightarrow$ SRC.

Reader's roadmap

We first construct the unconditional transition system and then expand the three clauses of the single condition H_{Q}. The following tables display the dependency order; they are not used as premises in any proof.

Chapter	Function	Main output
Chapter 1	Basic setup	Slice disks, ribbon disks, stable projections, double-decker data, height ordering, and high-dimensional projection language.
Chapter 2	Combinatorial target	Ribbon-normality, ribbon defects, source filling disks, and the clean ribbon criterion.
Chapter 3	Geometric preparation	Tubular neighborhoods, Morse normalization on filling disks, and certified Whitney–Roseman moves.
Chapter 4	Reductio seed	A putative counterexample yields a minimal projection datum with a non-empty irreducible defect core.
Chapter 5	Quenching engine, definitions	Quenching states, certified moves, quenching packages, boundary constraints, and high-dimensional encoded states.
Chapter 6	Quenching engine, theorems	The lexicographic energy, finite termination, the ZFC selector, terminal alternatives, liftability, and high-dimensional finite quenching.
Chapter 7	Terminal analysis	Blocked states, closed-core sublinks, and the insufficiency of ordinary projected linking numbers.
Chapter 8	Logical packaging	Universal terminal obstruction, WTQF–Milnor packaging, and the dependency chain leading to $\mathsf{C1}$.
Section 8.1–Section 9.5	Weak flow and clean branch	Certificate-suspension WTQF, the high-dimensional homological shell, Milnor readability, terminal non-vanishing, clean rigidity, and clean blocked-state exclusion.
Chapter 10	Completion condition and reduction	Unconditional relative local theorems, the three clauses of H_{Q}, and the implication $\mathsf{H}_{\mathrm{Q}} \Rightarrow$ SRC.

Table 1: Chapter-by-chapter roadmap for the proof.

Formal predicates and theorem schema

Let $\mathcal{K}$ denote the set of smooth oriented knot types in S^3. For $K \in \mathcal{K}$, define

$$\begin{aligned} \mathrm{Slice}(K) &:\Longleftrightarrow \exists\, \Phi : (D^2, \partial D^2) \hookrightarrow (B^4, S^3) \text{ smooth and proper with } \Phi(\partial D^2) = K, \\ \mathrm{Ribbon}(K) &:\Longleftrightarrow \exists\, \Phi \text{ as above such that } r \circ \Phi \text{ has no index-two interior critical point.} \end{aligned}$$

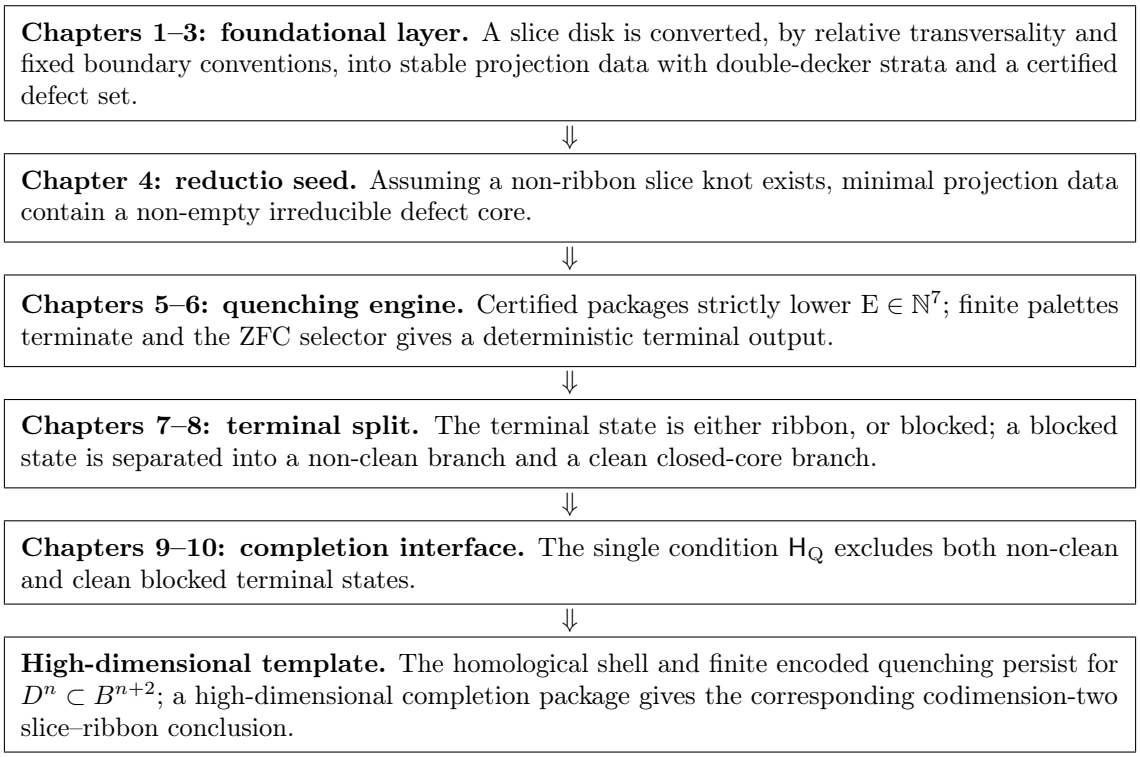


Figure 1: Global logic flow of the quenching reduction.

Reading issue	**Guide**
Core concept chain	Slice disk → stable projection → quenching state → well-founded energy → finite terminal output → terminal split → application of H_{Q} → ribbon terminal state.
Forward references	The quenching engine is proved before the technical clauses of H_{Q} are expanded in Chapters 9–10.
Quick route	Read this roadmap, Chapters 5–6 for the mechanism, the definition of H_{Q}, and Theorem 10.26.
Verification route	Read chapters in order, mark forward references to $\mathsf{C1}$, then reread Chapter 8 and Section 8.1 after Chapter 10.
High-dimensional route	Follow Proposition 1.21, Definition 5.3, Theorem 6.13, Theorem 8.7, and Theorem 10.33.

Table 2: Concept dependencies, forward references, and suggested reading routes.

Let $\mathfrak{S}$ be the encoded state set of Definition 5.2. For $\mathcal{S} \in \mathfrak{S}$ and $K \in \mathcal{K}$, the predicates

$$\mathrm{Realizes}(\mathcal{S}, K), \qquad \mathrm{Clean}(\mathcal{S}), \qquad \mathrm{Terminal}_{\mathfrak{P}}(\mathcal{S}), \qquad \mathrm{RibbonTerm}(\mathcal{S}), \qquad \mathrm{BlockedTerm}_{\mathfrak{P}}(\mathcal{S})$$

mean, respectively, that $\mathcal{S}$ is realized by a slice disk with boundary K, that its non-clean strata are absent, that no package in the declared package relation $\mathfrak{P}$ starts at $\mathcal{S}$, and that $\mathcal{S}$ lies in the ribbon or blocked part of the terminal partition. The definitions below prove

$$\forall \mathcal{S} \in \mathfrak{S}, \qquad \mathrm{Terminal}_{\mathfrak{P}}(\mathcal{S}) \Longrightarrow \big(\mathrm{RibbonTerm}(\mathcal{S}) \,\dot\vee\, \mathrm{BlockedTerm}_{\mathfrak{P}}(\mathcal{S})\big),$$

where $\dot\vee$ denotes exclusive disjunction.

Definition 0.1 (Quenching completion condition)**.** The *quenching completion condition* is the single assertion

$$\mathsf{H}_{\mathrm{Q}} \;:\Longleftrightarrow\; \mathsf{H}_{\mathrm{nc}} \wedge \mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}},$$

where the three displayed factors are respectively the projected-disk recovery clause of Assumption 10.5, the clean-core reduction clause of Assumption 8.11, and the projected-core applicability clause of Assumption 9.14. They are not additional global hypotheses: they are the three formally separated clauses of H_{Q}.

The exact final implication is

$$\mathsf{H}_{\mathrm{Q}} \Longrightarrow \forall K \in \mathcal{K}\,\big(\mathrm{Slice}(K) \Longrightarrow \mathrm{Ribbon}(K)\big).$$

The clauses of H_{Q} are used separately only to identify which terminal branch they exclude. The unconditional termination assertion is the separate statement

$$\neg\exists\,(\mathcal{S}_j)_{j\in\mathbb{N}} \subset \mathfrak{S} \quad \forall j \in \mathbb{N}, \quad \mathrm{E}(\mathcal{S}_{j+1}) < \mathrm{E}(\mathcal{S}_j),$$

proved from the well-foundedness of the lexicographic order on $\mathbb{N}^7$.

A certificate is a finite encoded record c together with a realization predicate $\mathrm{Valid}(c)$. Soundness theorems have the logical form

$$\mathrm{Valid}(c) \Longrightarrow \exists\, H : [0,1] \times B^4 \to B^4$$

with the stated ambient-isotopy, boundary-fixation, liftability, and endpoint properties. We prove soundness conditional on $\mathrm{Valid}(c)$; we do not prove that Valid is decidable.

For $n \geq 2$, the unconditional high-dimensional algebraic input is

$$H_n(B^{n+2}, S^{n+1}; \mathbb{Z}) = 0, \qquad \pi_n(B^{n+2}, S^{n+1}) = 0.$$

The high-dimensional slice–ribbon implication remains conditional on the completion package of Definition 10.32.

Global boundary convention. All disks are proper smooth disks in B^4 with boundary in $S^3 = \partial B^4$. Every isotopy, projection perturbation, and quenching move is fixed on a specified collar of the boundary unless explicitly stated otherwise. Thus the boundary knot, its orientation, and the chosen boundary projection convention are preserved throughout the certified evolution.

Admissible diagrammatic states. An admissible diagrammatic state is a stable projection datum for a properly embedded smooth disk, together with its double-decker structure, normalized height-ordering data, boundary collar, adapted finite cell structure, and recognized defect set. A state is *ribbon-normal* if all interval double-decker components have ribbon decker type; in such a state the only remaining ribbon defects are circular double-decker components and explicitly recognized stable singularities.

Trivial and boundary-degenerate cases. If the terminal projection has no ribbon defects, it is ribbon by the ribbon criterion below. This includes the embedded disk case $\Gamma = \emptyset$, whose boundary is the unknot. Nontrivial ribbon knots generally have nonempty double-decker sets consisting of ribbon singular arcs. Hence the correct terminal condition is not $\Gamma = \emptyset$, but $\mathrm{Def} = \emptyset$ after the adopted normalization.

1 Projection data and transversality conventions

The slice–ribbon conjecture, attributed to Fox, asks whether every smooth slice knot in the standard four-ball is ribbon [6]. A smooth slice disk is an embedded disk in B^4 bounded by a knot in $S^3 = \partial B^4$. A ribbon disk may be described either four-dimensionally, as a slice disk for which the radial Morse function has no local maxima, or three-dimensionally, as an immersed disk in S^3 whose singularities are all ribbon singularities. The implication

$$\text{ribbon} \quad \Longrightarrow \quad \text{slice}$$

is elementary: push the local lower sheet near each ribbon singularity slightly into the interior of B^4. The converse is the open content of the conjecture.

The predicates differ by quantified geometric data:

$$\mathrm{Slice}(K) \Longleftrightarrow \exists \Phi : D^2 \hookrightarrow B^4, \qquad \mathrm{Ribbon}(K) \Longleftrightarrow \exists \Phi : D^2 \hookrightarrow B^4 \text{ with } \mathrm{Crit}_2(r|_{\Phi(D^2)}) = \emptyset.$$

For a projection $p : B^4 \to B^3$, the forgotten coordinate defines a height function h. Hence every double-decker pair (x, y) satisfies $p(x) = p(y)$ and $h(x) \neq h(y)$. The encoded state records precisely these equalities, inequalities, and boundary data.

The first four chapters construct the foundational part of that approach. They do four things.

(1) It fixes the smooth category, the boundary conventions, and the Morse-theoretic definition of a ribbon disk.

(2) It replaces the informal phrase "the double-point set" by the correct *double-decker set*: a finite one-dimensional object in the source disk, paired by the equality of projected points and equipped with fourth-coordinate height ordering.

(3) It states the three-dimensional ribbon criterion in its correct form: not every absence of closed double curves is sufficient; every singular interval must also have ribbon decker type. Under a ribbon-normal projection convention, closed double-decker components are the remaining obstruction.

(4) It formulates a logically valid reductio seed. If a non-ribbon slice knot exists, then any minimal projection datum contains a non-empty finite ribbon-defect core. A later "topological quenching" theorem would have to produce certified complexity-decreasing moves for every such core. The existence of such a theorem would imply the slice–ribbon conjecture, but it is an additional theorem, not a consequence of the definitions.

Generic projections may contain double curves, isolated triple points, and branch points [4, 24, 1]. The implication

$$\gamma \subset \mathrm{Int}(D^2),\ \gamma \cong S^1 \Longrightarrow \exists E_\gamma \subset D^2,\ \partial E_\gamma = \gamma$$

does not imply the existence of a boundary-fixed ambient isotopy cancelling the projected component. We therefore distinguish the following predicates:

$$\text{projection data} \quad \Longrightarrow \quad \text{finite defect core} \quad \Longrightarrow \quad \text{certified simplification.}$$

The first two levels are established in the projection and reductio chapters; the third is isolated as the exact target for the quenching and obstruction chapters.

1.1 Basic spaces, maps, and transversality conventions

1.1.1 Knots, slice disks, and ambient category

Convention 1.1 (Smooth category)**.** All manifolds, embeddings, immersions, isotopies, and Morse functions are smooth of class C^∞. Manifolds with boundary are understood in the standard collar sense. Unless stated otherwise, isotopies of properly embedded disks in B^4 are ambient isotopies rel boundary collar.

Definition 1.2 (Knot)**.** An oriented knot is the image $K = f(S^1) \subset S^3$ of a smooth embedding $f : S^1 \hookrightarrow S^3$, equipped with the orientation induced from the standard orientation of S^1. Two oriented knots are equivalent if an orientation-preserving ambient isotopy of S^3 carries one to the other respecting orientations.

Definition 1.3 (Mirror, reverse, and convention for knots)**.** The reverse of an oriented knot is obtained by reversing the orientation of its source circle. The mirror is obtained by applying an orientation-reversing diffeomorphism of S^3. Sliceness and ribbonness are invariant under reversing orientation and under mirroring after applying the corresponding diffeomorphism to (B^4, S^3). Consequently the arguments below do not depend on a choice between a knot and its reverse or mirror, except where an oriented linking number is explicitly chosen.

Definition 1.4 (Smooth slice disk)**.** Let $K \subset S^3 = \partial B^4$ be a knot. A smooth slice disk for K is a proper smooth embedding

$$\Phi : (D^2, \partial D^2) \hookrightarrow (B^4, S^3)$$

such that $\Phi(\partial D^2) = K$. The knot K is smoothly slice if it admits a smooth slice disk. The source disk D^2 is identified with its image $D = \Phi(D^2)$ when no confusion can result.

Remark 1.5 (Orientations). The disk D^2 is orientable and carries its standard orientation. If K is oriented, the boundary orientation induced by the disk is required to agree with the orientation of K. The theory below can be repeated for unoriented knots by dropping this compatibility condition.

Definition 1.6 (Boundary collar equivalence)**.** Two slice disks for the same oriented boundary knot are boundary-collar equivalent if there is an ambient isotopy of B^4, fixed on a collar of the boundary knot in S^3, carrying one disk to the other. We perform all minimization and quenching procedures modulo this equivalence and stable projection equivalence.

1.1.2 Ribbon disks: Morse and immersed definitions

Let $r : B^4 \to [0,1]$ denote a smooth radial function with $r^{-1}(1) = S^3$ and with a single minimum at the origin. After a small isotopy of a properly embedded disk, $r|_D$ may be assumed Morse away from ∂D and constant on a boundary collar.

Definition 1.7 (Ribbon disk in B^4)**.** A slice disk $D \subset B^4$ is a ribbon disk if $r|_D$ is Morse on $\mathrm{Int}(D)$ and has no index-two critical points. Equivalently, $r|_D$ has local minima and saddles but no local maxima in the interior of D. A knot is ribbon if it bounds a ribbon disk.

Definition 1.8 (Ribbon singularity of an immersed disk)**.** Let $F : D^2 \looparrowright S^3$ be a self-transverse immersion. A connected component a of the image self-intersection set is a ribbon singularity if a is an embedded interval and

$$F^{-1}(a) = a_\partial \sqcup a_{\mathrm{Int}},$$

where $a_\partial \subset D^2$ is a properly embedded arc with endpoints on ∂D^2, while $a_{\mathrm{Int}} \subset \mathrm{Int}(D^2)$ is an embedded arc with endpoints in $\mathrm{Int}(D^2)$. The two arcs are called the boundary decker and the interior decker of the ribbon singularity.

Definition 1.9 (Ribbon immersed disk in S^3)**.** A self-transverse immersed disk $F : D^2 \looparrowright S^3$ is a ribbon immersed disk if every component of its image self-intersection set is a ribbon singularity in the sense of Definition 1.8. Its boundary knot $F(\partial D^2)$ is then called ribbon.

Definition 1.10 (Ribbon datum and defect-free datum)**.** A stable projection datum is a ribbon datum if, after deleting inessential stable artifacts by certified boundary-fixed moves, its immersed disk is a ribbon immersed disk in the sense of Definition 1.9. It is defect-free if its recognized defect set is empty and, when the state is not already clean, the required cleanification and normalization certificates have been supplied. In the clean ribbon-normal setting these two conditions agree by Proposition 2.3; outside that setting the normalization and cleanification certificates are part of the datum.

Proposition 1.11 (Equivalence of the two ribbon descriptions)**.** *For a knot $K \subset S^3$, the following are equivalent.*

(a) *K bounds a ribbon disk in B^4 in the Morse-theoretic sense of Definition 1.7.*

(b) *K bounds a ribbon immersed disk in S^3 in the sense of Definition 1.9.*

Proof. Assume first that $D \subset B^4$ is a ribbon disk and put $r|_D$ in Morse position. Reading the level sets of $r|_D$ from the interior toward S^3, index-zero critical points create trivial circles and index-one critical points attach bands. Since there are no index-two critical points, no component is capped off by a local maximum. Project the trace of these births and band attachments to the final boundary level S^3. The result is an immersed disk whose only self-intersections occur where a band passes through an earlier sheet. Each such intersection has one preimage arc lying on the passing band

with endpoints on the boundary of the disk and the other preimage arc lying in the interior sheet. Hence every singularity is ribbon.

Conversely, suppose $F : D^2 \looparrowright S^3$ has only ribbon singularities. Choose disjoint small three-balls around the singular intervals. In each ball push the interior decker sheet slightly into B^4, in a collar direction transverse to S^3, while keeping the boundary decker in S^3. Outside these balls push the interior of the immersed disk slightly into B^4 and keep the boundary fixed. The local model for a ribbon singularity ensures that the pushed sheets become disjoint; the endpoints on the boundary decker create only saddles in the radial movie. The resulting properly embedded disk has a Morse radial function built from minima and saddles and no local maxima. This proves the equivalence. This is the standard ribbon-disk correspondence; see, for example, [7, 12, 14]. □

1.1.3 Projection data for an embedded disk

Fix a smooth projection

$$\pi : B^4 \longrightarrow B^3$$

which, in a coordinate chart, forgets a fourth coordinate w. Write

$$\Phi = (p, h) : D^2 \longrightarrow B^3 \times \mathbb{R}, \qquad p = \pi \circ \Phi,$$

where h is the fourth-coordinate height along the slice disk. Locally, Φ is an embedding even when p is not.

Definition 1.12 (Stable and clean projection)**.** Let $D \subset B^4$ be a smooth slice disk.

(a) A projection datum (D, p, h) is *stable* if $p : D^2 \to B^3$ has only the standard stable singularities for maps of surfaces to three-manifolds: transverse double curves, isolated triple points, and isolated branch points, all disjoint from the boundary except for prescribed transverse boundary crossings.

(b) A projection datum is *clean* if, in addition, p is an immersion, has no triple points, and all self-intersections are transverse double curves. Clean projections are a convenient normal form for local arguments, but generic projections of embedded surfaces need not be clean without additional moves.

Definition 1.13 (Stable diagram equivalence)**.** Two stable projection data are stably diagram equivalent if they are related by a finite sequence of boundary-fixed ambient isotopies of the lifted disk, stable perturbations of the projection direction rel boundary collar, and Roseman-type local moves for surface diagrams, with all branch, triple, height-ordering, and boundary-collar data tracked. Equivalence is always understood in this certified sense, not merely as equality of the underlying immersed image.

Definition 1.14 (Boundary crossing convention)**.** Boundary crossings of the knot diagram are recorded separately from interior double-decker data. A legal interior move cannot create, remove, or reorder boundary crossings. We use only boundary-fixed moves, so the boundary knot diagram is fixed throughout every certified history.

Remark 1.15 (Role of triple and branch points). For knotted surfaces in four-space, broken surface diagrams and Roseman moves are the surface analogue of knot diagrams and Reidemeister moves.

The singular set of a generic standard projection may include double curves, isolated triple points, and branch points. Therefore any argument formulated only for clean diagrams must either justify a passage to a clean diagram or explicitly include triple and branch data in the defect set.

Lemma 1.16 (Relative transversality for projection data)**.** *Let $D \subset B^4$ be a smooth slice disk and fix a collar of ∂D on which a boundary projection convention is prescribed. After an arbitrarily small ambient isotopy supported away from the fixed boundary collar, the projection datum may be made stable. If a clean datum is required, then the absence of triple and branch points is an additional regularity condition rather than an automatic consequence of genericity.*

Proof. Apply the relative Thom transversality theorem to the map p and to its multijet extensions on D^2, $(D^2)^2 \setminus \Delta$, and $(D^2)^3\setminus$ partial diagonals. Transversality to the diagonal in $B^3 \times B^3$ gives transverse double curves; the corresponding triple condition has expected dimension zero and hence gives isolated triple points; the rank-drop condition for a map $2 \to 3$ gives isolated branch points in the stable list. The perturbation can be chosen rel boundary collar by the relative form of transversality. The final sentence follows because the dimension count permits isolated triple and branch points in stable diagrams, so removing them requires special geometric input or diagrammatic moves, not genericity alone. □

Remark 1.17 (Codimension-two projection data in dimension n). The same encoding has a formal codimension-two analogue. For a properly embedded disk $D^n \subset B^{n+2}$ with boundary in S^{n+1}, choose a projection $B^{n+2} \to B^{n+1}$ and record the forgotten coordinate as a height function. A clean projection has transverse double-decker locus of expected dimension $n-1$; higher multiple strata and rank-drop strata replace the isolated triple and branch points of the surface case. The certificate record is then a finite normal-cell record for these strata, their height orderings, and their boundary-collar behavior. Thus the four-dimensional definitions are the $n = 2$ instance of a dimension-independent codimension-two projection language.

1.1.4 The double-decker set

For a clean projection, the most useful object is not merely the image self-intersection locus in B^3, but the paired preimage locus in D^2.

Definition 1.18 (Double-decker set)**.** Let (D, p, h) be a clean projection datum. Define the ordered double-decker set by

$$\widetilde{\Gamma}(p) = \{(x, y) \in D^2 \times D^2 \mid x \neq y,\ p(x) = p(y)\}.$$

The involution $\tau(x, y) = (y, x)$ acts freely on $\widetilde{\Gamma}(p)$. The unordered double-decker set is

$$\Gamma(p) = \widetilde{\Gamma}(p)/\langle\tau\rangle.$$

The two source decker sets are the projections of $\widetilde{\Gamma}(p)$ to the first and second factors. The image double set is $p(\Gamma) = \{z \in B^3 : |p^{-1}(z)| = 2\}$.

Convention 1.19 (Normalized double-decker lift)**.** For each connected component $C \subset \Gamma(p)$ of a clean projection datum, choose once and for all a connected lift $\widetilde{C}^+ \subset \widetilde{\Gamma}(p)$ such that the height separation

$$\delta_C(x, y) = h(x) - h(y), \qquad (x, y) \in \widetilde{C}^+,$$

is positive. The first source projection of $\widetilde{C}^+$ is called the upper decker of C, and the second source projection is called the lower decker. Replacing $\widetilde{C}^+$ by $\tau(\widetilde{C}^+)$ reverses all signs and interchanges the words upper and lower, so the normalization is a recording convention rather than additional topology. A stable equivalence of normalized data is required either to preserve this convention or to transport it by rechoosing the positive lift in the final state.

Proposition 1.20 (Structure of the clean double-decker set)**.** *For a clean projection datum (D, p, h), $\Gamma(p)$ is a compact one-dimensional smooth manifold with boundary. Consequently each connected component of $\Gamma(p)$ is diffeomorphic either to a compact interval or to a circle. Boundary points of $\Gamma(p)$ occur precisely when at least one of the two source points lies on ∂D^2, under the fixed boundary-crossing convention.*

Proof. Consider the map

$$P : (D^2 \times D^2) \setminus \Delta \longrightarrow B^3 \times B^3, \qquad P(x, y) = (p(x), p(y)).$$

Clean self-transversality means that P is transverse to the diagonal Δ_{B^3}. Since $\dim(D^2 \times D^2) = 4$ and $\operatorname{codim}(\Delta_{B^3}) = 3$, the preimage $P^{-1}(\Delta_{B^3})$ is a one-dimensional submanifold of $(D^2 \times D^2) \setminus \Delta$, with boundary only when one of the factors lies on ∂D^2. The free involution τ preserves this submanifold, so the quotient is again a compact one-dimensional manifold with boundary. The classification of compact one-manifolds gives the interval-or-circle alternative. □

Proposition 1.21 (High-dimensional double-decker stratum)**.** *Let $n \geq 2$, and let (D^n, p, h) be a clean codimension-two projection datum for an embedded disk $D^n \subset B^{n+2}$, where $p : D^n \to B^{n+1}$ is self-transverse away from the diagonal. Then the ordered double-decker set*

$$\widetilde{\Gamma}_n(p) = \{(x, y) \in D^n \times D^n \mid x \neq y,\ p(x) = p(y)\}$$

is a smooth manifold of dimension $n - 1$, with boundary only from the fixed boundary convention, and the free involution $(x, y) \mapsto (y, x)$ gives an unordered double-decker stratum $\Gamma_n(p)$. The height separation $h(x) - h(y)$ is non-zero on each ordered clean stratum because the lifted map is embedded.

Proof. The map $P : (D^n \times D^n) \setminus \Delta \to B^{n+1} \times B^{n+1}$, $P(x, y) = (p(x), p(y))$, is transverse to the diagonal for a clean projection. The source has dimension $2n$, and the diagonal has codimension $n + 1$. Hence $P^{-1}(\Delta)$ has dimension $2n - (n + 1) = n - 1$. The involution is free away from the diagonal and preserves this preimage. If $h(x) = h(y)$ at a paired point, then the lifted points in $B^{n+1} \times \mathbb{R} \cong B^{n+2}$ would coincide, contradicting embeddedness of the lifted disk. This proves the stated high-dimensional form. □

Remark 1.22 (Dimension-independent combinatorial record). The change from $n = 2$ to general n changes the dimension of the double-decker strata, not the logic of the certificate record. Sheet-height passes, decker-stratum slides, empty bigon or empty ball cancellations, and finite palette expansions are still finite normal-cell operations equipped with height-ordering and boundary-fixation data. In dimension two the strata are arcs and circles; in dimension n they are $(n - 1)$-dimensional cells and their incidence strata. The energy function records the same kinds of data: non-clean strata, bad boundary behavior, closed or non-ribbon components, interaction cells, normal subdivision length, and active certificate records.

Definition 1.23 (Height ordering and signs)**.** Let C be a connected component of $\Gamma(p)$, and choose a lift $\widetilde{C} \subset \widetilde{\Gamma}(p)$. The height separation function on that lift is

$$\delta_C(x, y) = h(x) - h(y), \qquad (x, y) \in \widetilde{C}.$$

Because $\Phi = (p, h)$ is an embedding, $\delta_C \neq 0$. On a connected component of a clean datum, its sign is constant. The sign attached to a particular ordered lift changes under τ, but the constancy and the unordered upper/lower sheet relation do not. Unless explicitly stated otherwise, the sign recorded in a state is the sign after the positive-lift normalization of Convention 1.19; hence the normalized value is $+1$ for each clean component, while the induced upper/lower assignment is the meaningful height-ordering datum.

Proof of constancy. The function δ_C is continuous and never zero. Since $\widetilde{C}$ is connected after choosing a connected lift over C, δ_C has constant sign. □

Remark 1.24 (Crossing signs versus height signs). The height sign in Definition 1.23 is not the same as an oriented intersection sign. A broken surface diagram needs height ordering to say which sheet is broken; an oriented double curve may also carry a local sign derived from orientations. The present reductio framework uses height ordering because ribbonness is a decker-type condition.

2 Ribbonness as a three-dimensional combinatorial target

2.1 Ribbon decker type

The condition that the double-point set have no closed loops is insufficient in general. A clean immersed disk may have only interval components of $\Gamma(p)$ and still fail to be a ribbon disk if an interval has the wrong boundary behavior. The relevant condition is therefore a decker condition rather than a condition only on the image double set.

Definition 2.1 (Ribbon-type interval component)**.** Let $C \cong [0, 1]$ be an interval component of $\Gamma(p)$. The component C has ribbon decker type if, after choosing the two source projections of a lift of C, one source arc is properly embedded in D^2 with endpoints on ∂D^2, and the other source arc is embedded in $\mathrm{Int}(D^2)$. The former is the boundary decker; the latter is the interior decker.

Definition 2.2 (Ribbon-normal clean projection)**.** A clean projection datum (D, p, h) is ribbon-normal if every interval component of $\Gamma(p)$ has ribbon decker type. In a ribbon-normal clean projection, the only remaining non-ribbon components of $\Gamma(p)$ are circular components.

Cleanness hypothesis for the criterion. The next criterion concerns clean self-transverse immersions. For a general stable datum, interval decker type alone does not imply ribbonness. We include every branch point, triple point, and normalization failure in Def; hence $\mathrm{Def}(\mathcal{S}) = \emptyset$ implies cleanness only after the corresponding cleanification records have been incorporated into the state.

Proposition 2.3 (Clean ribbon criterion)**.** *Let $F : D^2 \looparrowright S^3$ be a clean self-transverse immersed disk. Then F is a ribbon immersed disk if and only if every connected component of its unordered double-decker set is an interval of ribbon decker type. Equivalently, F is ribbon if and only if it is ribbon-normal and $\Gamma(F)$ has no circular component.*

Proof. If F is ribbon, each image singularity is by definition an embedded interval whose preimage is the disjoint union of one properly embedded boundary decker and one interior decker. Thus every component of the double-decker set is an interval of ribbon decker type.

Conversely, if every double-decker component is an interval of ribbon decker type, then every component of the image self-intersection set is precisely a ribbon singularity in the sense of Definition 1.8. Therefore F is a ribbon immersed disk. The equivalent formulation follows directly from the definition of ribbon-normality and from Proposition 1.20. □

Corollary 2.4 (Closed components are obstructions only after normalization)**.** *For a ribbon-normal clean projection, the presence of a circular double-decker component obstructs ribbonness, and the absence of circular double-decker components implies ribbonness. Without the ribbon-normal hypothesis, absence of circular components alone does not imply ribbonness.*

Proof. The first sentence is exactly Proposition 2.3. For the second, consider an interval component for which both decker arcs meet the boundary, or neither decker has the boundary/interior form required in Definition 2.1. Such a component is not a ribbon singularity although it is not circular. □

Remark 2.5 (High-dimensional ribbon-normal target). For $D^n \subset B^{n+2}$, the ribbon target is formulated in the same Morse-theoretic way: the radial function on the embedded disk should have no interior local maxima. In projection language this means that the allowed self-intersection strata have the higher-dimensional analogue of ribbon decker type: one sheet is attached to the boundary-side movie data, and the other lies in the interior sheet being passed. Closed or non-ribbon double-decker strata, higher multiple strata, and failures of boundary-collar control are the high-dimensional defect strata. Thus the four-dimensional circular-loop obstruction is the low-dimensional shadow of a general closed-stratum obstruction in codimension two.

2.2 Elementary eliminable loops

Circular double-decker components are not all equal. Some are local artifacts that can be removed by an explicit diagrammatic isotopy; others are entangled with the rest of the diagram. The following definition includes only removals that are genuinely certified.

Definition 2.6 (Source circle and its filling disk)**.** Let $\gamma \subset \mathrm{Int}(D^2)$ be an embedded circle. Since D^2 is a disk, $D^2 \setminus \gamma$ has exactly two components, one of which has closure a disk disjoint from ∂D^2. This closure is denoted E_γ and called the source filling disk of γ.

Lemma 2.7 (Planar filling)**.** *Every embedded circle in* $\mathrm{Int}(D^2)$ *bounds a unique source filling disk. No loop theorem is needed for this statement.*

Proof. This is the Jordan curve theorem applied in the disk. The component not meeting ∂D^2 has closure a compact disk bounded by γ, and uniqueness follows from the two-component decomposition. □

Definition 2.8 (Certified elementary loop)**.** Let $C \cong S^1$ be a circular component of $\Gamma(p)$, and let γ_+ and γ_- be its two source decker circles. The component C is a certified elementary loop if at least one decker circle, say γ_+, satisfies the following conditions.

(a) The filling disk E_{γ_+} is disjoint from all source decker sets except along $\partial E_{\gamma_+} = \gamma_+$.

(b) The height separation between the two paired sheets along C is constant and the chosen sheet can be pushed across the projected sheet through a compactly supported isotopy of the lifted embedded disk rel boundary.

(c) During the isotopy no new defect of complexity greater than or equal to that of C is created.

Lemma 2.9 (Elementary loop removal)**.** *If C is a certified elementary loop, then there is a compactly supported isotopy of the embedded disk in B^4, fixed on a boundary collar, whose projected clean diagram removes C and strictly decreases the number of circular double-decker components.*

Proof. Condition (a) gives an innermost source disk on one sheet. Condition (b) is exactly the missing geometric certificate: it supplies a lifted isotopy in the fourth coordinate, so that the chosen sheet passes across the other sheet without creating an actual self-intersection of the embedded disk in B^4. In the projected diagram this is the surface analogue of removing an innermost kink by a local Roseman-type simplification. Condition (c) ensures that the local move decreases the chosen five-coordinate complexity. When this removal is used inside the quenching system, the certificate must also verify the seven-coordinate energy decrease required by Definition 6.8; equivalently, it is treated as a completed package whose final E is smaller. Since the isotopy is supported away from the boundary collar, the boundary knot is unchanged. □

Remark 2.10 (Limits of elementary loop removal). The existence of the planar disk E_γ in Lemma 2.7 does not imply that a circular double-decker component is certified elementary. The disk may contain other decker curves, the height ordering may obstruct the local push, or the simplification may trade one loop for several other defects. Thus the source-level fact that a closed curve bounds a disk does not by itself prove ribbonness.

2.3 Defect sets

Because generic projections may not be clean and clean projections may not be ribbon-normal, the finite obstruction set must include all non-ribbon features.

Definition 2.11 (Ribbon defects)**.** For a stable projection datum $\mathcal{D} = (D, p, h)$, the ribbon-defect set $\mathrm{Def}(\mathcal{D})$ consists of the following finite or compact one-dimensional features:

(a) branch points of the stable surface diagram;

(b) triple points of the stable surface diagram;

(c) interval double-decker components which do not have ribbon decker type;

(d) circular double-decker components;

(e) any recognized local configuration for which the chosen normalization convention fails, where recognized means recorded by the certificate-record set $\mathcal{A}$ of the current quenching state when the datum is being used inside the quenching formalism; see Definition 5.2.

For a ribbon-normal clean datum, $\mathrm{Def}(\mathcal{D})$ is exactly the set of circular double-decker components.

Definition 2.12 (Combinatorial complexity)**.** Fix once and for all a lexicographic complexity

$$\mathrm{Comp}(\mathcal{D}) = \big(b(\mathcal{D}), t(\mathcal{D}), a_{\mathrm{bad}}(\mathcal{D}), c(\mathcal{D}), i(\mathcal{D})\big) \in \mathbb{N}^5,$$

where b is the number of branch points, t the number of triple points, a_{bad} the number of non-ribbon interval components, c the number of circular double-decker components, and i is the interaction count later computed by the transverse counting rules of Definition 6.1 and Convention 6.2. At this stage i should be regarded as the fifth coordinate of a finite well-founded complexity, not as an independent geometric invariant. The order on $\mathbb{N}^5$ is lexicographic.

Remark 2.13 (Choice of complexity). The particular final coordinate i is not canonical. Its purpose is to make local simplifications strictly decreasing after branch, triple, bad-arc, and circle counts are fixed. Any well-founded complexity with this property can be used. Later refinements may replace i by signed intersections, nesting depth, Milnor-type link data, or handle data, provided that every certified move still strictly lowers the chosen complexity.

3 Four-dimensional smoothness and what projection data remembers

3.1 Normal bundles and height functions

Let $D \subset B^4$ be a smooth slice disk. Since D is properly embedded and orientable, its normal bundle is a smooth oriented rank-two disk bundle. Because D^2 is contractible, this bundle is trivial.

Proposition 3.1 (Tubular neighborhood)**.** *Every smooth slice disk $D \subset B^4$ has a tubular neighborhood diffeomorphic to $D^2 \times B^2$, with D identified with $D^2 \times \{0\}$. The trivialization may be chosen product-like near the boundary collar.*

Proof. The tubular neighborhood theorem gives a rank-two disk bundle over D^2. Oriented rank-two vector bundles over a contractible base are trivial because their classifying maps to $BSO(2)$ are null-homotopic. A collar-compatible trivialization is obtained by first fixing it near the boundary and then extending over the contractible disk. □

Definition 3.2 (Projection height on a source filling disk)**.** For an embedded source circle $\gamma \subset \mathrm{Int}(D^2)$, let E_γ be its source filling disk. The projection height on E_γ is the smooth function

$$h_\gamma = h|_{E_\gamma} : E_\gamma \to \mathbb{R}.$$

After an arbitrarily small perturbation rel ∂E_γ, it may be assumed Morse.

Lemma 3.3 (Morse normalization on filling disks)**.** *Let $\gamma \subset \mathrm{Int}(D^2)$ be an embedded source circle. The height function $h|_{E_\gamma}$ can be made Morse by an arbitrarily small perturbation of the projection direction or of the disk supported in a neighborhood of E_γ, without changing the boundary knot.*

Proof. Morse functions are residual in $C^\infty(E_\gamma, \mathbb{R})$, and the relative version allows the function to be fixed on a collar of ∂E_γ if desired. Such a perturbation is realized by a small perturbation of the fourth-coordinate direction or by a compactly supported ambient isotopy in the tubular neighborhood of Proposition 3.1. Since E_γ is contained in the interior of the slice disk, the boundary knot is unaffected. □

3.2 The correct fillability statement

The following lemma is intentionally modest. It is the rigorous statement that can be used without adding unproved cancellation assumptions.

Lemma 3.4 (Fillability with certified limitations)**.** *Let $C \cong S^1$ be a circular double-decker component of a clean projection datum, with source decker circles γ_+ and γ_-. Then:*

(a) *each $\gamma_\pm$ bounds a unique source filling disk $E_\pm \subset D^2$;*

(b) *the fourth-coordinate height restricts to smooth functions $h_\pm : E_\pm \to \mathbb{R}$, which may be made Morse rel boundary;*

(c) *the height separation $h(\gamma_+) - h(\gamma_-)$ along the paired component is nowhere zero and has constant sign;*

(d) *no cancellation, Whitney move, or ribbon simplification follows from* (a)–(c) *alone.*

Proof. Part (a) is Lemma 2.7. Part (b) is Lemma 3.3. Part (c) is Definition 1.23. For (d), observe that (a)–(c) are source and height facts only. A simplification of the projected diagram requires an ambient isotopy of the lifted embedded disk, or equivalently a Roseman/Whitney-type local certificate with the correct framing, disjointness, and height-ordering properties. These extra properties are not encoded merely by the existence of a source filling disk. Hence the asserted automatic cancellation is not a theorem. □

Remark 3.5 (Height separation rather than a distinguished plane). In a projection $B^4 \to B^3$, a point of the slice disk does not lie in the plane $w = 0$ merely because it projects to B^3. The projection forgets w; it does not set w equal to zero. Along a double curve there are two height values, not one. Arguments about closed double curves must therefore be formulated in terms of the height separation function.

3.3 Certified Whitney–Roseman moves

The slice disk is embedded in four dimensions, so its projection defects are shadows rather than actual self-intersections. Nevertheless, eliminating a projection defect still requires a certified isotopy.

Definition 3.6 (Certified projection move)**.** A certified projection move from a stable datum $\mathcal{D}$ to a stable datum $\mathcal{D}'$ consists of:

(a) an ambient isotopy $H_t : B^4 \to B^4$, fixed on a boundary collar, with $H_0 = \mathrm{id}$;

(b) a choice of projection direction after the isotopy, if needed, so that the final datum is stable;

(c) a proof that $\partial(H_1(D)) = K$ as an oriented boundary knot;

(d) a proof that $\mathrm{Comp}(\mathcal{D}') < \mathrm{Comp}(\mathcal{D})$, unless $\mathcal{D}'$ is already a ribbon datum.

Lemma 3.7 (Soundness of certified moves)**.** *A certified projection move preserves sliceness and the boundary knot. If it ends in a ribbon datum, then the boundary knot is ribbon. If it strictly lowers complexity, it cannot occur infinitely often without eventually leaving any fixed initial complexity class.*

Proof. The ambient isotopy carries a properly embedded disk to a properly embedded disk and is fixed near the boundary, so sliceness and the boundary knot are preserved. If the final projected datum is ribbon, Propositions 2.3 and 1.11 imply that the boundary knot is ribbon. Finally, lexicographic order on $\mathbb{N}^5$ is well-founded; there is no infinite strictly decreasing sequence. □

Definition 3.8 (Cleanification certificate)**.** A cleanification certificate for a stable projection datum $\mathcal{D}$ is a finite sequence of certified projection moves, possibly passing through controlled non-clean stable diagrams, whose final state is clean, whose support is disjoint from the boundary collar, and whose completed package is energy-legal after the energy of Definition 6.3 has been chosen. The certificate includes the required local Roseman-type data, the final stable perturbation, and the proof that the lifted surface remains a properly embedded disk throughout the package.

Remark 3.9 (Status of cleanification). For a stable state $\mathcal{S}$, genericity does not imply

$$\exists P \in \mathfrak{P}(\mathcal{S}) \quad \mathrm{Clean}(\mathrm{End}(P)).$$

Whenever such a package is used, its existence is either included in H_{nc} or stated as a separate hypothesis. Statements restricted to clean states explicitly assume $\mathrm{Clean}(\mathcal{S})$.

4 Reductio hypothesis and formal extraction of a minimal core

4.1 The conjecture and its negation

Conjecture 4.1 (Slice–ribbon conjecture)**.** *Every smooth slice knot in S^3 is ribbon.*

Assumption 4.2 (Reductio hypothesis)**.** Assume, for contradiction, that Conjecture 4.1 is false. Thus there exists a knot $K \subset S^3$ which bounds a smooth slice disk in B^4 but does not bound any ribbon disk.

Under Assumption 4.2, every projection datum for every slice disk bounded by K must contain at least one ribbon defect. Otherwise a defect-free clean ribbon-normal datum would give a ribbon immersed disk and hence a ribbon disk for K.

Definition 4.3 (Admissible projection class)**.** For a fixed slice knot K, let $\mathcal{P}(K)$ be the set of all stable projection data $\mathcal{D} = (D, p, h)$, where $D \subset B^4$ ranges over smooth slice disks bounded by K, modulo ambient isotopy fixed on a boundary collar and stable diagram equivalence. Only data for which $\mathrm{Comp}(\mathcal{D})$ is finite are included.

Lemma 4.4 (Existence of a minimal datum)**.** *If K is slice, then $\mathcal{P}(K)$ is non-empty. For every non-empty subset of $\mathcal{P}(K)$, the set of complexities has a lexicographically minimal element. In particular, under Assumption 4.2 there exists a stable projection datum $\mathcal{D}_{\min}$ for K with minimal complexity.*

Proof. Sliceness gives at least one smooth slice disk. By Lemma 1.16, after a small rel-boundary perturbation it has a stable projection datum, so $\mathcal{P}(K) \neq \emptyset$. The complexity takes values in $\mathbb{N}^5$ with lexicographic order. This order is well-founded, hence every non-empty subset has a minimal element. □

4.2 Irreducible defect cores

Definition 4.5 (Local reduction system)**.** A local reduction system $\mathcal{Q}$ is a specified collection of certified projection moves, each supported in a compact ball in the target diagram and each strictly decreasing Comp unless it reaches a ribbon datum. Examples include certified elementary loop removals in the sense of Lemma 2.9 and any additional certified Roseman simplifications chosen in later palettes.

Definition 4.6 (Irreducible core)**.** Let $\mathcal{D}$ be a stable projection datum. Starting from $\mathrm{Def}(\mathcal{D})$, repeatedly apply all moves in $\mathcal{Q}$ that are certified and supported on the current defect configuration. Since each move strictly lowers Comp, every such sequence terminates. A terminal non-empty defect configuration is called a $\mathcal{Q}$-irreducible ribbon-defect core and is denoted $\mathcal{R}_{\mathcal{Q}}(\mathcal{D})$. The core need not be unique unless $\mathcal{Q}$ is confluent.

Lemma 4.7 (Existence, finiteness, and non-uniqueness)**.** *For every stable projection datum $\mathcal{D}$ and every local reduction system $\mathcal{Q}$, at least one terminal $\mathcal{Q}$-irreducible core exists. It is finite as a combinatorial diagram. In general it is not canonically unique.*

Proof. Apply any certified move available in $\mathcal{Q}$. The complexity strictly decreases in the well-founded set $\mathbb{N}^5$, so the process terminates after finitely many steps. The remaining stable diagram has finitely many branch points, triple points, non-ribbon intervals, circular components, and recognized interactions by the definition of stable finite-complexity data. Non-uniqueness is the usual phenomenon for terminating rewriting systems without confluence: different available local moves may lead to different terminal configurations. □

Proposition 4.8 (Non-empty core under the reductio hypothesis)**.** *Assume Assumption 4.2. Let $\mathcal{D}_{\min}$ be a minimal projection datum for the counterexample knot K. For any local reduction system $\mathcal{Q}$, no sequence of certified $\mathcal{Q}$-moves from $\mathcal{D}_{\min}$ can end in a ribbon datum. Hence every terminal $\mathcal{Q}$-irreducible core extracted from $\mathcal{D}_{\min}$ is non-empty.*

Proof. If a certified sequence ended in a ribbon datum, Lemma 3.7 would imply that K is ribbon, contradicting Assumption 4.2. If a terminal core were empty, then all ribbon defects would have been removed. The resulting clean ribbon-normal datum would be ribbon by Proposition 2.3; again K would be ribbon. Therefore the terminal core must be non-empty. □

4.3 Structure and limitations of rigid cores

A rigid singularity core need not be a system of mutually linked closed loops. Pairwise linking number is neither necessary nor sufficient: Whitehead-type links have zero pairwise linking but can be nontrivial, and some obstruction patterns are not closed loops at all but bad decker intervals, branch points, or triple-point configurations.

Definition 4.9 (Interaction graph of a core)**.** Let $\mathcal{R}$ be a finite irreducible core in a stable projection datum. Its interaction graph $G(\mathcal{R})$ has one vertex for each defect component and an edge whenever two components interact by one of the following mechanisms:

(a) a source filling disk for one circular decker component meets another decker component;

(b) two image components have nontrivial linking or nontrivial higher-order link interaction in the complement of the boundary diagram;

(c) the height ordering of one component obstructs a certified move for another;

(d) the components meet at a triple or branch configuration in a stable diagram.

Proposition 4.10 (Minimal-core structure theorem)**.** *Assume Assumption 4.2, and let $\mathcal{R}$ be a $\mathcal{Q}$-irreducible core extracted from a minimal datum $\mathcal{D}_{\min}$. Then:*

(a) *$\mathcal{R}$ is non-empty and finite;*

(b) *no component of $\mathcal{R}$ supports a certified move in $\mathcal{Q}$ by itself;*

(c) *if $\mathcal{D}_{\min}$ is not clean, then $\mathcal{R}$ may contain branch points, triple points, or failed cleanification data, and no closed-loop-only conclusion follows;*

(d) *if $\mathcal{D}_{\min}$ is clean and ribbon-normal, then every component of $\mathcal{R}$ is a circular double-decker component;*

(e) *every circular source decker in $\mathcal{R}$ bounds a source filling disk, but these filling disks need not be free of other defect data;*

(f) *no general theorem forces all circular components of $\mathcal{R}$ to have non-zero pairwise linking number.*

Proof. Part (a) is Proposition 4.8 and the finiteness part of Lemma 4.7. Part (b) is the definition of terminal irreducibility. Part (c) follows because branch points, triple points, and failed normalization or cleanification data are explicitly included in Definition 2.11; without cleanness the defect set is not a collection only of double-decker circles. Part (d) follows from Definition 2.11: for a clean ribbon-normal datum the only defects are circular double-decker components. Part (e) follows from Lemmas 3.4 and 2.7. For (f), pairwise linking number is only one possible obstruction to local simplification. There are nontrivial links with zero pairwise linking numbers, and there are diagrammatic obstructions involving source-disk nesting, triple points, branch points, or height ordering rather than ordinary pairwise linking. Therefore non-zero pairwise linking cannot be a forced property of every minimal core. □

5 Certified topological quenching data and operations

We define topological quenching as the binary relation generated by valid finite package records. If $\mathcal{S} \to \mathcal{S}'$, then the package certificate contains a boundary-fixed ambient isotopy of the lifted disk and satisfies $\mathrm{E}(\mathcal{S}') < \mathrm{E}(\mathcal{S})$. No analytic flow equation is part of the definition.

This distinction is indispensable. A purely pictorial move on the projected double curves is not automatically realized by an isotopy of the slice disk. Conversely, an isotopy of the disk may pass through non-clean intermediate projections with branch or triple events. Thus the evolution must be defined at the level of stable projection data, not only at the level of clean double curves.

The same point is dimension-independent. For a codimension-two disk $D^n \subset B^{n+2}$, a quenching state records the projected $(n-1)$-dimensional double-decker strata, higher multiple strata, height orderings, boundary collars, and finite normal-cell certificates. The local names used below—sheet-height pass, decker-arc slide, and empty-bigon cancellation—are the $n = 2$ surface names for operations whose high-dimensional versions are sheet-height passes, decker-stratum slides, and empty normal-ball cancellations. The validity condition is unchanged: the completed package must be boundary-fixed, liftable to an embedded isotopy in B^{n+2}, and strictly decreasing for the chosen lexicographic energy.

Initialization from projection data. Given a stable projection datum $\mathcal{D} = (D, p, h)$, define $\mathcal{S}(\mathcal{D})$ by the fixed subdivision rule, the defect set $\mathrm{Def}(\mathcal{D})$, the empty initial certificate queue, and the normalized height-ordering record of Convention 1.19. This assignment is well defined on stable-equivalence classes. If $\mathcal{D}$ is non-clean, its branch and triple strata remain in $\mathrm{Def}(\mathcal{S}(\mathcal{D}))$ until a valid cleanification package is applied.

In the high-dimensional version the phrase "source decker graph" is replaced by the finite normal-cell complex carried by $\Gamma_n(p)$ and its higher singular strata. Finiteness is a statement about the encoded cell structure and certificate records, not about the analytic function space of all maps. This is why the ZFC selector and finite-palette arguments below extend formally once the relevant high-dimensional local packages have been supplied.

Definition 5.1 (Boundary collar and fixed frame)**.** For a slice disk $D \subset B^4$ bounded by K, fix a collar

$$\nu(\partial D) \cong \partial D \times [0, \varepsilon]$$

and a global oriented trivialization of the normal bundle

$$\tau : \nu_D \cong D^2 \times \mathbb{R}^2,$$

whose existence follows from Proposition 3.1. The fixed normal frame on the boundary collar is the restriction of τ, and the boundary projection convention is chosen relative to this frame. Consequently

$$e(\nu_D, \tau|_{\partial D}) = 0.$$

A move is *boundary-fixed* if its supporting ambient isotopy is the identity on $\nu(\partial D)$. In particular, the boundary knot, the chosen diagram, and the extendible boundary normal frame are unchanged throughout the move.

Definition 5.2 (Quenching state)**.** A quenching state is a tuple

$$\mathcal{S} = (D, p, h, \Sigma, \mathrm{Def}, \mathbf{w}, \mathcal{N}, \mathcal{A}),$$

where (D, p, h) is a stable projection datum for a slice disk, Σ is the associated finite singular surface diagram in B^3, $\mathrm{Def} = \mathrm{Def}(D, p, h)$ is the ribbon-defect set of Definition 2.11, $\mathbf{w}$ records the normalized upper/lower decker choices of Convention 1.19 and Definition 1.23, $\mathcal{N}$ is the adapted normal cell structure, and $\mathcal{A}$ is the finite set of certificate records currently recognized by the

certificate system. Two states are considered equivalent if they differ by a boundary-fixed ambient isotopy and by stable diagram equivalence preserving the defect data, normalized height-ordering data, and recognized certificate records.

Definition 5.3 (High-dimensional quenching state)**.** For $n \geq 2$, a high-dimensional codimension-two quenching state is a tuple

$$\mathcal{S}_n = (D^n, p, h, \Sigma_n, \mathrm{Def}_n, \mathbf{w}_n, \mathcal{N}_n, \mathcal{A}_n),$$

where (D^n, p, h) is a stable projection datum for a properly embedded disk $D^n \subset B^{n+2}$, $p : D^n \to B^{n+1}$, Σ_n is the finite normal-cell record of the projected multiple-point strata, Def_n is the set of non-clean, non-ribbon, closed, or boundary-degenerate defect cells, $\mathbf{w}_n$ records the sheet-height ordering on every clean double-decker cell, $\mathcal{N}_n$ is a finite source normal-cell subdivision adapted to all strata and boundary collars, and $\mathcal{A}_n$ is the finite set of recognized certificate records. Two such states are equivalent when they are related by boundary-fixed ambient isotopy of the lifted disk and by stable high-dimensional diagram equivalence preserving the finite defect, height-ordering, normal-cell, and certificate data.

Remark 5.4 (Why the high-dimensional state is still finite)*.* Although the geometric strata of $\mathcal{S}_n$ have positive dimension, the state is an encoded finite object: it records a finite cell complex, incidence data, height-ordering data, and finitely many certificate records. The analytic assertion that this finite record is realized by a smooth projected disk is part of certificate validity, while the selector and termination arguments use only the encoded finite record.

Definition 5.5 (Faces, arcs, and vertices)**.** For a stable state $\mathcal{S}$, the combinatorial surface diagram has:

(a) vertices: branch points, triple points, endpoints of boundary double-decker intervals, and any additional recognized endpoints used by the chosen cell subdivision;

(b) arcs: connected components of the double-decker set after removing the vertices;

(c) faces: connected components of D^2 after removing the source decker graph and the boundary collar.

Faces are source faces. Their projections may overlap in B^3, and their boundaries may contain decker arcs. A face is not automatically movable merely because it is a component of this complement.

Convention 5.6 (Normal cell structure)**.** After each stable state is obtained, choose a finite cell subdivision of D^2 adapted to the source decker graph, the boundary collar, and all recognized disks used by local certificates. Arcs and strips appearing in legal moves are required to be normal with respect to this subdivision. This convention removes irrelevant metric choices and makes the auxiliary complexity below discrete.

Definition 5.7 (The three quenching constraints)**.** A quenching move must satisfy the following constraints.

(A) *Boundary invariance*: the move is boundary-fixed in the sense of Definition 5.1.

(B) *Topological preservation*: the source remains a single smooth disk, and the lifted surface remains a properly embedded disk in B^4.

(C) *Discrete monotonicity*: the move strictly decreases a specified well-founded integer complexity.

Remark 5.8 (Use of discrete rather than metric length). A real-valued length functional on decker arcs is useful for intuition but not for finite termination. A strictly decreasing sequence of non-negative real numbers can be infinite. The termination proof below therefore uses only lexicographically ordered integer data. If a geometric length is later introduced, it must be converted into a discrete normal length or used only inside a certified finite search.

5.1 Certified local operations and legality criteria

This section defines the allowed local operations. Each operation is a *certified* move: the definition includes the local geometric hypotheses needed to realize the move by a boundary-fixed isotopy of the embedded disk. Without such hypotheses, the projected local picture has no independent mathematical force.

Existential status of certificates. A certificate is a finite geometric tuple c with a predicate $\mathrm{Valid}(c)$. We prove $\mathrm{Valid}(c) \Rightarrow \mathrm{Sound}(c)$ and termination for any selected sequence of valid energy-decreasing packages. We do not provide a decision procedure for Valid.

5.1.1 Operation I: certified sheet-height pass

Definition 5.9 (Sheet-height pass certificate)**.** Let F be a source face of a quenching state $\mathcal{S}$. A sheet-height pass certificate for F is a finite tuple consisting of:

(a) a compact subsurface $F_0 \subset \mathrm{Int}(F)$ with smooth boundary, separated from the fixed boundary collar;

(b) a four-dimensional product neighborhood $U \cong F_0 \times B^2$ in which the disk is $F_0 \times \{0\}$;

(c) a target height function h' on F_0, equal to h near ∂F_0, such that the homotopy from h to h' keeps the lifted sheet disjoint from all other lifted sheets;

(d) a stable perturbation of the final projection and a verified comparison

$$\mathrm{E}(\mathcal{S}') < \mathrm{E}(\mathcal{S})$$

for the integer energy of Definition 6.3.

The tuple is valid precisely when these data are realized by a compactly supported ambient isotopy of the lifted disk, followed by the stated stable perturbation. The resulting certified move is called a sheet-height pass.

Remark 5.10 (Certified face lifting). An entire face whose boundary contains double-decker arcs cannot in general be lifted while those boundary arcs remain fixed. The boundary conditions and adjacent sheets may obstruct such a motion. The certificate in Definition 5.9 is precisely the extra data that makes the move legitimate.

Lemma 5.11 (Soundness of sheet-height passes)**.** *A certified sheet-height pass is realized by a boundary-fixed ambient isotopy of the slice disk and preserves the boundary knot.*

Proof. The product neighborhood identifies the local disk with the zero section over F_0. Since $h' = h$ near ∂F_0, interpolate by a compactly supported smooth section of the normal bundle. The disjointness clause in the certificate says that the resulting lifted surface remains embedded during the interpolation. Extending the isotopy by the identity outside U and on the boundary collar gives a boundary-fixed ambient isotopy by the isotopy extension theorem. Hence the boundary knot is unchanged. □

5.1.2 Operation II: certified decker-arc slide

Definition 5.12 (Decker-arc slide certificate)**.** Let a be a normal source decker arc whose endpoints are vertices of the adapted cell structure. A decker-arc slide certificate from a to a normal arc a' is a finite tuple consisting of:

(a) an embedded strip $B \subset D^2 \setminus \nu(\partial D)$ with boundary $a \cup a' \cup e_0 \cup e_1$, where e_0, e_1 are small endpoint arcs;

(b) disjointness of $\mathrm{Int}(B)$ from the source decker graph, except for explicitly listed transverse crossings that are part of a stable Roseman event;

(c) a compatible height-ordering extension over the corresponding lifted sheets, ensuring that the slide is realized by an embedded isotopy in B^4;

(d) a proof that the final state is stable after perturbation and that either the integer energy decreases, or the slide is a preparatory submove inside a quenching package whose net energy decreases.

The tuple is valid precisely when it determines a boundary-fixed ambient isotopy of the lifted disk realizing the slide, possibly as a submove of a larger package.

Lemma 5.13 (Soundness of certified arc slides)**.** *A certified decker-arc slide preserves the boundary knot and the topological type of the source disk. If it is not itself energy-decreasing, it is legal only as part of a finite package with certified net energy decrease.*

Proof. The strip gives a standard isotopy of an arc on the source disk rel endpoints. The height-ordering clause is the condition that lifts this source isotopy to an ambient isotopy of the embedded disk in B^4. The support is disjoint from the boundary collar, so the boundary knot is fixed. The final clause is part of the definition and prevents infinite sequences of energy-neutral slides. □

Remark 5.14 (Slides do not preserve linking in general). If closed image curves are present, a slide can change ordinary pairwise linking numbers by passing through other sheets or through a stable triple event. Therefore pairwise linking is not used as an unconditional Lyapunov function. It may be included only as a certified auxiliary weight for a restricted move class.

5.1.3 Operation III: certified empty-bigon cancellation

Definition 5.15 (Empty-bigon cancellation certificate)**.** An empty-bigon cancellation certificate is a finite tuple consisting of two paired decker arcs a_1, a_2, two embedded disks $B_{\mathrm{src}} \subset D^2$, $B_{\mathrm{img}} \subset B^3$, and the following verification data:

(a) the boundary of B_{src} is made from the source arcs corresponding to a_1, a_2 and small endpoint arcs;

(b) the boundary of B_{img} is the projected bigon between the corresponding sheets;

(c) the interiors of both bigons are free of source decker data, image singular data, and boundary-collar data, except for explicitly certified harmless stable events;

(d) the height ordering along the two sides is opposite in exactly the way needed for a local cancellation;

(e) the local isotopy removes the indicated pair of defect arcs or vertices and the final stable state has strictly smaller integer energy.

The tuple is valid precisely when the listed data are realized by a boundary-fixed ambient isotopy of the lifted disk. The move determined by such a certificate is a certified empty-bigon cancellation.

Lemma 5.16 (Soundness of empty-bigon cancellation)**.** *A certified empty-bigon cancellation is realized by a boundary-fixed ambient isotopy of the embedded disk. It removes the certified local defect and creates no boundary change.*

Proof. The source and image bigons give the usual local model for a Reidemeister-II or Whitney-bigon type simplification of a projected sheet diagram. The emptiness assumptions prevent uncontrolled intersections with other sheets. The height-ordering condition ensures that the two projected intersections are shadows of sheets that can be separated in the fourth coordinate during the isotopy. Since the supporting neighborhood avoids the boundary collar, the isotopy fixes the boundary knot. The final energy decrease is part of the certificate. □

Remark 5.17 (Relation with the Whitney trick). The cancellation above is not the unrestricted four-dimensional Whitney trick. The double curves in a projection of an embedded disk are not actual self-intersections of the disk in B^4. The relevant object is a certified local isotopy of the projection data. Framing, emptiness, and height-ordering hypotheses are therefore built into the certificate.

5.1.4 Quenching packages

Definition 5.18 (Quenching package)**.** A quenching package is a finite sequence of certified sheet-height passes, decker-arc slides, stable perturbations, and empty-bigon cancellations

$$\mathcal{S} = \mathcal{S}_0 \longrightarrow \mathcal{S}_1 \longrightarrow \cdots \longrightarrow \mathcal{S}_m = \mathcal{S}'$$

such that every submove is boundary-fixed and sound, and the net energy satisfies $\mathrm{E}(\mathcal{S}') < \mathrm{E}(\mathcal{S})$. Intermediate slides are allowed to be energy-neutral, or even to increase an auxiliary linking weight, only inside the package and only when the net integer energy decreases.

Proposition 5.19 (Soundness of packages)**.** *Every quenching package is realized by a boundary-fixed ambient isotopy of the slice disk, followed by a stable perturbation of the projection. It preserves sliceness, the boundary knot, and the source-disk topology.*

Proof. Each certified submove is sound by Lemmas 5.11, 5.13 and 5.16. Concatenating the ambient isotopies gives a boundary-fixed ambient isotopy from the initial embedded disk to the final embedded disk. A final stable perturbation is performed rel boundary collar by Lemma 1.16. Thus the boundary knot and the fact that the source is a properly embedded disk are preserved. □

6 Energy, termination, and liftability

The energy must be discrete. A weighted sum of double points, closed loops, and pairwise linking numbers is useful as motivation, but it is not robust enough for a proof: pairwise linking may be undefined for non-closed components, may vanish for nontrivial Whitehead-type configurations, and need not be monotone under slides. The quenching theory therefore uses an integer lexicographic energy adapted to the certified move system.

From this chapter onward, the seven-tuple energy E refines and replaces the five-tuple complexity Comp of Definition 2.12. The first five coordinates of E correspond to the branch, triple, bad-arc, circular-component, and interaction coordinates of Comp. The last two coordinates, normal preparation length and certificate queue length, are added to control bounded backtracking and certificate bookkeeping. A move that strictly lowers Comp by decreasing one of the first five coordinates is admitted as a quenching step only when its certificate also proves that the completed package strictly lowers E; in particular, the termination theorem below uses E, not Comp, as the final Lyapunov function.

Definition 6.1 (Interaction and preparation indices)**.** For a quenching state $\mathcal{S}$, let:

(a) $b(\mathcal{S})$ be the number of branch points;

(b) $t(\mathcal{S})$ be the number of triple points;

(c) $a(\mathcal{S})$ be the number of interval double-decker components that are not of ribbon decker type;

(d) $c(\mathcal{S})$ be the number of circular double-decker components;

(e) $i(\mathcal{S})$ be the interaction number of the defect set, computed by Convention 6.2 from the currently recognized source filling disks, certified strips, and remaining source decker graph in the chosen normal cell structure;

(f) $n(\mathcal{S})$ be the normal preparation length, namely the total number of normal elementary arc segments, counted by Convention 6.2, in the currently recognized strips and filling disks;

(g) $q(\mathcal{S}) = |\mathcal{A}(\mathcal{S})|$ be the queue length, where $\mathcal{A}(\mathcal{S})$ is the finite set of certificate records carried by the state, namely the currently recognized but not yet completed or abandoned certificate candidates.

All these numbers are finite non-negative integers.

Convention 6.2 (Energy computation rules)**.** The last three energy coordinates are computed as follows.

(a) Before counting intersections, all recognized filling disks and strips are put, by an arbitrarily small perturbation inside their certificate neighborhoods and rel their prescribed boundaries, in transverse position with the source decker graph and with each other. This perturbation is part of the certificate record and does not change the stable equivalence class of the quenching state.

(b) The interaction number $i(\mathcal{S})$ is the number of transverse interior intersection points among the recognized filling disks, certified strips, and the remaining source decker graph. Boundary coincidences, shared prescribed endpoints, and arcs lying entirely in the decker graph are not counted; if such coincidences are relevant, the certificate must first subdivide or perturb them into transverse interior intersections.

(c) The normal preparation length $n(\mathcal{S})$ counts only normal elementary arc segments of recognized strips and filling disks that are not contained in the source decker graph. Repeated traversal of the same normal segment by distinct certificate records is counted with multiplicity.

(d) The queue length $q(\mathcal{S})$ is a function of the state, not of an external clock. A state transition may add certificate records to $\mathcal{A}$, complete records, abandon records, or replace one record by finitely many subrecords; the new set $\mathcal{A}(\mathcal{S}')$ is part of the target state. Thus $q(\mathcal{S}') = |\mathcal{A}(\mathcal{S}')|$ is well defined. Such a bookkeeping transition is a legal quenching step only when the completed package has lower total energy.

Definition 6.3 (Integer quenching energy)**.** The integer energy of a quenching state is

$$\mathrm{E}(\mathcal{S}) = \big(b(\mathcal{S}), t(\mathcal{S}), a(\mathcal{S}), c(\mathcal{S}), i(\mathcal{S}), n(\mathcal{S}), q(\mathcal{S})\big) \in \mathbb{N}^7,$$

ordered lexicographically. A state is lower energy than another state if its vector is smaller in this lexicographic order.

Definition 6.4 (High-dimensional quenching energy)**.** For a high-dimensional state $\mathcal{S}_n$, choose a finite list of defect coordinates

$$\mathrm{E}_n(\mathcal{S}_n) = (s_1(\mathcal{S}_n), \dots, s_r(\mathcal{S}_n), i(\mathcal{S}_n), \nu(\mathcal{S}_n), q(\mathcal{S}_n)) \in \mathbb{N}^{r+3},$$

ordered lexicographically. Here the coordinates s_j count the finitely many declared types of non-clean strata, non-ribbon double-decker cells, closed defect cells, boundary-collar degeneracies, and higher multiple-point cells used by the chosen high-dimensional palette; i counts certified interaction cells after transverse normal-cell preparation; ν counts the total normal subdivision length or volume of recognized certificate supports; and q is the certificate-queue length. A high-dimensional package is energy-legal if it strictly lowers E_n.

Remark 6.5 (The four-dimensional energy as the first case). For $n = 2$, the coordinates (b, t, a, c, i, n, q) of Definition 6.3 are the specialized version of Definition 6.4: branch points, triple points, bad arcs, closed circles, interaction points, normal preparation length, and certificate queue length. In higher dimensions, the names of the first coordinates change because the singular strata have higher dimension, but the well-founded lexicographic mechanism is identical.

Lemma 6.6 (Well-definedness of the energy)**.** *For every quenching state,* $\mathrm{E}(\mathcal{S})$ *is a finite element of* $\mathbb{N}^7$*. The lexicographic order on the set of such energies is well-founded.*

Proof. Stability gives finitely many branch points, triple points, and double-decker components. The adapted normal cell structure is finite, and Convention 6.2 counts only finitely many transverse intersections, normal segments, and certificate records. Thus the energy lies in $\mathbb{N}^7$. The lexicographic product of finitely many copies of the well-ordered set $\mathbb{N}$ is well-founded: an infinite strictly decreasing sequence would force an infinite strictly decreasing sequence in the first coordinate that changes infinitely often, or eventually in the second coordinate, and so on, which is impossible after finitely many coordinates. □

Lemma 6.7 (High-dimensional energy well-foundedness)**.** *For every fixed high-dimensional palette and every high-dimensional quenching state $\mathcal{S}_n$, the energy $\mathrm{E}_n(\mathcal{S}_n)$ of Definition 6.4 is a finite element of $\mathbb{N}^{r+3}$. The lexicographic order on these energies is well-founded.*

Proof. By Remark 5.4, the state is represented by a finite normal-cell complex and finitely many certificate records. Each coordinate of E_n counts a finite set of cells, incidences, subdivisions, or records in that finite encoding. Hence $\mathrm{E}_n(\mathcal{S}_n) \in \mathbb{N}^{r+3}$. A finite lexicographic product of copies of $\mathbb{N}$ is well-founded, exactly as in the proof of Lemma 6.6. □

Definition 6.8 (Energy-legal move)**.** A certified local move or quenching package is energy-legal if its initial and final states satisfy

$$\mathrm{E}(\mathcal{S}') < \mathrm{E}(\mathcal{S}).$$

Only energy-legal packages are steps of the quenching evolution.

Proposition 6.9 (Monotonicity of legal operations)**.** *Every quenching step strictly lowers the integer energy. In particular, a step may contain preparatory slides whose local pictorial effect is not monotone, but the package as a whole is monotone.*

Proof. This is exactly the legality condition in Definitions 6.8 and 5.18. The point is not circular: the certificates are local geometric data, while energy-legality is an additional admission rule for the algorithm. A pictorial operation not satisfying this rule is not a quenching step, even if it is an ambient isotopy. □

Remark 6.10 (Optional linking weight). If a restricted clean ribbon-normal class contains oriented closed image components, one may add an auxiliary term

$$\ell(\mathcal{S}) = \sum_{i<j} |\operatorname{lk}(\gamma_i, \gamma_j)|$$

after the primary defect counts. This is harmless only when every allowed move has a certificate proving that the enlarged vector decreases. It is not part of the unconditional energy because ordinary pairwise linking does not detect all rigid configurations and is not monotone under all diagram slides.

6.1 Finite termination and terminal states

Definition 6.11 (Certified package relation and selected evolution)**.** Let $\mathfrak{P}$ be a set of valid completed quenching packages. For $\mathcal{S} \in \mathfrak{S}$, define

$$\mathfrak{P}(\mathcal{S}) = \{P \in \mathfrak{P} : \operatorname{Init}(P) = \mathcal{S},\ \mathrm{E}(\operatorname{End}(P)) < \mathrm{E}(\mathcal{S})\}.$$

A selector for $\mathfrak{P}$ is a function $\sigma_{\mathfrak{P}}$ such that

$$\sigma_{\mathfrak{P}}(\mathcal{S}) \in \mathfrak{P}(\mathcal{S}) \quad \text{if } \mathfrak{P}(\mathcal{S}) \neq \emptyset,$$

and $\sigma_{\mathfrak{P}}(\mathcal{S})$ is the constant package at $\mathcal{S}$ otherwise. The selected evolution is defined recursively by

$$\mathcal{S}_{j+1} = \mathrm{End}(\sigma_{\mathfrak{P}}(\mathcal{S}_j)).$$

If $\mathfrak{P}$ is generated by a finite palette $\mathcal{Q}$, we write $\mathfrak{P}_{\mathcal{Q}}$, $\sigma_{\mathcal{Q}}$, and $\mathrm{Term}_{\mathcal{Q},\sigma}(\mathcal{S}_0)$.

Theorem 6.12 (Finite termination)**.** *Let $\mathfrak{P}$ be any set of valid energy-legal completed packages, and let $\sigma_{\mathfrak{P}}$ be any selector. For every initial state $\mathcal{S}_0$, there exists $N \in \mathbb{N}$ such that*

$$\mathcal{S}_{N+k} = \mathcal{S}_N \qquad \textit{for every } k \in \mathbb{N}.$$

Thus every selected evolution has finitely many nonconstant steps. No finiteness assumption on $\mathfrak{P}$ is required.

Proof. Assume, for contradiction, that infinitely many selected steps are nonconstant. Let $j_0 < j_1 < j_2 < \cdots$ be their indices. By the definition of $\mathfrak{P}(\mathcal{S})$,

$$\mathrm{E}(\mathcal{S}_{j_{m+1}}) < \mathrm{E}(\mathcal{S}_{j_m}) \qquad (m \in \mathbb{N}).$$

This is an infinite strictly decreasing sequence in $\mathbb{N}^7$ with lexicographic order, contradicting Lemma 6.6. Hence only finitely many nonconstant steps occur. After the last such step the selector returns the constant package, so the sequence is constant. □

Theorem 6.13 (High-dimensional finite quenching theorem)**.** *Fix $n \geq 2$, a high-dimensional encoded state space, and any set $\mathfrak{P}_n$ of boundary-fixed, liftable, E_n-legal completed packages. Every selected high-dimensional evolution is eventually constant. Its terminal output is either high-dimensionally ribbon terminal or blocked relative to $\mathfrak{P}_n$.*

Proof. Choose a selector as in Theorem 6.14. If infinitely many selected packages were nonconstant, their endpoint energies would form an infinite strictly decreasing sequence in $\mathbb{N}^{r+3}$, contrary to Lemma 6.7. Hence the history is eventually constant at a state $\mathcal{S}_N$ with $\mathfrak{P}_n(\mathcal{S}_N) = \emptyset$. If $\mathrm{Def}_n(\mathcal{S}_N) = \emptyset$ after the high-dimensional ribbon-normal convention, the state is ribbon terminal; otherwise it is blocked relative to $\mathfrak{P}_n$. □

Theorem 6.14 (ZFC selector and terminal-output certificate)**.** *Let $\mathfrak{P}$ be any set of valid energy-legal completed packages on the encoded state set $\mathfrak{S}$. In ZFC there exists a selector*

$$\sigma_{\mathfrak{P}}(\mathcal{S})$$

which chooses an element of $\mathfrak{P}(\mathcal{S})$ whenever that set is non-empty and chooses the constant package otherwise. For every initial state $\mathcal{S}_0$, the resulting endpoint iteration has a unique eventual value

$$\mathcal{S}_0 \longrightarrow \mathcal{S}_1 \longrightarrow \cdots \longrightarrow \mathcal{S}_N = \mathrm{Term}_{\mathfrak{P},\sigma}(\mathcal{S}_0),$$

and $\mathfrak{P}(\mathcal{S}_N) = \emptyset$. The selector need not be computable.

Proof. By hypothesis, $\mathfrak{P}$ is a set. The well-ordering theorem gives a well-order $\preceq$ on $\mathfrak{P}$. Define $\sigma_{\mathfrak{P}}(\mathcal{S})$ to be the $\preceq$-least element of $\mathfrak{P}(\mathcal{S})$ when $\mathfrak{P}(\mathcal{S}) \neq \emptyset$, and the constant package otherwise. This satisfies the selector axioms of Definition 6.11. Eventual constancy follows from Theorem 6.12. At the eventual value $\mathcal{S}_N$, the selector is constant; by construction this is equivalent to $\mathfrak{P}(\mathcal{S}_N) = \emptyset$. Uniqueness holds after $\sigma_{\mathfrak{P}}$ is fixed because the recursion defining $\mathcal{S}_{j+1}$ is deterministic. □

Remark 6.15 (Scope of the ZFC selector). Theorem 6.14 proves only the existence of a choice function on a declared set of valid packages. It does not prove membership in $\mathfrak{P}$, validity of a proposed certificate, or the external package H_{nc}. If $\mathfrak{P} = \mathfrak{P}_{\mathcal{Q}}$ comes from a finite palette, the preceding theorem specializes to the original finite-palette algorithm.

Definition 6.16 (Quenched terminal state)**.** A state $\mathcal{S}$ is $\mathfrak{P}$-terminal if $\mathfrak{P}(\mathcal{S}) = \emptyset$. It is ribbon terminal if $\mathrm{Clean}(\mathcal{S})$ and $\mathrm{Def}(\mathcal{S}) = \emptyset$. It is blocked terminal relative to $\mathfrak{P}$ if it is $\mathfrak{P}$-terminal and either $\neg\,\mathrm{Clean}(\mathcal{S})$ or $\mathrm{Def}(\mathcal{S}) \neq \emptyset$. For $\mathfrak{P} = \mathfrak{P}_{\mathcal{Q}}$, these predicates are denoted $\mathcal{Q}$-terminal and $\mathcal{Q}$-blocked.

Proposition 6.17 (Terminal alternative)**.** *For every package relation $\mathfrak{P}$, selector $\sigma_{\mathfrak{P}}$, and initial state $\mathcal{S}_0$, the eventual value $\mathcal{S}_N$ in Theorem 6.12 satisfies exactly one of the following:*

(a) $\mathrm{RibbonTerm}(\mathcal{S}_N)$, *equivalently* $\mathrm{Clean}(\mathcal{S}_N) \wedge \mathrm{Def}(\mathcal{S}_N) = \emptyset$;

(b) $\mathrm{BlockedTerm}_{\mathfrak{P}}(\mathcal{S}_N)$, *equivalently* $\mathfrak{P}(\mathcal{S}_N) = \emptyset$ *and* $\neg\,\mathrm{Clean}(\mathcal{S}_N) \vee \mathrm{Def}(\mathcal{S}_N) \neq \emptyset$.

Proof. By Theorem 6.14, $\mathfrak{P}(\mathcal{S}_N) = \emptyset$. The proposition is therefore the Boolean partition

$$\big(\mathrm{Clean}(\mathcal{S}_N) \wedge \mathrm{Def}(\mathcal{S}_N) = \emptyset\big) \,\dot{\vee}\, \big(\neg\,\mathrm{Clean}(\mathcal{S}_N) \vee \mathrm{Def}(\mathcal{S}_N) \neq \emptyset\big).$$

The two formulas are negations of one another, hence are mutually exclusive and exhaustive. In the first case Proposition 2.3 identifies the normalized projection with a ribbon immersed disk. □

Remark 6.18 (High-dimensional terminal alternative). The same terminal alternative is available in the high-dimensional encoded system of Theorem 6.13. A terminal state is ribbon terminal when all high-dimensional defect cells vanish after the ribbon-normal convention of Remark 2.5; otherwise it is blocked relative to the chosen finite palette. No analytic compactness theorem is used in this discrete alternative: terminality is a statement about the absence of recognized energy-legal packages in the finite encoded state.

Remark 6.19 (Ribbon terminal does not mean empty double set). A ribbon terminal state need not have $\Gamma = \emptyset$. A nontrivial ribbon knot typically has a ribbon immersed disk with ribbon singular arcs. The correct ribbon condition is absence of ribbon defects, not absence of all double-decker components.

Lemma 6.20 (Blocked cores are local minima only for the chosen palette)**.** *A blocked terminal state is irreducible only relative to $\mathcal{Q}$ and* E*. It need not be globally minimal among all possible projection data for the same slice disk or for the same boundary knot.*

Proof. Terminality says exactly that no move in the fixed finite palette has an energy-legal certificate at the state. A different palette, a different complexity, or a non-local isotopy may contain a simplification not visible to $\mathcal{Q}$. Hence terminality is relative. □

6.2 Four-dimensional liftability and invariant preservation

Theorem 6.21 (Liftability of the quenching evolution)**.** *Let*

$$\mathcal{S}_0 \to \mathcal{S}_1 \to \cdots \to \mathcal{S}_N$$

be a certified quenching evolution. Then there is a boundary-fixed ambient isotopy $H_t : B^4 \to B^4$ and a sequence of stable projection directions or stable perturbations rel boundary collar realizing the successive states. In particular, every state is the projection data of a smooth properly embedded disk in B^4 bounded by the original knot K.

Proof. Each step is a quenching package. By Proposition 5.19, each package is realized by a boundary-fixed ambient isotopy of the embedded disk, followed by a stable perturbation of the projection rel boundary collar. Concatenating finitely many such isotopies gives the required ambient isotopy from the initial disk to the final disk. Since all perturbations are rel boundary collar, the boundary knot and its diagram remain fixed. □

Corollary 6.22 (Preservation of sliceness and boundary knot)**.** *Certified quenching preserves smooth sliceness, the source-disk topology, and the oriented boundary knot.*

Proof. By Theorem 6.21, every intermediate and final surface is obtained from the original properly embedded disk by ambient isotopy rel boundary collar. Ambient isotopy preserves being a smooth properly embedded disk and preserves the boundary knot with orientation. □

Proposition 6.23 (Endpoint preservation certificate)**.** *Let*

$$\mathcal{S}_0 \to \mathcal{S}_1 \to \cdots \to \mathcal{S}_N$$

be any completed certified quenching history, with each package supported away from the fixed boundary collar. Then there is a canonical boundary identification under which the initial and terminal boundary knots are identical as oriented framed boundary data. More precisely, the concatenated ambient isotopy supplied by Theorem 6.21 restricts to the identity on the chosen collar $\nu(\partial D)$, and the terminal disk is a smooth proper disk with the same boundary embedding, the same boundary orientation, and the same adopted boundary projection convention as the initial disk.

Proof. Each certified submove in a package is boundary-fixed by Definitions 5.1, 5.7 and 5.18. Therefore the ambient isotopy realizing that package is the identity on $\nu(\partial D)$. The concatenation of finitely many isotopies that are identity on the same collar is again identity on that collar. The stable perturbations allowed after packages are also performed rel boundary collar by Lemma 1.16 and Proposition 5.19. Hence no endpoint of the boundary circle is moved, no orientation convention is reversed, and no boundary projection convention is changed. The terminal object is still a proper smooth disk by Corollary 6.22, so the two ends of the certified history agree in the stated boundary-fixed sense. □

Remark 6.24 (Scope of endpoint preservation). Proposition 6.23 proves equality of the initial and terminal boundary data for all completed certified histories. It permits interior projection freedom, temporary certified singular events, and bounded-backtracking packages without changing the boundary knot. It does not assert that the interior freedom is large enough to remove every defect.

Remark 6.25 (Non-preserved diagrammatic quantities). The projection diagram, the number of double-decker components, pairwise linking numbers among image curves, and the chosen cell structure need not be preserved. These are diagrammatic quantities, not invariants of the slice disk. The algorithm is allowed to change them only through certified packages with net energy decrease.

6.3 Strategy completeness and bounded backtracking

The finite termination theorem is unconditional for any fixed certified palette. It does not imply that the palette finds every possible simplification. A greedy rule can stop at a local minimum even when a different, temporarily more complicated route would eventually simplify the diagram. This section records the correct completeness statement.

Definition 6.26 (Bounded-backtracking package)**.** A bounded-backtracking package of depth B is a quenching package whose intermediate states are allowed to have energy at most B lexicographic levels above the initial state in coordinates after the first coordinate that eventually decreases, but whose final state has strictly smaller energy than the initial state. The depth B and the permitted intermediate certificates are part of the finite palette.

Lemma 6.27 (Termination with bounded backtracking)**.** *If bounded-backtracking packages are admitted only as finite packages with certified net energy decrease, then Theorem 6.12 remains valid.*

Proof. The termination proof uses only the energy of completed algorithmic steps. Intermediate states inside a finite package are not algorithmic stopping points. Since every completed package has lower final energy, the same well-foundedness argument applies. □

Definition 6.28 (Existentially complete certified palette)**.** A certified palette $\mathcal{Q}$ is *existentially complete* for slice disks if its finite list includes both ordinary quenching packages and any cleanification package types being allowed, and if, for every quenching state $\mathcal{S}$ arising from a smooth slice disk, one of the following holds:

(a) $\mathcal{S}$ is clean and $\mathrm{Def}(\mathcal{S}) = \emptyset$;

(b) a valid $\mathcal{Q}$-cleanification package starts at $\mathcal{S}$;

(c) an energy-legal $\mathcal{Q}$-package starts at $\mathcal{S}$.

It is *constructively complete* if an effective procedure outputs the required certificate. We use "complete" without qualification to mean existentially complete.

Remark 6.29 (Logical strength of completeness). Existential completeness is a strong global hypothesis. By Theorem 6.30, its existence implies the slice–ribbon conjecture; equivalently, by Corollary 6.31, if the conjecture is false then no such palette exists. The cleanification component in an existentially complete palette is an additional hypothesis, not a genericity formality; its difficulty may be comparable to the original problem. The converse implication, that the conjecture alone produces a finite existentially complete palette, is not proved here and would require an additional finite-encoding theorem. Thus the framework is a precise reformulation target, not a claim that completeness is easier than the original conjecture.

Theorem 6.30 (Conditional reduction to complete quenching)**.** *If there exists an existentially complete certified palette $\mathcal{Q}$ for slice disks, including cleanification certificates whenever non-clean stable states are allowed, then the slice–ribbon conjecture is true.*

Proof. Let K be a smooth slice knot and choose a stable quenching state $\mathcal{S}_0$ for a slice disk bounded by K. Run the certified quenching algorithm with the existentially complete palette $\mathcal{Q}$. By Theorem 6.12, it terminates after finitely many steps at a $\mathcal{Q}$-terminal state $\mathcal{S}_N$. If $\mathcal{S}_N$ is not clean, completeness supplies a valid cleanification package, contradicting terminality. If $\mathcal{S}_N$ is clean but $\mathrm{Def}(\mathcal{S}_N) \neq \emptyset$, completeness supplies an energy-legal $\mathcal{Q}$-package out of $\mathcal{S}_N$, again contradicting terminality. Hence the terminal state is clean and defect-free. By the ribbon criterion for defect-free normalized diagrams, K bounds a ribbon immersed disk in S^3, and by Proposition 1.11 it is ribbon. Since K was arbitrary, the slice–ribbon conjecture follows. □

Corollary 6.31 (Contrapositive form)**.** *If the slice–ribbon conjecture is false, then no certified palette is existentially complete in the sense of Definition 6.28. For every palette $\mathcal{Q}$, some non-ribbon slice counterexample has a blocked $\mathcal{Q}$-terminal state with non-empty defect set or with missing cleanification data.*

Proof. This is the contrapositive of Theorem 6.30, together with the terminal alternative Proposition 6.17. □

Remark 6.32 (Conditional status of strategy completeness)*.* The theorem above is conditional. Completeness cannot be inferred from greediness, from the existence of fourth-dimensional room, or from the fact that every source loop bounds a disk. It is a separate theorem that must be proved for a specific palette of certified moves, and constructively finding the relevant certificates is an additional computational topology problem not addressed here.

The methodological status is therefore deliberately modest: finite termination is proved for every fixed finite certified palette, but the existence of an existentially complete palette is an open completion problem. Constructing such a palette, or proving that no blocked terminal state can arise from slice disks, is precisely what remains beyond the purely formal termination theorem.

Remark 6.33 (High-dimensional finite termination)*.* The termination proof is insensitive to the dimension of the projected strata. For a fixed high-dimensional palette, one replaces counts of branch points, triple points, arcs, and circles by counts of normal cells in the corresponding non-clean, non-ribbon, interaction, subdivision, and certificate-queue strata. If every completed package strictly lowers the chosen lexicographic vector in $\mathbb{N}^r$, then well-foundedness gives finite termination exactly as in Theorem 6.12. Thus the high-dimensional difficulty is not termination; it is the geometric construction and verification of the local packages in the palette.

6.4 Summary and bridge to obstruction theory

The following theorem records the precise output of the quenching chapters and the obstruction-theoretic input that remains.

Theorem 6.34 (Reductio seed and quenching terminal theorem)**.** *Assume that the slice–ribbon conjecture is false. Then there exist a knot $K \subset S^3$, a smooth slice disk $D \subset B^4$ bounded by K, and a minimal stable quenching state $\mathcal{S}_{\min}$ such that:*

(1) *K is not ribbon and* $\mathrm{Def}(\mathcal{S}_{\min}) \neq \emptyset$*;*

(2) *for every finite certified palette $\mathcal{Q}$, the quenching algorithm starting from $\mathcal{S}_{\min}$ terminates after finitely many steps;*

(3) *the terminal state is not ribbon terminal unless the reductio hypothesis is contradicted;*

(4) *consequently, under the reductio hypothesis, every chosen palette produces a blocked terminal state with a non-empty finite ribbon-defect core;*

(5) *in the clean ribbon-normal case this blocked core consists of circular double-decker components, each of whose source decker circles bounds a source filling disk with a Morse height function after perturbation;*

(6) *in the non-clean case the blocked core may include branch points, triple points, or missing cleanification certificates;*

(7) *the existence of such filling disks is not itself contradictory. A contradiction requires either an existentially complete certified palette, including cleanification where needed, or an independent theorem excluding all blocked terminal states of slice disks.*

Proof. Choose K from Assumption 4.2. Since K is slice, Lemma 4.4 gives a minimal stable projection datum, hence a minimal quenching state $\mathcal{S}_{\min}$. If it were clean and defect-free, Propositions 2.3 and 1.11 would imply that K is ribbon, contrary to Assumption 4.2. For any finite palette, finite termination is Theorem 6.12. A ribbon terminal state would again imply that K is ribbon, because all quenching moves preserve the boundary knot by Corollary 6.22; hence, under the reductio hypothesis, the terminal state must be blocked. Finiteness of the remaining core follows from stability and Lemma 4.7. The clean ribbon-normal description follows from Definition 2.11; the non-clean alternative follows from Remark 3.9. The filling-disk and Morse-height assertions follow from Lemmas 3.3, 3.4 and 2.7. The final clause is Lemma 3.4(d) and Remark 6.32. □

Definition 6.35 (Terminal obstruction theorem)**.** A terminal obstruction theorem for a palette $\mathcal{Q}$ is a theorem asserting that no blocked $\mathcal{Q}$-terminal state can arise as the projection data of a smooth slice disk in B^4.

Theorem 6.36 (Terminal-obstruction route to slice–ribbon)**.** *If there exists a finite certified palette $\mathcal{Q}$ for which a terminal obstruction theorem holds and whose terminal ribbon states are clean in the sense of Definition 6.16, then the slice–ribbon conjecture is true.*

Proof. Let K be a smooth slice knot and run the $\mathcal{Q}$-quenching algorithm from any stable state for a slice disk bounded by K. By Theorem 6.12, the algorithm terminates. By Theorem 6.21, the terminal state is still projection data of a smooth slice disk. The terminal obstruction theorem rules out the blocked case. Therefore the terminal state is ribbon terminal, and Propositions 2.3 and 1.11 imply that K is ribbon. □

Bridge to terminal obstruction theory

The quenching chapters establish a rigorous finite evolution theory, but they intentionally stop short of an unconditional proof of the slice–ribbon conjecture. The obstruction chapters must attack blocked

terminal states, and the final bookkeeping isolates the non-clean blocked-state condition: such states must be cleanified by valid certificates or excluded by a certified non-clean obstruction. Chapter 10 later discharges that condition relative to standard local inputs. Existential palette completeness and terminal obstruction theorems are retained as packages built from that same condition, rather than as independent open endpoints. Any argument that a blocked core is impossible must use a genuine four-dimensional constraint beyond the planar fact that each source circle bounds a disk.

7 Terminal states and projected linking

The preceding chapters prove that a certified quenching algorithm terminates. Termination alone gives a dichotomy: either the terminal state is ribbon, or a blocked defect core remains. The purpose of the present chapter is to analyze what can be concluded from a blocked terminal state without adding false topological assertions. In particular, ordinary pairwise linking numbers of projected closed double-decker curves do not supply an automatic contradiction with sliceness.

Definition 7.1 (Blocked critical state)**.** Fix a finite certified palette $\mathcal{Q}$. A quenching state $\mathcal{S}$ is a $\mathcal{Q}$-blocked critical state if:

(a) $\mathrm{Def}(\mathcal{S}) \neq \emptyset$;

(b) no energy-legal $\mathcal{Q}$-package begins at $\mathcal{S}$;

(c) $\mathcal{S}$ is still realized as the stable projection data of a smooth properly embedded disk in B^4, with boundary fixed.

If $\mathcal{S}$ is clean and ribbon-normal, its defect set consists of circular double-decker components; otherwise branch points, triple points, and non-ribbon interval components may also occur.

Remark 7.2 (Clean versus non-clean blocked states). The linking analysis below applies only to clean ribbon-normal blocked states, where the remaining closed core components have well-defined image representatives after small perturbation. A non-clean blocked state may be blocked because branch points, triple points, or missing cleanification certificates remain. Such states require a separate obstruction theory and are not reduced here to ordinary linking questions.

Remark 7.3 (Blocked is relative). The adjective blocked is relative to the palette $\mathcal{Q}$, the chosen normal cell structure, and the energy E. It does not mean that no ambient isotopy of the disk can simplify the projection. It means only that no certified package from the chosen finite list applies with net energy decrease.

Definition 7.4 (Closed-core sublink)**.** For a clean ribbon-normal blocked state, let

$$\mathcal{R}_{\mathrm{cl}}(\mathcal{S}) = \{C_1, \dots, C_m\}$$

be the set of circular double-decker components. Choosing one image representative for each component gives an immersed or embedded closed curve in B^3, after a small perturbation if necessary. When these representatives are pairwise disjoint and oriented, their pairwise linking numbers define a symmetric integer matrix

$$L(\mathcal{S}) = \big(\mathrm{lk}(C_i, C_j)\big)_{i \neq j}.$$

This matrix is called the ordinary linking matrix of the closed-core sublink, when it is defined.

Remark 7.5 (When the linking matrix is undefined). The ordinary linking matrix is not intrinsic to an arbitrary surface diagram. It may be undefined or choice-dependent if image curves have self-intersections, pass through triple or branch events, are not oriented, or are not pairwise disjoint after the chosen perturbation. Even when defined, it is a diagrammatic invariant of a selected representative, not an invariant of the slice disk.

Lemma 7.6 (Terminality does not force pairwise linking)**.** *Let $\mathcal{S}$ be a clean ribbon-normal $\mathcal{Q}$-blocked critical state for some finite palette $\mathcal{Q}$. Terminality does not imply that the entries of $L(\mathcal{S})$, when defined, are all non-zero. It also does not imply that every proper sublink has non-zero total linking with its complement.*

Proof. Terminality is a negative statement about the absence of certified energy-decreasing packages in the fixed palette. Pairwise linking is a homological measurement of selected closed curves in the three-dimensional projection. There is no formal implication from the absence of certified moves to non-vanishing of those homological numbers. A palette may be too small to remove an unlinked circular component, or the obstruction to a move may be height-ordering, source-disk intersections, triple events, branch events, or lack of a certificate rather than ordinary linking. Conversely, links with all pairwise linking numbers zero, such as Whitehead-type configurations, can still be diagrammatically nontrivial. Thus pairwise non-zero linking cannot be deduced from blocked terminality alone. □

Definition 7.7 (Linking-critical state)**.** A clean ribbon-normal blocked state is called linking-critical if its ordinary linking matrix is defined and has at least one non-zero entry. It is strongly linking-critical if every non-empty proper sublink has non-zero linking interaction with its complement in a specified certified sense. Strong linking-criticality is an additional hypothesis, not a consequence of terminality.

Proposition 7.8 (Local minimality is palette-local)**.** *In a $\mathcal{Q}$-blocked critical state, any local minimality statement about linking, interaction number, or normal length is valid only for moves belonging to $\mathcal{Q}$. It does not imply global minimality among all diagrams of the same slice disk.*

Proof. By definition, a blocked state admits no energy-legal $\mathcal{Q}$-package. If a different move outside $\mathcal{Q}$, a larger bounded-backtracking package, or a different projection direction lowers a chosen quantity, this does not contradict $\mathcal{Q}$-terminality. Hence all such minimality claims are relative to the specified finite certified move system. □

7.1 Four-dimensional constraints and projected linking

A strong projected-vanishing principle would assert that two closed double-decker components coming from a slice disk necessarily have zero projected linking number. This assertion is false in the form needed for an unconditional proof. The correct four-dimensional fact is subtler: two disjoint curves that bound disjoint surfaces in B^3 have zero linking in B^3, but disjoint surfaces in B^4 may project to intersecting surfaces in B^3, and those projected intersections are precisely what linking can measure.

7.1.1 Planar fillings and nesting

Lemma 7.9 (Simultaneous planar fillings)**.** *Let $\gamma_1, \dots, \gamma_m \subset \mathrm{Int}(D^2)$ be pairwise disjoint embedded circles. Each γ_i bounds a unique source disk $E_i \subset D^2$. The disks E_i are either disjoint or nested: for $i \neq j$, either $E_i \cap E_j = \emptyset$, $E_i \subset E_j$, or $E_j \subset E_i$.*

Proof. Each circle bounds a unique interior disk by the Jordan curve theorem. For two disjoint simple closed curves in a disk, their bounded complementary disks are either disjoint or one curve lies inside the disk bounded by the other. This gives the stated nesting alternative. □

Remark 7.10 (Nesting of source disks). The phrase "two disjoint closed curves in a disk bound disjoint disks" is false without excluding nesting. If γ_2 lies inside E_1, then $E_2 \subset E_1$. For linking arguments one must either choose innermost curves, pass to annular regions between nested curves, or explicitly account for intersections of fillings with other decker data.

7.1.2 Classical linking facts

Lemma 7.11 (Linking via a Seifert surface)**.** *Let $C_1, C_2 \subset S^3$ be disjoint oriented closed curves, and let $F_1 \subset S^3$ be an oriented compact surface with $\partial F_1 = C_1$, transverse to C_2. Then*

$$\mathrm{lk}(C_1, C_2) = F_1 \cdot C_2,$$

the algebraic intersection number of F_1 with C_2. The same statement holds in a three-ball after closing it in S^3, provided the curves lie in the interior.

Proof. This is the standard definition of linking number by a Seifert surface. If F_1 is replaced by another Seifert surface, the difference is a closed two-cycle in S^3, whose algebraic intersection with C_2 is zero because $H_2(S^3) = 0$. Thus the number is well-defined. □

Lemma 7.12 (Disjoint three-dimensional fillings imply zero linking)**.** *Let $C_1, C_2 \subset B^3$ be disjoint oriented closed curves. If C_1 bounds an oriented compact surface $F_1 \subset B^3 \setminus C_2$, then $\mathrm{lk}(C_1, C_2) = 0$.*

Proof. After closing B^3 to S^3, Lemma 7.11 gives $\mathrm{lk}(C_1, C_2) = F_1 \cdot C_2$. Since F_1 is disjoint from C_2, the intersection number is zero. □

Remark 7.13 (Surface–surface intersection in three-space). The linking number of two curves in S^3 is computed by intersecting a Seifert surface for one curve with the other curve, not by an intrinsic algebraic intersection number of two spanning surfaces. Two surfaces in a three-manifold generically intersect in curves, not isolated signed points. Therefore formulas of the form $\mathrm{lk}(C_1, C_2) = F_1 \cdot F_2$ require additional choices and are not the standard linking formula.

7.1.3 Limits of projected-linking vanishing

Proposition 7.14 (No automatic vanishing of projected linking)**.** *Let $S_1, S_2 \subset B^4$ be disjoint smoothly embedded compact surfaces and let $p : B^4 \to B^3$ be a projection. It is not generally true that two disjoint closed curves lying on the projected surfaces $p(S_1)$ and $p(S_2)$, even if they are projections of disjoint curves on S_1 and S_2, must have zero linking in B^3.*

Proof. Work in a coordinate ball $B^3 \times (-\varepsilon, \varepsilon) \subset B^4$, with projection to the first factor. Let $H_1 \cup H_2 \subset B^3$ be a Hopf link, and choose pairwise disjoint narrow annuli $A_1, A_2 \subset B^3$ containing H_1, H_2, respectively. Define

$$S_1 = A_1 \times \{-\varepsilon/2\}, \qquad S_2 = A_2 \times \{\varepsilon/2\}.$$

These are disjoint embedded annuli in B^4. Their core curves $\widehat{H}_i = H_i \times \{\pm\varepsilon/2\} \subset S_i$ are disjoint in B^4, while their projections are exactly the Hopf link components $H_1, H_2 \subset B^3$, whose linking number is ± 1. Thus four-dimensional disjointness of the lifted surfaces and lifted curves does not force zero linking of their projected curves.

Conceptually, linking in three dimensions measures the obstruction to finding a spanning surface for one curve disjoint from the other curve inside the same three-dimensional space. A surface or annulus in four dimensions has an extra normal direction and does not provide such a three-dimensional disjoint Seifert surface. □

Corollary 7.15 (Certified constraint for double-decker loops)**.** *For closed double-decker components of a projected slice disk, sliceness alone does not imply that all pairwise projected linking numbers vanish. A vanishing conclusion is valid only if one supplies, for each pair, a three-dimensional Seifert surface for one projected curve disjoint from the other projected curve, or an equivalent certified condition.*

Proof. This follows from Lemma 7.12 and Proposition 7.14. The source filling disk lies in the slice disk in B^4, and its projection to B^3 may intersect the other projected curve or may fail to be an embedded Seifert surface disjoint from that curve. Therefore no vanishing follows without the extra three-dimensional disjointness certificate. □

Definition 7.16 (Linking obstruction certificate)**.** A linking obstruction certificate for a clean blocked state consists of two closed core components C_i, C_j, an orientation choice, and two verified statements:

(a) a diagrammatic calculation showing $\mathrm{lk}(C_i, C_j) \neq 0$;

(b) an independent four-dimensional or three-dimensional theorem implying $\mathrm{lk}(C_i, C_j) = 0$ for this specific certified configuration.

Only the combination of (a) and (b) yields a contradiction.

Remark 7.17 (Required content of a terminal obstruction). For the reductio program, it is not enough to show that a blocked state has nonzero linking in some examples. One must prove that every blocked state produced by an existentially complete palette carries a certified invariant that sliceness forces to vanish, or else prove directly that such states cannot be projections of embedded slice disks. This is exactly the terminal obstruction theorem isolated in Definition 6.35.

8 Derived terminal-obstruction packages and logical dependencies

The preceding linking analysis shows that projected-linking vanishing is not available as an unconditional substitute for a terminal obstruction theorem. This chapter therefore treats the former

palette-completeness, universal-obstruction, and WTQF–Milnor languages as derived theorem packages: once the non-clean branch is discharged by the standard local inputs and the clean branch is handled by terminal Milnor rigidity, the three formulations express the same exclusion of slice-realized blocked terminal states.

Definition 8.1 (Universal terminal obstruction)**.** Let $\mathcal{Q}$ be a finite certified palette. A universal terminal obstruction for $\mathcal{Q}$ is an assignment to every blocked $\mathcal{Q}$-terminal state $\mathcal{S}$ of a certified quantity $I(\mathcal{S})$ such that:

(a) $I(\mathcal{S})$ is well-defined for the terminal diagram and invariant under the allowed stable equivalences;

(b) blocked terminality implies $I(\mathcal{S}) \neq 0$;

(c) realizability as the projection of a smooth slice disk implies $I(\mathcal{S}) = 0$.

Remark 8.2 (Two forms of terminal obstruction). A universal terminal obstruction is a strengthened, certificate-valued form of the palette-specific terminal obstruction theorem of Definition 6.35. The latter may be a direct impossibility theorem saying that blocked terminal states cannot be projections of smooth slice disks. The former gives a specific invariant $I(\mathcal{S})$ whose non-vanishing follows from blocked terminality and whose vanishing follows from slice-disk realizability. Hence any universal terminal obstruction implies the corresponding terminal obstruction theorem, but the converse need not provide an explicit invariant.

Theorem 8.3 (Contradiction from a universal terminal obstruction)**.** *If there exists a finite certified palette $\mathcal{Q}$ and a universal terminal obstruction for $\mathcal{Q}$, then the slice–ribbon conjecture is true.*

Proof. Let K be a smooth slice knot and run the $\mathcal{Q}$-quenching algorithm from a stable state associated to a slice disk bounded by K. By Theorem 6.12, the algorithm terminates. By Theorem 6.21, the terminal state is still realized by a smooth slice disk. If the terminal state is ribbon, then K is ribbon by Propositions 2.3 and 1.11. If it is blocked, the universal obstruction gives both $I(\mathcal{S}) \neq 0$ and $I(\mathcal{S}) = 0$, a contradiction. Hence the terminal state must be ribbon, and K is ribbon. Since K was arbitrary, the conjecture follows. □

Remark 8.4 (Two terminal routes inside the completion condition). The logical alternatives can be summarized as

$$\begin{aligned} \text{stable slice data} &\xrightarrow{\text{certified quenching}} \text{terminal state}, \\ \text{existential palette completeness} &\Longrightarrow \text{ribbon terminal}, \\ \text{universal obstruction} &\Longrightarrow \text{blocked impossible}. \end{aligned}$$

The first route proves that an outgoing certified package exists unless the state is already clean and defect-free. The second route allows blocked states as formal outputs but proves that no such output is realizable by a smooth slice disk.

8.1 Weak topological quenching flow as a theorem package

The preceding obstruction formalism can also be phrased in a more flexible "weak flow" language. This subsection records the precise version that is logically safe. It does not introduce a differential

equation, and it does not assert that arbitrary self-crossing homotopies of projected curves preserve the data needed for ribbonness. The freedom belongs only to certified intermediate paths in the space of projections; the terminal conclusion may use only data that persist under liftability, compactness, and certified invariance.

Let $L \subset S^3$ be a clean terminal sublink with certified basing and indeterminacy data. Classical Milnor theory supplies the implication

$$L \text{ bounds pairwise disjoint smooth disks in } B^4 \Longrightarrow \overline{\mu}_I(L) = 0$$

for every certified multi-index I. Conversely, Theorem 8.12 proves that vanishing of the complete certified Milnor family produces a further energy-decreasing package. Hence blocked terminality implies $\overline{\mu}_I(L) \neq 0$ for at least one certified multi-index I, by Theorem 8.13.

Definition 8.5 (Weak topological quenching flow)**.** Let $\mathcal{X}$ be a path space of boundary-fixed smooth projection data $p : D^2 \to B^3$, enlarged if necessary to allow finitely many controlled non-stable intermediate singularities. A weak topological quenching flow, abbreviated WTQF, is a semigroup homotopy

$$\Phi : [0, \infty) \times \mathcal{X} \to \mathcal{X}, \qquad \Phi_0 = \mathrm{id},$$

equipped with the following certified data.

(a) *Boundary fixation*: every Φ_t is the identity on the fixed boundary collar convention, so the boundary knot and its orientation are unchanged.

(b) *Fiberwise liftability*: for every initial state arising from a slice disk, there is a continuous family $F_t : D^2 \to B^4$, fixed on the boundary collar, such that $\pi \circ F_t = \Phi_t$. At regular times F_t is an embedded slice disk, and at controlled singular times it is interpreted as the prescribed limit of embedded lifted disks.

(c) *Weak monotonicity*: whenever defects remain, the flow admits arbitrarily short forward intervals whose endpoints do not increase the primary defect vector (b, t, a, c), after replacing the endpoint by a stable representative. Temporary increases inside the interval are allowed only as certified intermediate events and are not terminal data.

(d) *Terminal compactness*: every forward orbit has a sequence of stable times $t_j \to \infty$ whose states converge, in a topology specified as part of the WTQF certificate, to a stable terminal state $\mathcal{S}_\infty$.

The liftability clause is deliberately fiberwise: it asserts that each projected path is the shadow of a controlled family over the disk. It does not say that an arbitrary property of the three-dimensional base projection is preserved through singular times.

Definition 8.6 (High-dimensional WTQF)**.** For $n \geq 2$, a high-dimensional WTQF for codimension-two slice disks is defined by replacing (D^2, B^3, B^4, S^3) in Definition 8.5 by $(D^n, B^{n+1}, B^{n+2}, S^{n+1})$. Thus the state space consists of boundary-fixed projection data $p : D^n \to B^{n+1}$, enlarged by controlled non-stable intermediate strata, and a flow certificate consists of boundary fixation, fiberwise liftability by maps $(D^n, \partial D^n) \to (B^{n+2}, S^{n+1})$, weak monotonicity of a finite normal-cell defect vector, and terminal compactness in a declared topology. The flow is again a certificate-suspension object, not a PDE.

Theorem 8.7 (High-dimensional homological shell)**.** *For every $n \geq 2$,*

$$H_n(B^{n+2}, S^{n+1}; \mathbb{Z}) = 0, \qquad \pi_n(B^{n+2}, S^{n+1}) = 0.$$

Consequently, a boundary-fixed lifted high-dimensional WTQF history has constant relative homology class and constant relative homotopy class in degree n, and both classes are zero.

Proof. For homology, the long exact sequence of the pair (B^{n+2}, S^{n+1}) contains

$$H_n(B^{n+2}; \mathbb{Z}) \longrightarrow H_n(B^{n+2}, S^{n+1}; \mathbb{Z}) \longrightarrow H_{n-1}(S^{n+1}; \mathbb{Z}).$$

The first group is zero because the ball is contractible, and the last group is zero for $n \geq 2$ because it is a reduced homology group below the top dimension of S^{n+1}. Hence the middle group is zero.

For homotopy, the long exact homotopy sequence contains

$$\pi_n(B^{n+2}) \longrightarrow \pi_n(B^{n+2}, S^{n+1}) \longrightarrow \pi_{n-1}(S^{n+1}).$$

Again $\pi_n(B^{n+2}) = 0$, and $\pi_{n-1}(S^{n+1}) = 0$ because S^{n+1} is n-connected. Thus $\pi_n(B^{n+2}, S^{n+1}) = 0$. A lifted WTQF history is a homotopy of pairs, so its relative homology and homotopy classes are constant; since the target groups vanish, those classes are zero. □

Remark 8.8 (Exact consequence of the relative-group calculation). If $F : [0,1] \times (D^n, \partial D^n) \to (B^{n+2}, S^{n+1})$ is a homotopy of pairs, then

$$[F_0] = [F_1] = 0 \quad \text{in} \quad \pi_n(B^{n+2}, S^{n+1})$$

and the induced relative homology classes are equal to zero. No conclusion about the complement group or a Milnor-type invariant follows from these equalities alone.

Remark 8.9 (Compactness as certificate data). The compactness clause cannot be replaced by the sentence "B^4 is compact." Arzela–Ascoli type compactness requires equicontinuity or uniform C^k bounds, and C^∞ compactness requires stronger estimates. Likewise, a bounded integer defect vector gives only finitely many defect counts, not compactness of maps. Therefore terminal compactness is part of the WTQF certificate and must be proved for any proposed weak flow.

Definition 8.10 (Milnor obstruction certificate for a terminal core)**.** Let $\mathcal{S}$ be a clean ribbon-normal state with non-empty closed core, and let $L = L_{\mathcal{I}} \subset \mathcal{R}_{\mathrm{cl}}(\mathcal{S})$ be a non-empty innermost closed-core sublink. A Milnor reduction certificate for L consists of:

(a) oriented representatives, meridians, preferred longitudes, basing paths, lower-order indeterminacy data, and any string-link representatives and Habegger–Lin orbit data needed to specify a complete link-homotopy obstruction for the certified closed link;

(b) a finite generic link-homotopy realization rule: every ambient isotopy and every permitted self-crossing change in the classified link homotopy is represented by a finite sequence of certified sheet-height passes, decker slides, and stable perturbations, all fixed on the boundary collar;

(c) a terminal cancellation rule: if the realized link homotopy carries the selected closed core to an unlink in a union of disjoint source neighborhoods, certified empty-bigon or empty-ball cancellations remove at least one closed core component;

(d) an endpoint comparison proving that the concatenation in (b)–(c) preserves the earlier clean coordinates $b = t = a = 0$ and strictly lowers the closed-core coordinate c, hence strictly lowers E;

(e) the geometric rigidity and filling data needed to apply classical Milnor concordance invariance to a slice-realized terminal core.

The certificate separates two implications: clauses (a)–(d) convert Milnor triviality into a legal outgoing package, whereas clause (e) converts slice realization into Milnor vanishing.

Assumption 8.11 (Global clean Milnor-reduction input)**.** For every clean ribbon-normal state $\mathcal{S}$ with non-empty selected innermost core $L_{\mathcal{I}}$, if every certified Milnor invariant in the declared family vanishes, then $L_{\mathcal{I}}$ admits a reduction certificate satisfying Definition 8.10(a)–(d), and the resulting package belongs to the declared global package relation. Denote this assertion by $\mathsf{H}_{\mathrm{red}}$.

This is a special assertion about the terminal cores produced by the quenching system, not a consequence of general link-homotopy classification. Habegger–Lin classify links through string links modulo group actions, and equality of the ordinary closed-link Milnor invariants alone is not a complete link-homotopy criterion in general [10, 16].

Theorem 8.12 (Milnor-trivial clean cores admit a legal reduction)**.** *Assume* $\mathsf{H}_{\mathrm{red}}$*. Let* $\mathcal{S}$ *be clean and ribbon-normal with* $\mathcal{R}_{\mathrm{cl}}(\mathcal{S}) \neq \emptyset$*. Suppose*

$$\overline{\mu}_I(L_{\mathcal{I}}) = 0 \qquad \text{for every } I \in \mathcal{I}_L.$$

Then the clean-domain certified package relation contains a package P *with*

$$\mathrm{Init}(P) = \mathcal{S}, \qquad \mathrm{E}(\mathrm{End}(P)) < \mathrm{E}(\mathcal{S}).$$

Proof. By $\mathsf{H}_{\mathrm{red}}$, simultaneous vanishing supplies a reduction certificate and a link homotopy of the certified based representative to the unlink. Choose a generic representative of that link homotopy. Compactness of its parameter interval and genericity imply that it decomposes into finitely many ambient-isotopy intervals and finitely many isolated self-crossing changes.

Clause (b) of Definition 8.10 replaces each interval and crossing event by a finite boundary-fixed liftable certificate record. Concatenating these records gives a finite clean-domain package from $\mathcal{S}$ to a state in which the selected closed core is an unlink in disjoint source neighborhoods. Clause (c) then cancels at least one closed component. Throughout the completed package the branch, triple, and non-ribbon interval coordinates remain zero. The endpoint therefore satisfies

$$(b, t, a)(\mathrm{End}(P)) = (0, 0, 0) = (b, t, a)(\mathcal{S}), \qquad c(\mathrm{End}(P)) < c(\mathcal{S}).$$

By lexicographic ordering, $\mathrm{E}(\mathrm{End}(P)) < \mathrm{E}(\mathcal{S})$. Thus P is an energy-legal outgoing package. □

Theorem 8.13 (Blocked clean terminality forces Milnor non-vanishing)**.** *Assume* $\mathsf{H}_{\mathrm{red}}$*. Let* $\mathcal{S}_\infty$ *be a clean ribbon-normal blocked terminal state with non-empty closed core, and let* $L_{\mathcal{I}}$ *be the innermost closed-core sublink selected by Definition 8.10. Then*

$$\exists I \in \mathcal{I}_L, \qquad \overline{\mu}_I(L_{\mathcal{I}}) \neq 0.$$

Proof. Assume the contrary. Then every invariant in the complete certified family vanishes. By Theorem 8.12, an energy-legal package starts at $\mathcal{S}_\infty$. This contradicts terminality. Hence at least one certified Milnor invariant is non-zero. □

Remark 8.14 (High-dimensional Milnor-type terminal invariants). The clean terminal obstruction is not restricted to pairwise linking in S^3. In higher dimensions, a replacement invariant must satisfy three explicit predicates: algebraic readability, non-vanishing on the selected terminal core, and vanishing on the certified slice-link class. Koschorke and Haefliger invariants are possible inputs [11, 15]; we do not prove that they satisfy the required terminal non-vanishing condition for the states considered here.

Theorem 8.15 (WTQF limit homology inheritance and its exact scope)**.** *Let $\Phi_t : D^2 \to B^3$, $0 \le t < \infty$, be a WTQF orbit for which the liftability and compactness certificates supply a continuous lifted extension*

$$F : [0,\infty] \times (D^2, \partial D^2) \longrightarrow (B^4, S^3), \qquad F_t = F(t,\cdot),$$

where $[0,\infty]$ is the one-point compactification of $[0,\infty)$, $\pi \circ F_t = \Phi_t$ for finite t, and the boundary collar is fixed. Put $F_\infty = F(\infty,\cdot)$ and $\Phi_\infty = \pi \circ F_\infty$. Then:

(a) *the relative class $(F_t)_*[D^2] \in H_2(B^4, S^3; \mathbb{Z})$ is independent of $t \in [0,\infty]$;*

(b) *this common relative class is zero;*

(c) *if $\gamma \subset \mathrm{Int}(D^2)$ is an embedded closed source decker component and $E_\gamma \subset D^2$ is the disk bounded by γ, then $F_{\infty\#}[E_\gamma]$ is a singular two-chain in B^4 with boundary $F_{\infty\#}[\gamma]$, and $\Phi_{\infty\#}[E_\gamma]$ is a singular two-chain in B^3 with boundary $\Phi_{\infty\#}[\gamma]$;*

(d) *the conclusions in* (a)–(c) *do not imply that the projected closed-core sublink represented by $\Phi_\infty(\gamma)$ has vanishing pairwise linking numbers or vanishing Milnor $\overline{\mu}$-invariants.*

Proof. Choose a relative singular fundamental cycle $c \in C_2(D^2, \partial D^2; \mathbb{Z})$ representing $[D^2]$. For any $s,t \in [0,\infty]$, choose a path $\alpha : [0,1] \to [0,\infty]$ from s to t. The map

$$H : [0,1] \times (D^2, \partial D^2) \longrightarrow (B^4, S^3), \qquad H(u,x) = F(\alpha(u), x),$$

is a homotopy of maps of pairs from F_s to F_t. The prism-chain identity gives

$$F_{t\#}c - F_{s\#}c = \partial P(c) + P(\partial c)$$

for the usual prism chain $P(c)$. Since ∂c lies in $C_1(\partial D^2)$ and the boundary collar maps to S^3, the term $P(\partial c)$ lies in $C_2(S^3)$ and is zero in the relative chain complex $C_*(B^4, S^3)$. Thus $F_{s\#}c$ and $F_{t\#}c$ define the same class in $H_2(B^4, S^3; \mathbb{Z})$. This proves (a), including the terminal value $t = \infty$.

For (b), the long exact sequence of the pair (B^4, S^3) contains

$$H_2(B^4; \mathbb{Z}) \longrightarrow H_2(B^4, S^3; \mathbb{Z}) \longrightarrow H_1(S^3; \mathbb{Z}).$$

Both outside groups vanish, because B^4 is contractible and $H_1(S^3; \mathbb{Z}) = 0$. Hence $H_2(B^4, S^3; \mathbb{Z}) = 0$, so the common class from (a) is zero.

For (c), the Jordan curve theorem gives the unique disk $E_\gamma \subset D^2$ bounded by γ, after choosing the component of $D^2 \setminus \gamma$ not containing ∂D^2. With the induced boundary orientation, $\partial[E_\gamma] = [\gamma]$ as singular chains in D^2. Naturality of the boundary operator gives

$$\partial F_{\infty\#}[E_\gamma] = F_{\infty\#}[\gamma], \qquad \partial \Phi_{\infty\#}[E_\gamma] = \Phi_{\infty\#}[\gamma].$$

Thus the lifted curve bounds a singular two-chain in B^4, and the projected curve bounds a singular two-chain in B^3.

Finally, (d) is the same complement-level distinction as in Proposition 7.14 and Corollary 7.15. A two-chain bounding one projected component may intersect the other projected components, and Milnor invariants are invariants of the link group or link-homotopy class in the three-dimensional complement, not merely of the existence of singular fillings in B^3 or B^4. Concretely, a Hopf link in a three-ball has components that bound disks whose interiors meet the other component, and the corresponding disks may also be regarded as singular fillings in B^4; nevertheless its length-two Milnor invariant is its linking number ± 1. Therefore the homological inheritance proved above is unconditional, but Milnor vanishing requires the additional liftability-and-filling theorem demanded in Definition 8.10. □

Theorem 8.16 (WTQF Milnor terminal obstruction from $\mathsf{C1}$)**.** *Assume* $\mathsf{C1} \wedge \mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$. *Then no blocked WTQF terminal state is realized by smooth slice-disk data.*

Proof. Let $\mathcal{S}_\infty$ be a blocked WTQF terminal state realized by slice data. If $\neg\,\mathrm{Clean}(\mathcal{S}_\infty)$, $\mathsf{C1}$ gives either an outgoing energy-legal package or a certificate that $\mathcal{S}_\infty$ is not slice-realizable. The first contradicts terminality and the second contradicts $\mathrm{Realizes}(\mathcal{S}_\infty, K)$. Hence $\mathrm{Clean}(\mathcal{S}_\infty)$. The clean case contradicts Theorem 9.16. □

Theorem 8.17 (WTQF route to terminal obstruction)**.** *If there exists a WTQF satisfying terminal compactness, fiberwise liftability, and weak monotonicity, then* H_{Q} *implies that no blocked WTQF terminal state is realized by smooth slice-disk data.*

Proof. By Corollary 10.24, H_{nc} implies $\mathsf{C1}$. The WTQF hypotheses preserve slice realization and the fixed boundary by Theorems 9.1 and 9.2. Apply Theorem 8.16. □

Remark 8.18 (Scope of intermediate WTQF freedom). For a continuous lifted family of maps of pairs, Theorem 8.15 proves invariance of the relative class and existence of the stated singular chains. Formally,

$$\mathrm{WTQF}(F) \Longrightarrow \big([F_t] \text{ is constant in relative homology}\big),$$

but

$$\mathrm{WTQF}(F) \not\Longrightarrow \overline{\mu}_I(L_\infty) = 0.$$

Proposition 8.19 (Insufficiency of projected-linking contradiction)**.** *We do not derive either of the following statements:*

(a) *every blocked terminal state has a pair of closed core components with non-zero pairwise linking;*

(b) *every pair of closed double-decker components in a projected slice disk has zero pairwise linking.*

Therefore their combination cannot be used as an unconditional proof of the slice–ribbon conjecture.

Proof. Statement (a) is ruled out by Lemma 7.6: blocked terminality is palette-relative and does not force ordinary linking. Statement (b) is ruled out by Proposition 7.14 and Corollary 7.15: four-dimensional disjointness of source fillings does not imply three-dimensional zero linking of their projected boundaries. Since both required implications fail, projected linking alone does not give an unconditional contradiction. □

Corollary 8.20 (Correct terminal alternative under a reductio hypothesis)**.** *Assume the slice–ribbon conjecture is false, and fix a finite certified palette $\mathcal{Q}$. The quenching algorithm applied to a minimal counterexample state terminates in a blocked $\mathcal{Q}$-terminal state. This conclusion is consistent with all results proved here unless $\mathcal{Q}$ is existentially complete, including needed cleanification certificates, or a universal terminal obstruction is supplied.*

Proof. The first sentence is Theorem 6.34. The second follows from Theorems 6.30, 6.36 and 8.3: one of the stated additional inputs is exactly what is needed to rule out the blocked case. □

Definition 8.21 (Logical dependency chain)**.** The logical hypotheses are the assertions

$$\mathsf{T},\ \mathsf{D},\ \mathsf{R},\ \mathsf{Q},\ \mathsf{F},\ \mathsf{A},\ \mathsf{C1},\ \mathsf{H}_{\mathrm{red}},\ \mathsf{H}_{\mathrm{app}},$$

where:

(i) T is the relative transversality input that produces stable projection data from a smooth slice disk;

(ii) D is the defect dictionary: double-decker data, ribbon-normality, and the equivalence between defect-free normalized data and ribbonness;

(iii) R is the reductio extraction of a non-empty finite defect core from any non-ribbon slice counterexample;

(iv) Q is certified quenching: every allowed local package, including any cleanification package needed for non-clean states, is boundary-fixed, liftable, and energy-legal;

(v) F is finite termination for every fixed finite certified palette;

(vi) A is the terminal alternative: the output is ribbon terminal or blocked terminal;

(vii) $\mathsf{C1}$ is the non-clean blocked-state condition: every non-clean blocked terminal state arising from slice data is either cleanified by valid boundary-fixed certificates or excluded by a certified non-clean terminal obstruction.

(viii) $\mathsf{H}_{\mathrm{red}}$ is the global clean-core reduction input of Assumption 8.11.

(ix) $\mathsf{H}_{\mathrm{app}}$ is the projected-core Milnor-applicability input of Assumption 9.14.

Their logical use is the single implication

$$(\mathsf{T} \wedge \mathsf{D} \wedge \mathsf{R} \wedge \mathsf{Q} \wedge \mathsf{F} \wedge \mathsf{A} \wedge \mathsf{C1} \wedge \mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}) \Longrightarrow \mathrm{SRC}.$$

No implication $\mathsf{T} \Rightarrow \mathsf{D}$, for example, is asserted; the symbols denote jointly required premises. We prove $\mathsf{C1}$ from H_{nc}. The clean terminal contradiction additionally uses $\mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$.

Remark 8.22 (Conditional theorem packages from C1). Under C1, existential palette completeness, universal terminal obstruction, and the WTQF–Milnor formulation are equivalent presentations of the same terminal exclusion, because the clean branch is already excluded internally. We prove C1 from H_{nc}.

Remark 8.23 (Unconditional auxiliary certificates). The following implications are unconditional: selector existence, finite termination, endpoint preservation, no recurrence, algebraic readability of already defined Milnor data, lifted-source rigidity, and singular-chain fillings. Milnor reduction and terminal non-vanishing require $\mathsf{H}_{\mathrm{red}}$, while projected-core vanishing requires $\mathsf{H}_{\mathrm{app}}$. The purely formal selector and homology statements do not imply C1. In symbols,

$$(\mathsf{F} \wedge \mathsf{A} \wedge \text{homology/readability}) \not\Longrightarrow \mathsf{C1}.$$

Lemma 8.24 (Soundness of the dependency chain)**.** *The assertions* $\mathsf{T}, \mathsf{D}, \mathsf{R}, \mathsf{Q}, \mathsf{F}, \mathsf{A}$ *and the clean terminal exclusion theorem are established by the cited definitions and theorems. We prove* $\mathsf{H}_{\mathrm{nc}} \Rightarrow \mathsf{C1}$.

Proof. The implication T is Lemma 1.16. The dictionary D is supplied by Definitions 2.1, 2.2, 2.11, 1.18 and 1.23, Convention 1.19, and Propositions 2.3 and 1.11. The reductio extraction R is Assumption 4.2, Lemmas 4.4 and 4.7, and Propositions 4.8 and 4.10. Certified quenching Q is defined in Definitions 5.2, 5.7 to 6.8 and 5.18 and is sound by Lemmas 5.11, 5.13 and 5.16, Propositions 5.19, 6.23 and 8.32, and Theorem 6.21. Finite termination F is Convention 6.2, Lemma 6.6, Proposition 6.9, and Theorems 6.12, 6.14 and 8.26. The terminal alternative A is Definition 6.16 and Proposition 6.17. The clean branch is closed by Theorems 8.12, 8.13, 9.15 and 9.16 only under $\mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$. The condition C1 is isolated in Definition 8.34. Its relative discharge is Theorem 10.23 and Corollary 10.24. □

Theorem 8.25 (Logical dependency theorem)**.** *Assume* H_{Q}. *Then*

$$\forall K \in \mathcal{K}, \qquad \mathrm{Slice}(K) \Longrightarrow \mathrm{Ribbon}(K).$$

Proof. Let $K \in \mathcal{K}$ satisfy $\mathrm{Slice}(K)$, and choose a slice disk. Relative transversality gives an initial state $\mathcal{S}_0$ with $\mathrm{Realizes}(\mathcal{S}_0, K)$. Let $\mathfrak{P}$ contain every valid energy-legal package admitted by the definitions, by H_{nc}, and by $\mathsf{H}_{\mathrm{red}}$, and choose $\sigma_{\mathfrak{P}}$ by Theorem 6.14. By Theorem 6.12, the selected history has an eventual value $\mathcal{S}_N$, and soundness gives $\mathrm{Realizes}(\mathcal{S}_N, K)$.

By Proposition 6.17, either $\mathrm{RibbonTerm}(\mathcal{S}_N)$ or $\mathrm{BlockedTerm}_{\mathfrak{P}}(\mathcal{S}_N)$. In the first case, Propositions 2.3 and 1.11 imply $\mathrm{Ribbon}(K)$. In the second case, if $\neg\,\mathrm{Clean}(\mathcal{S}_N)$, H_{nc} supplies a package in $\mathfrak{P}(\mathcal{S}_N)$, contradicting $\mathrm{Terminal}_{\mathfrak{P}}(\mathcal{S}_N)$. If $\mathrm{Clean}(\mathcal{S}_N)$, Theorem 9.16 gives a contradiction. Therefore only the ribbon case occurs. □

8.2 Imported control and coarse-existence barriers

Before invoking the standard local inputs of Chapter 10, one might try to close C1 by using a deliberately coarse existence principle: allow extra upward and downward moves in the middle of an evolution, suppress the precise invariant, and ask only for existence of a separating obstruction or a complete palette. The following results record the exact logical content of such coarse tools in the present framework. Dynamics, rewriting theory, group theory, and set theory are useful, but they do not by themselves remove the need for certified cleanification or certified exclusion of non-clean blocked terminal states.

Theorem 8.26 (Discrete dynamical no-recurrence certificate)**.** *Fix a finite certified palette $\mathcal{Q}$ and a selector $\sigma_{\mathcal{Q}}$ as in Theorem 6.14. Let $R_{\mathcal{Q},\sigma}$ be the selected endpoint map, with terminal states fixed by the constant package. Then* E *is a strict Lyapunov function for the non-terminal part of the selected discrete evolution:*

$$\mathcal{S} \neq R_{\mathcal{Q},\sigma}(\mathcal{S}) \quad \Longrightarrow \quad \mathrm{E}(R_{\mathcal{Q},\sigma}(\mathcal{S})) < \mathrm{E}(\mathcal{S}).$$

Consequently every forward orbit is eventually constant at a $\mathcal{Q}$-terminal state, no non-terminal state lies on a periodic orbit, and in the certificate-suspension model no non-terminal history is recurrent under positive time-shift.

Proof. The displayed inequality is exactly the energy-legality of the selected nonconstant package. An infinite forward orbit with infinitely many nonconstant steps would give an infinite strictly decreasing sequence in the well-founded lexicographic order on $\mathbb{N}^7$, contradicting Lemma 6.6. Hence every forward orbit has only finitely many nonconstant steps and is eventually constant. A non-terminal periodic orbit would give a finite strict chain

$$\mathrm{E}(\mathcal{S}_0) > \mathrm{E}(\mathcal{S}_1) > \cdots > \mathrm{E}(\mathcal{S}_m) = \mathrm{E}(\mathcal{S}_0),$$

which is impossible. In the suspension model, positive time-shift of a non-terminal history eventually enters its constant terminal tail, so it cannot return to the same non-terminal history in the compact-open history space. Thus the only recurrent histories are constant terminal histories. □

Remark 8.27 (Dynamical limitation). Theorem 8.26 is the precise dynamical content of the quenching scheme: the energy is a Lyapunov function and the transient dynamics has no recurrence. This is not enough to prove ribbonness, because a blocked terminal state is itself a fixed point of the selected dynamics. Dynamics eliminates cycles; it does not identify the terminal fixed-point set with the ribbon subset.

Theorem 8.28 (Milnor algebraic readability certificate)**.** *Let $L = C_1 \cup \cdots \cup C_m \subset S^3$ be an oriented smooth sublink represented by pairwise disjoint closed-core curves of a clean ribbon-normal terminal state. Suppose the certificate record fixes meridians, preferred longitudes, basing paths, component order, and the lower-order vanishing or indeterminacy data required to define a chosen finite list of Milnor $\overline{\mu}$-invariants. Then each invariant in that list is a well-defined algebraic quantity of the certified link group*

$$\pi_1(S^3 \setminus L),$$

read from the Magnus expansion of the certified longitude words modulo the standard lower-order indeterminacy. The value is invariant under the declared stable diagram equivalences and boundary-fixed isotopies, provided the basing and indeterminacy conventions are transported as part of the certificate.

Proof. For a based oriented link, meridians normally generate the link group and preferred longitudes determine elements in the lower central quotients used in Milnor's construction. A Wirtinger presentation obtained from a diagram gives words for the certified longitudes. The Magnus expansion of these words has integer coefficients indexed by multi-indices, and the corresponding $\overline{\mu}$-values are taken modulo the usual indeterminacy generated by lower-order coefficients and changes of choices. By hypothesis, those lower-order choices have been fixed or shown to vanish in the certified range, so the specified residue classes are well-defined.

Stable diagram equivalence and boundary-fixed isotopy transport meridians, longitudes, basing paths, and lower central quotients by canonical isomorphisms of based link groups. Since the certificate transports these choices, the Magnus coefficients in the certified indeterminacy range are unchanged. Thus the Milnor data are algebraically readable and invariant under the allowed equivalences. □

Remark 8.29 (Milnor readability is not Milnor obstruction). Theorem 8.28 alone closes only the group-theoretic readability part of the Milnor route. The implication from blocked terminality to non-vanishing additionally uses Theorem 8.12, and the implication from slice realization to vanishing additionally uses Theorems 9.13 and 9.15.

Proposition 8.30 (Set-theoretic obstruction-existence barrier)**.** *Fix a finite certified palette $\mathcal{Q}$. Let $\mathfrak{B}_{\mathcal{Q}}$ be the set of blocked $\mathcal{Q}$-terminal states in the encoded state space, modulo stable equivalence, and let $\mathfrak{S}_{\mathcal{Q}}$ be the subset of terminal states realized by smooth slice disks with the prescribed boundary convention. A set-valued or two-valued universal terminal obstruction in the sense of Definition 8.1 exists for $\mathcal{Q}$ if and only if*

$$\mathfrak{B}_{\mathcal{Q}} \cap \mathfrak{S}_{\mathcal{Q}} = \emptyset.$$

Thus a purely set-theoretic existence argument for an obstruction is exactly as strong as a terminal obstruction theorem; it does not by itself prove the required disjointness.

Proof. If a universal terminal obstruction I exists, then every state in $\mathfrak{B}_{\mathcal{Q}}$ has $I \neq 0$, while every state in $\mathfrak{S}_{\mathcal{Q}}$ has $I = 0$. No state can satisfy both conclusions, so the two sets are disjoint.

Conversely, if the two sets are disjoint, define a two-valued function on stable-equivalence classes by setting $I(\mathcal{S}) = 1$ on $\mathfrak{B}_{\mathcal{Q}}$ and $I(\mathcal{S}) = 0$ on $\mathfrak{S}_{\mathcal{Q}}$, extending arbitrarily to the remaining terminal classes. This function is well-defined on the quotient and satisfies the formal nonzero/zero clauses. Hence the only missing content in such a coarse obstruction is the disjointness statement itself, namely that no blocked terminal state is slice-realizable. □

Proposition 8.31 (Coarse complete-palette barrier)**.** *Any unconditional theorem asserting the existence of a finite certified palette with the following coarse property is precisely an existential completeness theorem in the sense of Definition 6.28: every slice-realized state that is not already clean and defect-free admits either a valid cleanification package or an energy-lowering package. Consequently such a theorem implies the slice–ribbon conjecture by Theorem 6.30.*

Proof. The displayed coarse property is exactly the trichotomy in Definition 6.28: a slice-realized state is either clean and defect-free, admits cleanification, or admits an energy-legal outgoing package. Therefore the palette is existentially complete. The implication to the slice–ribbon conjecture is Theorem 6.30. □

Proposition 8.32 (Intermediate freedom is absorbed by certificates)**.** *Suppose a proposed evolution segment between two admissible states uses arbitrary intermediate upward moves, downward moves, projection changes, controlled singularities, or four-dimensional lift/drop operations, but its endpoints are boundary-fixed liftable states and the final endpoint has strictly lower integer energy. If the segment is finite and its liftability, boundary fixation, and final energy comparison are verified, then it is already a quenching package after adjoining its finite record to the palette. If one of these verifications is absent, the segment cannot be used as a sound quenching step in this framework.*

Proof. The definition of a quenching package allows finite certified interpolation, preparatory moves that are not individually energy-decreasing, stable perturbations, and bounded backtracking, provided the completed package is boundary-fixed, liftable, and has lower final energy. Hence a proposed segment satisfying these endpoint and liftability clauses is not a new kind of argument; it is a candidate certified package or a finite palette enlargement. Conversely, without boundary fixation the boundary knot may change; without liftability the projected motion need not come from an embedded disk in B^4; without a finite record it is not part of a finite certified palette; and without final energy decrease it does not enter the termination proof. Therefore intermediate freedom is useful only after it is converted into the same certificate data already required by the framework. □

Remark 8.33 (Limits of coarse control). The preceding propositions explain the logical limit of coarse existence. A nonconstructive selector, a Lyapunov function, a set-theoretic separator, or an unspecified amount of intermediate geometric freedom can simplify formal organization, but none of them proves the substantive non-clean condition isolated here. That condition is certified non-clean blocked-state exclusion: every non-clean blocked terminal state arising from slice data must either admit a valid boundary-fixed cleanification package or be excluded by a certified non-clean terminal obstruction. Its relative proof is supplied in Chapter 10.

Definition 8.34 (Core non-clean blocked-state condition)**.** The core non-clean blocked-state condition of the framework, denoted $\mathsf{C1}$, is the following statement. For every blocked terminal state $\mathcal{S}_\infty$ arising from smooth slice-disk data in the certified quenching or WTQF state space, if $\mathcal{S}_\infty$ is non-clean, then one of the following certified alternatives holds:

(a) a finite boundary-fixed cleanification package starts at $\mathcal{S}_\infty$, is liftable through the prescribed interpolation data, and is energy-legal after completion; or

(b) a certified non-clean terminal obstruction proves that $\mathcal{S}_\infty$ is not realized by any smooth slice disk with the fixed boundary convention.

Thus $\mathsf{C1}$ is exactly the exclusion of non-clean blocked terminal states as possible final outputs of slice-realized histories. It is not discharged by the selector, endpoint, no-recurrence, Milnor-readability, or innermost singular-filling certificates; it is discharged later, relative to standard local inputs, by Corollary 10.24.

8.3 Logical structure of the quenching reduction

Definition 8.35 (Former completion packages)**.** Define the following three statements for a declared package relation $\mathfrak{P}$:

(a) EP($\mathfrak{P}$): every slice-realized state has a finite $\mathfrak{P}$-history ending at a ribbon terminal state;

(b) UTO($\mathfrak{P}$): no $\mathfrak{P}$-blocked terminal state is slice-realizable;

(c) WMC($\mathfrak{P}$): every WTQF terminal state is either ribbon terminal, excluded by $\mathsf{C1}$, or excluded by the clean terminal Milnor contradiction under $\mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$.

Definition 8.36 (Two-stage terminal obstruction theorem)**.** The terminal obstruction is divided into the following two statements:

(a) *non-clean stage*: $\mathsf{C1}$, the exclusion or certified cleanification of every non-clean blocked terminal state, proved in Theorem 10.23 and Corollary 10.24 from Assumption 10.5;

(b) *clean stage*: under $\mathsf{H}_{\mathrm{red}}$, Theorem 8.13 derives a non-trivial complete link-homotopy obstruction from blocked terminality; under $\mathsf{H}_{\mathrm{app}}$, Theorem 9.15 derives Milnor vanishing for the projected terminal sublink.

Thus the non-clean stage depends on H_{nc}, and the clean stage depends on $\mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$.

Theorem 8.37 (Completion packages as consequences)**.** *Assume H_{Q}, and let $\mathfrak{P}$ contain all valid energy-legal packages supplied by its three clauses. Then*

$$\mathrm{EP}(\mathfrak{P}), \qquad \mathrm{UTO}(\mathfrak{P}), \qquad \mathrm{WMC}(\mathfrak{P})$$

all hold.

Proof. By Corollary 10.24, $\mathsf{H}_{\mathrm{nc}} \Rightarrow \mathsf{C1}$. Hence a slice-realized $\mathfrak{P}$-terminal state cannot be non-clean: $\mathsf{C1}$ would produce an element of $\mathfrak{P}(\mathcal{S})$. The clean blocked case is excluded by Theorem 9.16. Thus no blocked terminal state is slice-realizable, proving $\mathrm{UTO}(\mathfrak{P})$. The WTQF restatement is $\mathrm{WMC}(\mathfrak{P})$. Finally, Theorem 6.12 gives a terminal state from every initial slice-realized state; UTO forces that terminal state to be ribbon, which is $\mathrm{EP}(\mathfrak{P})$. □

Theorem 8.38 (Core-certificate integrity theorem)**.** *The final implication uses $\mathsf{C1}$ for the non-clean branch and $\mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$ for the clean branch. Selector existence, no-recurrence, endpoint preservation, relative homology, and singular fillings alone imply none of these geometric inputs.*

Proof. Theorem 10.23 and Corollary 10.24 prove $\mathsf{C1}$ from H_{nc}. The clean exclusion follows from Theorems 8.12, 8.13, 9.15 and 9.16 under $\mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$. □

Corollary 8.39 (Former multiple certificates collapse)**.** *Under H_{Q}, EP, UTO, and WMC hold for the global package relation $\mathfrak{P}$.*

Proof. This is Theorem 8.37. □

9 Strict weak-flow construction and Milnor applicability

The weak-flow language of Section 8.1 becomes fully formal only after one separates two issues. First, there is a strict semigroup construction obtained by suspending the discrete certified quenching algorithm together with its certified interpolation paths. Second, after $\mathsf{C1}$ removes the non-clean terminal branch, the clean closed-core link requires the reduction and applicability inputs $\mathsf{H}_{\mathrm{red}}$ and $\mathsf{H}_{\mathrm{app}}$. The first point is formal and unconditional for a chosen certified selector; the second is conditional.

9.1 Existence as a certificate suspension

Theorem 9.1 (Existence of the certificate-suspension weak flow)**.** *Fix a finite certified palette $\mathcal{Q}$. Let $\mathcal{X}_{\mathcal{Q}}$ be the boundary-fixed liftable quenching state space equipped with its declared compact-open C^∞ topology on each stable stratum, enlarged by the controlled non-stable paths appearing inside certificates. Choose a deterministic certificate selector which, for each state $\mathcal{S}$, chooses one energy-legal completed $\mathcal{Q}$-package when such a package is recognized and otherwise chooses the constant terminal package. Let $R_{\mathcal{Q}}(\mathcal{S})$ be the endpoint of the selected package and let*

$$\eta_{\mathcal{S}} : [0,1] \longrightarrow \mathcal{X}_{\mathcal{Q}}$$

be its certified boundary-fixed liftable interpolation path, constant when no package is selected.

For each $\mathcal{S}$, define the concatenated history $\gamma_{\mathcal{S}} : [0,\infty) \to \mathcal{X}_{\mathcal{Q}}$ by

$$\gamma_{\mathcal{S}}(n+s) = \eta_{R^n_{\mathcal{Q}}(\mathcal{S})}(s), \qquad n \in \{0,1,2,\dots\}, \quad 0 \le s \le 1,$$

with the evident endpoint identification. Let $\widehat{\mathcal{X}}_{\mathcal{Q}}$ be the set of all time shifts of such histories, viewed as a subspace of $C([0,\infty),\mathcal{X}_{\mathcal{Q}})$ with the compact-open topology. Then

$$\Theta_t(\gamma)(u) = \gamma(t+u), \qquad t,u \ge 0,$$

defines a continuous one-parameter semigroup on $\widehat{\mathcal{X}}_{\mathcal{Q}}$. Moreover, for the initial history $\gamma_{\mathcal{S}}$,

$$\gamma_{\mathcal{S}}(n) = R^n_{\mathcal{Q}}(\mathcal{S}), \qquad \gamma_{\mathcal{S}}(n+s) = \eta_{R^n_{\mathcal{Q}}(\mathcal{S})}(s),$$

so integer time records the selected discrete quenching algorithm and non-integer time records the certified interpolation. If the selected packages are energy-legal in the sense of Definition 6.8, then each history is eventually constant by Theorem 6.12; hence terminal compactness holds for this suspension model.

Proof. The deterministic selector is part of the suspension data; in an effective finite-encoding version it can be supplied by ordering the recognized packages lexicographically, and when no legal package is recognized the constant package is selected. The endpoint relation

$$\eta_{R^n_{\mathcal{Q}}(\mathcal{S})}(1) = R^{n+1}_{\mathcal{Q}}(\mathcal{S}) = \eta_{R^{n+1}_{\mathcal{Q}}(\mathcal{S})}(0)$$

is part of the meaning of a completed certified package, so the paths $\eta_{R^n_{\mathcal{Q}}(\mathcal{S})}$ concatenate to a continuous history $\gamma_{\mathcal{S}}$. Boundary fixation and liftability of every point of the path are also part of the certificate data.

The shift operator Θ_t is the ordinary translation semiflow on the compact-open path space $C([0,\infty),\mathcal{X}_{\mathcal{Q}})$. It satisfies $\Theta_0 = \mathrm{id}$ and

$$\Theta_{s+t}(\gamma)(u) = \gamma(s+t+u) = \Theta_s(\Theta_t(\gamma))(u)$$

for all $s,t,u \ge 0$. Continuity is the standard compact-open continuity of precomposition by the continuous addition map $(t,u) \mapsto t+u$ on compact subsets of $[0,\infty)$. The set $\widehat{\mathcal{X}}_{\mathcal{Q}}$ is invariant by construction, since shifting a shifted concatenated history gives another shifted concatenated history. The displayed integer-time and interpolation formulas are exactly the definition of $\gamma_{\mathcal{S}}$. Finally, every nonconstant selected package strictly lowers E, and E is well founded by Theorem 6.12; after the terminal time the selected package is constant. □

This theorem is the precise replacement for the stronger assertion that the bare reduction map on the original state space is automatically a continuous semiflow. Such a bare-space assertion can fail when the lexicographically first available certificate changes discontinuously across strata. The strict object used here is therefore the certificate-suspension flow on histories, together with the observation map $\gamma \mapsto \gamma(0)$.

9.2 Applicability and topological rigidity

Theorem 9.2 (Applicability and rigidity on slice histories)**.** *Let Θ be the certificate-suspension weak flow of Theorem 9.1. Suppose the initial state $\mathcal{S}$ is the stable projection datum of a smooth slice disk $F_0 : D^2 \hookrightarrow B^4$, and suppose every selected nonconstant package is one of the sound packages realized by Proposition 5.19 and Theorem 6.21. Then the observed history $t \mapsto \gamma_{\mathcal{S}}(t)$ has the following properties.*

(a) *Each observed state is boundary-fixed and fiberwise liftable. At certified embedded stages it is the projection of a smooth properly embedded slice disk with the same boundary knot; at controlled singular stages it is the prescribed continuous limit of such lifted disks.*

(b) *The lifted maps form homotopies of pairs $(D^2, \partial D^2) \to (B^4, S^3)$, fixed on the boundary collar. Consequently the boundary-fixed homotopy class is independent of t; after choosing a boundary basepoint, the associated relative difference class lies in $\pi_2(B^4, S^3)$ and is zero. The relative homology class $(F_t)_*[D^2] \in H_2(B^4, S^3; \mathbb{Z})$ is also independent of t, and this target group is zero.*

(c) *The terminal observed state exists and is liftable. Its terminal lift satisfies the homological inheritance conclusions of Theorem 8.15; for innermost clean closed-core components it also satisfies the source-level singular-filling conclusion of Theorem 9.5.*

(d) *A conclusion that all Milnor $\overline{\mu}$-invariants of a terminal closed-core sublink vanish requires Milnor applicability data beyond singular-chain fillings. For the clean terminal branch used in the final proof, these data are supplied by clean terminal rigidity in Theorems 9.15 and 9.16: the selected terminal sublink is link-concordant to the unlink because it bounds the disjoint embedded slice-link fillings required by the classical Milnor-concordance vanishing theorem. The singular-chain fillings, endpoint preservation, and relative homology conclusions in* (b)–(c) *are not by themselves such data.*

Proof. Part (a) is exactly the soundness content of the selected certificates. By Proposition 5.19, Theorem 6.21, and Corollary 6.22, each completed package is realized by a boundary-fixed ambient isotopy of the lifted disk, followed when necessary by a stable perturbation rel boundary collar. Hence the boundary knot and the four-dimensional liftability data persist through the selected interpolation path. The controlled singular times are not arbitrary maps; they are the prescribed limits included in the certificate.

For (b), restrict attention first to one certified interval. Its lifted interpolation is a continuous family of maps of pairs, hence a homotopy of pairs between any two times in that interval. Concatenating finitely many certified intervals gives the same conclusion for arbitrary finite times, and the eventual constant tail gives it for all later times. The long exact homotopy sequence of (B^4, S^3) contains

$$\pi_2(B^4) \longrightarrow \pi_2(B^4, S^3) \longrightarrow \pi_1(S^3),$$

and both outside groups vanish. Thus $\pi_2(B^4, S^3) = 0$. The corresponding homology calculation is the one used in Theorem 8.15: the segment

$$H_2(B^4; \mathbb{Z}) \longrightarrow H_2(B^4, S^3; \mathbb{Z}) \longrightarrow H_1(S^3; \mathbb{Z})$$

has zero outside groups, so $H_2(B^4, S^3; \mathbb{Z}) = 0$. Therefore all finite-time lifts lie in the same boundary-fixed homotopy class, the associated relative difference class is zero, and the constant relative homology class is the zero class.

Part (c) follows from the final sentence of Theorem 9.1: the energy-legal selected history is eventually constant. The terminal lift is therefore a continuous map of pairs, and Theorem 8.15 applies directly. When the terminal state is clean ribbon-normal and innermost closed-core components are selected, Theorem 9.5 applies to the terminal lift and gives simultaneous source disks and singular chain fillings.

For (d), the relevant classical input is a complement-level or concordance-level Milnor-vanishing theorem, not merely the statement that a component bounds a singular two-chain in B^4. Slice links with disjoint embedded slice disks, or links concordant to the unlink, lie in the standard vanishing class for Milnor $\overline{\mu}$-invariants. By contrast, Proposition 7.14, Corollary 7.15, and Theorems 8.15 and 9.5 show that singular fillings and relative homology zero do not control linking or higher Milnor data in the three-dimensional complement. Thus the weak flow is applicable and topologically rigid in the stated homotopy, homology, endpoint, and singular-filling senses, and the clean terminal Milnor-vanishing input is supplied later by Theorems 9.15 and 9.16. □

9.3 Innermost cores and geometric Milnor certificates

Definition 9.3 (Innermost closed-core component)**.** Let $\mathcal{S}$ be a clean ribbon-normal state and let $\mathcal{R}_{\mathrm{cl}}(\mathcal{S}) = \{C_1, \dots, C_m\}$ be its closed-core sublink. Choose source decker circles $\gamma_i \subset D^2$ representing the components, and let $E_i \subset D^2$ be the source disk bounded by γ_i, chosen as the component of $D^2 \setminus \gamma_i$ not containing ∂D^2. A component C_i is *innermost* if $\mathrm{Int}(E_i)$ contains no source circle γ_j representing another closed-core component. The set of innermost components is denoted $\mathcal{I}(\mathcal{R}_{\mathrm{cl}}(\mathcal{S}))$.

Lemma 9.4 (Innermost source disks are disjoint)**.** *If C_i and C_j are distinct innermost closed-core components of a clean ribbon-normal state, then their source filling disks satisfy $E_i \cap E_j = \emptyset$.*

Proof. By Lemma 7.9, two disks bounded by disjoint source circles in D^2 are either disjoint or nested. If $E_i \subset E_j$, then $\gamma_i \subset \mathrm{Int}(E_j)$, so C_j is not innermost. The case $E_j \subset E_i$ is symmetric. Hence nesting is impossible and the disks are disjoint. □

Theorem 9.5 (Innermost singular-filling certificate)**.** *Let $\mathcal{S}_\infty$ be a clean ribbon-normal terminal state with terminal lift represented by a continuous map of pairs*

$$F_\infty : (D^2, \partial D^2) \longrightarrow (B^4, S^3),$$

and let $\mathcal{I} \subset \mathcal{R}_{\mathrm{cl}}(\mathcal{S}_\infty)$ be the set of innermost closed-core components, represented by source circles γ_i and source disks E_i. Then $\{E_i\}_{C_i \in \mathcal{I}}$ is a finite pairwise disjoint family in D^2, and each component has canonical singular fillings

$$F_{\infty\#}[E_i] \in C_2(B^4; \mathbb{Z}), \qquad (\pi \circ F_\infty)_\#[E_i] \in C_2(B^3; \mathbb{Z}),$$

with boundaries

$$\partial F_{\infty\#}[E_i] = F_{\infty\#}[\gamma_i], \qquad \partial(\pi \circ F_\infty)_\#[E_i] = (\pi \circ F_\infty)_\#[\gamma_i].$$

The construction is simultaneous over all innermost components and is boundary-collar disjoint in the source. It gives a genuine unconditional source-filling and singular-chain certificate, but it does not assert that the image chains are embedded, pairwise disjoint in B^4, or disjoint from the other projected core components in B^3.

Proof. The source circles of distinct closed-core components are disjoint in the source double-decker graph after the clean ribbon-normal perturbation. By Lemma 9.4, the innermost disks E_i are pairwise disjoint. They lie in the interior of the source disk, hence away from the fixed boundary collar.

Give each E_i the orientation whose boundary orientation is the chosen orientation of γ_i. Since E_i is an ordinary compact disk in the source, its singular fundamental chain satisfies $\partial[E_i] = [\gamma_i]$. Naturality of the singular boundary operator gives

$$\partial F_{\infty\#}[E_i] = F_{\infty\#}\partial[E_i] = F_{\infty\#}[\gamma_i],$$

and the same calculation after composing with π gives the projected equality in B^3. Finiteness follows from finiteness of the closed-core set. Nothing in this chain-level calculation implies injectivity of $F_\infty|_{E_i}$, local flatness, smoothness, or disjointness of the image supports; those are precisely the additional clauses isolated in Definitions 9.6 and 9.9. □

Definition 9.6 (Innermost geometric rigidity certificate)**.** Let $\mathcal{S}_\infty$ be a clean ribbon-normal terminal state with terminal lift F_∞, possibly arising as the endpoint of a certificate-suspension history, and let $\mathcal{I} \subset \mathcal{R}_{\mathrm{cl}}(\mathcal{S}_\infty)$ be a non-empty set of innermost closed-core components with source disks E_i. The word clean is essential here: branch points, triple points, and non-ribbon interval endpoints have already been removed or excluded before this certificate is invoked. An innermost geometric rigidity certificate for $\mathcal{I}$ consists of:

(a) a verification that F_∞ is injective on $\bigsqcup_{C_i \in \mathcal{I}} E_i$; explicitly, if $x, y \in \bigsqcup E_i$ satisfy $F_\infty(x) = F_\infty(y)$, then the trivial case $x = y$ is separated from the nontrivial case $x \neq y$, and the latter is excluded as a source double-decker point of the clean terminal projection;

(b) a verification that each restricted image $F_\infty(E_i)$ is locally flat or has a boundary-fixed smooth embedded replacement disk with the same boundary curve and with the boundary-fixed homotopy class preserved;

(c) a verification that the replacement disks can be chosen pairwise disjoint;

(d) the orientation and lower-order indeterminacy data needed for the Milnor $\overline{\mu}$-invariants under consideration.

The certificate may be proved by showing that $\mathrm{Int}(E_i)$ contains no relevant source decker data and that all possible inter-disk intersections have certified separating neighborhoods, but these facts are certificate data; they are not consequences of palette-relative blockedness alone. If Whitney approximation is used in clause (b), it is the boundary-fixed version: the smoothing is performed

within the fixed boundary homotopy class of $F_\infty|_{\gamma_i}$. Since $\pi_2(B^4, S^3) = 0$ by the calculation in Theorem 9.2, the relative homotopy class contributes no additional ambiguity, and the Milnor data are transported through the resulting link concordance or link homotopy as part of the certificate.

Lemma 9.7 (Boundary separation under a rigidity certificate)**.** *If $\mathcal{I}$ admits an innermost geometric rigidity certificate, then the corresponding boundary curves $F_\infty(\gamma_i) \subset S^3$ bound pairwise disjoint smooth embedded disks $\Delta_i \subset B^4$.*

Proof. By Lemma 9.4, the source disks are pairwise disjoint. The rigidity certificate supplies injectivity of the terminal lift on their union and supplies either local flatness with a boundary-fixed smoothing or explicit smooth embedded replacement disks. The injectivity clause is understood nontrivially: equality $F_\infty(x) = F_\infty(y)$ with $x = y$ is harmless, while equality with $x \neq y$ would give a source double-decker point in the clean terminal projection and is excluded by the certificate. Because the certified images or replacements are compact and pairwise disjoint, the certificate's separating-neighborhood clause gives pairwise disjoint neighborhoods in B^4. Perform the boundary-fixed smoothing or use the replacement disks inside these neighborhoods. The boundary-fixed Whitney approximation, when used, preserves the boundary homotopy class; the relative group $\pi_2(B^4, S^3)$ is zero by Theorem 9.2, so no additional relative homotopy ambiguity is introduced. The resulting disks Δ_i have the prescribed boundaries and remain pairwise disjoint. □

Remark 9.8 (Blockedness and rigidity certificates). The innermost condition proves only a planar separation statement. It does not prove that F_∞ is injective on $\bigsqcup E_i$, that the terminal limit is smoothly embedded there, or that simultaneous disjoint smoothings exist. A claim that any remaining interior defect could be removed would require either an existentially complete palette or a specific certified move. Therefore the geometric part can be closed by an innermost rigidity certificate, but the certificate is still an additional mathematical input unless it is proved for the chosen palette or WTQF.

Definition 9.9 (Geometric Milnor applicability certificate)**.** Let $\mathcal{S}_\infty$ be a clean ribbon-normal terminal state with an oriented closed-core sublink

$$L_\infty = C_1 \cup \cdots \cup C_m \subset B^3 \subset S^3$$

represented by pairwise disjoint smooth curves. A geometric Milnor applicability certificate for L_∞ consists of:

(a) selected source circles $\gamma_i \subset D^2$ projecting to C_i, together with either compact source disks $P_i \subset D^2$ satisfying $\partial P_i = \gamma_i$ after the prescribed orientation convention, or explicitly certified replacement disks already constructed in the lifted four-dimensional disk;

(b) in the source-disk case, a proof that the terminal lift F_∞ restricts to a smooth embedding on each P_i, after a boundary-fixed smoothing if the terminal map was obtained as a controlled continuous limit; in the replacement case, the same smooth embeddedness is part of the replacement certificate;

(c) a proof that the resulting embedded disks, denoted $\Delta_i \subset B^4$, are pairwise disjoint and have $\partial \Delta_i = C_i \subset S^3$;

(d) the lower-order vanishing or indeterminacy choices needed to read the relevant Milnor $\overline{\mu}$-invariants of L_∞;

(e) when the obstruction argument uses a link-homotopy from L_∞ to another representative, a finite realization certificate showing that the link-homotopy is implemented in the projection framework by certified sheet-height passes, arc slides, and stable perturbations, with the different components supported in disjoint source neighborhoods whenever innermost source disks are used.

When these data are supplied, L_∞ is a classical smooth slice link in the sense needed for Milnor concordance invariance.

Theorem 9.10 (Geometric rigidity implies classical Milnor applicability)**.** *If a terminal closed-core sublink L_∞ admits a geometric Milnor applicability certificate in the sense of Definition 9.9, then the classical Milnor-vanishing input required in Definition 8.10(e) applies to L_∞. In particular, every Milnor $\overline{\mu}$-invariant of L_∞ in the certified indeterminacy range is zero.*

Proof. By Definition 9.9(b)–(c), the components of L_∞ bound pairwise disjoint smooth embedded disks $\Delta_1, \dots, \Delta_m \subset B^4$. Therefore L_∞ is a smooth slice link and is smoothly concordant to the unlink by deleting small interior disks from the Δ_i and reading the remaining annuli in $S^3 \times [0,1]$. Milnor $\overline{\mu}$-invariants in the certified range are invariant under link concordance, with the usual lower-order indeterminacy conventions. Since the unlink has all such invariants equal to zero, the same invariants of L_∞ vanish. If the obstruction proof also uses link-homotopy classification, clause (e) supplies the separate finite realization data needed to interpret that link homotopy inside the certified quenching language. This is a direct use of the classical Milnor theory cited in Definition 8.10; no singular-filling or generalized Milnor theorem is being invoked. □

Theorem 9.11 (Innermost rigidity certificate gives Milnor applicability)**.** *Let $\mathcal{S}_\infty$ be a clean ribbon-normal terminal state arising as a certified WTQF terminal lift from slice data. If a non-empty innermost closed-core sublink $L_{\mathcal{I}}$ admits an innermost geometric rigidity certificate, then $L_{\mathcal{I}}$ admits a geometric Milnor applicability certificate in the sense of Definition 9.9. Consequently the classical Milnor-vanishing theorem applies to $L_{\mathcal{I}}$.*

Proof. The selected source circles and their disks are those in Definition 9.3. The required disjoint smooth slice disks are provided by Lemma 9.7. The orientation and lower-order indeterminacy data are part of the rigidity certificate. For the Milnor-vanishing conclusion used here, concordance to the unlink is obtained directly from the disjoint slice disks, so no separate link-homotopy realization is needed; if a later obstruction proof invokes link-homotopy classification instead, the realization data required in Definition 9.9(e) must be supplied as part of that proof. Thus the data satisfy Definition 9.9, and Theorem 9.10 gives the classical Milnor-vanishing conclusion. □

Definition 9.12 (Clean terminal rigidity)**.** Let $\mathcal{S}_\infty$ be a clean ribbon-normal blocked terminal state arising from slice data, and let $L_{\mathcal{I}}$ be a non-empty innermost closed-core sublink selected by the terminal Milnor obstruction certificate. The state is *lifted-source rigid on $L_{\mathcal{I}}$* if the terminal lift maps the selected source disks smoothly and injectively to pairwise disjoint disks with boundary the lifted source curves. This property does not identify those lifted boundaries with the projected curves defining $L_{\mathcal{I}}$.

Theorem 9.13 (Clean terminal states are rigid on innermost cores)**.** *Let $\mathcal{S}_\infty$ be a clean ribbon-normal terminal state obtained from a smooth slice disk by a finite certified history, and let $L_{\mathcal{I}}$ be its non-empty innermost closed-core sublink. Then the selected source disks have pairwise disjoint*

smooth embedded images under the terminal lift, with boundaries equal to the lifted source curves $\widetilde{L}_{\mathcal{I}} = F_\infty(\bigsqcup_i \gamma_i)$.

Proof. By Theorem 6.21 and Proposition 6.23, the terminal lift is represented by a smooth proper embedding

$$F_\infty : (D^2, \partial D^2) \hookrightarrow (B^4, S^3)$$

with the original boundary knot. Let $E_1, \dots, E_m \subset \mathrm{Int}(D^2)$ be the innermost source disks. They are pairwise disjoint by Lemma 9.4. Since F_∞ is injective on all of D^2, its restriction to $\bigsqcup_i E_i$ is injective. Since F_∞ is smooth and proper, each restriction $F_\infty|_{E_i}$ is a smooth embedded disk. Injectivity of F_∞ and disjointness of the source disks imply

$$F_\infty(E_i) \cap F_\infty(E_j) = \emptyset \qquad (i \neq j).$$

Thus the lifted source curves $\widetilde{L}_{\mathcal{I}}$ bound pairwise disjoint smooth disks $F_\infty(E_i)$. Their boundaries are the curves $F_\infty(\gamma_i)$ in the interior lifted disk, not automatically the projected curves $p_\infty(\gamma_i) \subset B^3$. Therefore this theorem supplies lifted-source rigidity but does not by itself supply a classical slice-link certificate for the projected core. □

Assumption 9.14 (Projected-core Milnor-applicability input)**.** For every clean ribbon-normal blocked terminal state reached from slice data, the selected projected core $L_{\mathcal{I}} = p_\infty(\bigsqcup_i \gamma_i) \subset B^3 \subset S^3$ admits a geometric Milnor applicability certificate in the sense of Definition 9.9. Equivalently, the certificate includes a boundary-controlled bridge from the projected curves $p_\infty(\gamma_i)$ to the lifted curves $F_\infty(\gamma_i)$, or directly supplies pairwise disjoint smooth disks in a four-ball with boundary $L_{\mathcal{I}}$. Denote this assertion by $\mathsf{H}_{\mathrm{app}}$.

Theorem 9.15 (Projected-core applicability supplies Milnor vanishing)**.** *Assume* $\mathsf{H}_{\mathrm{app}}$. *Let* $\mathcal{S}_\infty$ *be a clean ribbon-normal blocked terminal state obtained by a finite certified history from slice data. Then every Milnor* $\overline{\mu}$*-invariant of the selected innermost closed-core sublink* $L_{\mathcal{I}}$ *in the certified indeterminacy range vanishes.*

Proof. By $\mathsf{H}_{\mathrm{app}}$, $L_{\mathcal{I}}$ has a geometric Milnor applicability certificate. Apply Theorem 9.10. □

Theorem 9.16 (Clean blocked terminal exclusion)**.** *Assume* $\mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$. *Then no clean ribbon-normal blocked terminal state obtained by a finite certified history from smooth slice-disk data exists.*

Proof. Suppose $\mathcal{S}_\infty$ is such a state. Since it is blocked, clean, and ribbon-normal, its closed core is non-empty. By Theorem 8.13, there exists a certified multi-index I such that

$$\overline{\mu}_I(L_{\mathcal{I}}) \neq 0.$$

By Theorem 9.15, the same invariant satisfies

$$\overline{\mu}_I(L_{\mathcal{I}}) = 0.$$

Contradiction. □

Remark 9.17 (Role of terminal rigidity). The implication from source-disk filling to Milnor vanishing factors as

$$\text{singular source fillings} \not\Rightarrow \text{embedded disjoint slice disks},$$
$$\text{clean-terminal rigidity} \Rightarrow \text{embedded disjoint slice disks} \Rightarrow \overline{\mu} = 0.$$

The non-vanishing conclusion is Theorem 8.13, obtained from terminality by contraposition of Theorem 8.12.

Corollary 9.18 (WTQF route completed by terminal Milnor rigidity)**.** *Assume* H_{Q}*. Then every certificate-suspension WTQF history arising from slice data has a ribbon terminal value.*

Proof. Theorems 9.1 and 9.2 give a terminal state with the same boundary knot and slice realization class. Corollary 10.24 excludes the non-clean blocked case under H_{nc}, and Theorem 9.16 excludes the clean blocked case under $\mathsf{H}_{\mathrm{red}} \wedge \mathsf{H}_{\mathrm{app}}$. The terminal alternative leaves only the ribbon case. □

9.4 Conclusion on weak-flow legality

Remark 9.19 (Formal scope of the weak-flow construction). The WTQF is the suspension of a discrete relation, not a solution of a differential equation. Its unconditional consequences are the semigroup law, boundary-fixed liftability, and constancy of the specified relative homotopy and homology classes. Exclusion of all blocked terminal states is exactly the use of H_{Q}.

9.5 Certificate solvability levels

For an encoded record c, let $\mathrm{Valid}(c)$ denote geometric realizability by the prescribed boundary-fixed isotopy or lifted homotopy. We prove implications of the form

$$\mathrm{Valid}(c) \Longrightarrow \mathrm{Sound}(c)$$

and termination for any set of valid energy-decreasing records. We do not prove that $\mathrm{Valid}(c)$ is decidable. Markov-type undecidability results [20] preclude an unrestricted decidability assertion without additional hypotheses; this observation is independent of the geometric existence inputs isolated above.

10 The completion condition and final reduction

This chapter proves the relative branch-cancellation, Roseman/movie, and framed cusp-merge theorems. It then proves three unconditional structural results for branch-free source graphs and isolates the projected-disk endpoint-recovery clause H_{nc}. Together with the clean-core clauses $\mathsf{H}_{\mathrm{red}}$ and $\mathsf{H}_{\mathrm{app}}$, this forms the single quenching completion condition H_{Q}. The chapter concludes with the reduction $\mathsf{H}_{\mathrm{Q}} \Rightarrow$ SRC.

10.1 Standard non-clean inputs

Convention 10.1 (Relative category for the external inputs)**.** Fix a collar

$$C = \partial D^2 \times [0, \varepsilon] \subset D^2$$

and a smooth proper embedding

$$\Phi : (D^2, \partial D^2) \hookrightarrow (B^4, S^3)$$

whose restriction to C is the prescribed boundary model. Put $p = \pi \circ \Phi$, where $\pi : B^4 \to B^3$ is the fixed projection. A deformation is *relative to the boundary collar*, abbreviated *rel* C, if it is represented by an ambient isotopy

$$\Psi : [0,1] \times B^4 \longrightarrow B^4$$

such that, for every $s \in [0,1]$,

$$\Psi_s|_{\nu(\Phi(C))} = \mathrm{id}, \qquad \Psi_s|_{S^3} = \mathrm{id}, \qquad \Phi_s = \Psi_s \circ \Phi,$$

for a fixed neighborhood $\nu(\Phi(C))$. If $U \Subset \mathrm{Int}(D^2)$ is a compact support, we additionally require $\Psi_s = \mathrm{id}$ outside a compact neighborhood of $\Phi(U)$. Endpoints are required to be stable projections; intermediate maps may cross only the finitely many codimension-one singularities specified by the relevant Roseman or movie theorem.

Theorem 10.2 (Relative Carter–Saito branch cancellation)**.** *Let $\Phi : (D^2, \partial D^2) \hookrightarrow (B^4, S^3)$ have stable projection $p = \pi \circ \Phi$, with the extendible collar frame of Definition 5.1. Then there is an ambient isotopy rel C, supported in the interior, whose final stable projection has no branch points.*

Proof. For an oriented surface projection, the algebraic number of branch points equals the normal Euler number, with the local signs used by Carter–Saito [2, Theorem 1]. The identical obstruction calculation for a surface with fixed boundary framing gives

$$\sum_{b \in \mathrm{Br}(p)} \mathrm{sign}(b) = e(\nu_D, \tau|_{\partial D}).$$

By Definition 5.1, the boundary frame is the restriction of a global normal trivialization, so the right-hand side is zero. Hence the positive and negative branch points can be paired.

Fix an opposite-sign pair b_+, b_-. The branch and triple preimages are finite and the source double-decker locus is a finite one-complex. General position in the connected surface $\mathrm{Int}(D^2)$ therefore gives an embedded joining arc

$$\gamma : [0,1] \hookrightarrow \mathrm{Int}(D^2)$$

from b_+ to b_-, disjoint from the remaining branch and triple preimages and transverse to the double-decker locus. Carter–Saito's branch-transport construction moves one branch point along γ; passage across another sheet is their Lemma 5. Their local opposite-sign cancellation then removes the pair.

All charts used in the construction lie in an arbitrarily small neighborhood of the compact arc $\gamma([0,1])$. Choose

$$\gamma([0,1]) \subset U \Subset \mathrm{Int}(D^2), \qquad U \cap C = \emptyset.$$

The local deformation is fixed near $\Phi(\partial U)$. The relative isotopy extension theorem extends it to an ambient isotopy of B^4 equal to the identity on S^3, on a neighborhood of $\Phi(C)$, and outside a compact neighborhood of $\Phi(U)$. A final stable perturbation is chosen rel C by Lemma 1.16. Thus the cancellation is boundary-fixed.

Each cancellation removes exactly two branch points. Since the branch set is finite, iteration terminates and yields a branch-free stable projection. □

Theorem 10.3 (Relative finite Roseman and movie decomposition)**.** *Let $\Phi_0, \Phi_1 : (D^2, \partial D^2) \hookrightarrow (B^4, S^3)$ have stable endpoint projections and be joined by a smooth ambient isotopy rel C, supported over $U \Subset \mathrm{Int}(D^2)$. Then, after a perturbation fixed on*

$$(D^2 \times \{0,1\}) \cup (C \times [0,1])$$

and outside $U \times [0,1]$, the projected isotopy decomposes into finitely many Roseman moves and planar isotopies. Any associated movie decomposes into finitely many Carter–Saito movie moves and level-preserving changes.

Proof. Write $\Phi_s = \Psi_s \circ \Phi_0$ and $p_s = \pi \circ \Phi_s$. Apply relative multijet transversality to the one-parameter family p_s, fixing the endpoint slices, the collar cylinder, and the complement of $U \times [0,1]$. The perturbed family is transverse to every codimension-one stratum of the discriminant of surface projections and avoids all strata of codimension at least two. Hence p_s is stable except at isolated parameters, with exactly one codimension-one event at each exceptional parameter.

The exceptional parameter set is a closed zero-dimensional submanifold of the compact interval $[0,1]$, hence is finite. Roseman's local classification identifies each event with one of the seven Roseman move types [24, Theorem 1]. Since the family is unchanged outside U, every move is supported in U, and ordering the exceptional parameters gives the required finite decomposition.

Choose a generic movie height function, fixed near the collar and the endpoint movies. The Carter–Saito movie theorem converts the resulting generic two-parameter data into finitely many movie moves, with level-preserving changes on intervals containing no exceptional event [3, 4]. The lifted isotopy transports the broken-sheet order, branch signs, and triple-point labels. This proves the relative statement. □

Theorem 10.4 (Relative framed cusp merge)**.** *Let $f : X^4 \to B^2$ be a stable map occurring in an auxiliary certificate, and let $c_-, c_+ \in \mathrm{Crit}(f)$ be indefinite cusps contained in an interior support. Suppose there is a framed joining curve λ from c_- to c_+ satisfying the embedding, endpoint-sector, singular-fiber-disjointness, and kernel-framing hypotheses of Behrens–Hayano. Then every neighborhood $V \Subset \mathrm{Int}(X)$ of λ supports a cusp-merge homotopy $f_s : X \to B^2$ such that*

$$f_0 = f, \qquad f_s|_{X \setminus V} = f|_{X \setminus V}, \qquad \mathrm{Crit}(f_1) \cap V = \big(\mathrm{Crit}(f) \cap V\big) \setminus \{c_-, c_+\}.$$

If V is disjoint from the boundary collar, the homotopy is relative to that collar.

Proof. Theorem A of Behrens–Hayano identifies elementary cusp merges with homotopy classes of framed joining curves satisfying precisely these hypotheses [5]. Apply their construction inside a coordinate neighborhood V_0 with

$$\lambda([0,1]) \subset V_0 \Subset V.$$

The elementary model is constant near ∂V_0. Extending it by the constant homotopy f on $X \setminus V_0$ therefore gives a smooth global homotopy supported in V. If V is disjoint from the fixed collar, every member of the homotopy equals f there. □

Assumption 10.5 (Restricted projected-disk triple transfer)**.** The sole remaining non-clean hypothesis is

$$\mathsf{H}_{\mathrm{nc}} := \mathsf{H}_{\mathrm{ct}}^{\mathrm{rel}}.$$

It asserts that every slice-realized stable state $\mathcal{S}$ with $b(\mathcal{S}) = 0$ and $t(\mathcal{S}) > 0$ admits a finite boundary-fixed certified package P, lifted through properly embedded disks rel C, whose endpoint $\mathcal{S}' = \mathrm{End}(P)$ satisfies

$$b(\mathcal{S}') = 0, \qquad t(\mathcal{S}') < t(\mathcal{S}), \qquad \mathrm{E}(\mathcal{S}') < \mathrm{E}(\mathcal{S}).$$

No claim is made that a prescribed triple point is locally removable or that each elementary submove decreases t.

Remark 10.6 (Why the remaining condition is difficult). Condition H_{nc} is a package-level existence assertion, not a universal local cancellation rule. A universal monotone triple-point elimination theorem is too strong in view of positive triple-point phenomena for knotted surfaces. Two weaker constructions remain conceivable: a finite classification of the encoded branch-free configurations followed by casewise packages, or a temporary four-dimensional or immersed WTQF excursion followed by a separate endpoint-embedding recovery theorem. Neither construction is established here. The next three theorems prove the unconditional source-graph and tubular-neighborhood statements available to such a program and isolate the missing recovery step.

10.2 Three unconditional structural theorems for a possible replacement of condition C

Theorem 10.7 (Circuit decomposition of a branch-free source decker graph)**.** *Let $G \subset \mathrm{Int}(D^2)$ be the finite source decker graph of a stable projected disk after subdivision at all triple-point preimages. If G has no branch vertices and no boundary vertices, then every vertex has even valence and*

$$E(G) = E(T_1) \,\dot\cup\, \cdots \,\dot\cup\, E(T_r),$$

where each $T_j : S^1 \to G$ is a closed edge trail. Smoothing the finitely many vertex visits produces immersed closed curves in D^2 whose union traverses every edge exactly once.

Proof. Regular source double-curve vertices have valence two, while a triple-point preimage is the transverse meeting of two source double-curve branches and hence has valence four. Branch and boundary vertices are excluded. Thus all vertices have even valence.

In a connected component containing an edge, start along an unused edge and continue through an unused incident edge. Every arrival at a vertex consumes one edge and every departure consumes a second. Even valence prevents the trail from stopping away from its initial vertex. Finiteness forces a closed trail. Removing its edges leaves even valence at every vertex. Induction on the number of edges, followed componentwise, gives the stated decomposition. Finally, smooth each incoming–outgoing pair inside pairwise disjoint vertex disks. This changes no edge away from the vertex disks and gives the required immersed circuits. □

Theorem 10.8 (Separation in a three-dimensional source thickening)**.** *Let $\gamma_j : S^1 \looparrowright \mathrm{Int}(D^2)$, $1 \le j \le r$, be finitely many smooth immersed closed curves with transverse multiple intersections and no coincident arcs. For every $\varepsilon > 0$ there are pairwise disjoint smooth embeddings*

$$\widehat{\gamma}_j : S^1 \hookrightarrow \mathrm{Int}(D^2) \times (-\varepsilon, \varepsilon)$$

such that $\mathrm{pr}_{D^2} \circ \widehat{\gamma}_j = \gamma_j$. Any prescribed strict local ordering of the branches can be realized.

Proof. The inverse image of the finite multiple-point set is finite. Assign distinct numbers in $(-\varepsilon, \varepsilon)$ to the branch preimages over each multiple point, respecting the prescribed order. On pairwise disjoint small parameter intervals around these preimages choose smooth height functions with the assigned central values and value zero near the interval endpoints; extend them by zero elsewhere. Put

$$\widehat{\gamma}_j(u) = (\gamma_j(u), z_j(u)).$$

Two distinct branch preimages of a multiple point have different heights. Away from those preimages, the projected curves have neither multiple points nor coincident arcs. Hence the union of the lifted curves is injective. It is an immersion, and compact injective immersions of circles into a Hausdorff manifold are embeddings. Distinct components are disjoint by the same injectivity argument. □

Theorem 10.9 (Relative four-dimensional tubular realization)**.** *Let $\Phi : (D^2, \partial D^2) \hookrightarrow (B^4, S^3)$ be a smooth proper embedding with the fixed extendible boundary frame. Every resolved link*

$$\widehat{L} \subset \mathrm{Int}(D^2) \times (-\varepsilon, \varepsilon)$$

from Theorem 10.8, for sufficiently small ε, has a pairwise disjoint smooth realization $L^4_\Phi \subset \mathrm{Int}(B^4)$ in an arbitrarily small tubular neighborhood of $\Phi(D^2)$. Every component is null-homologous in B^4 and bounds an integral singular two-chain.

Proof. The normal bundle of the embedded disk is an oriented rank-two bundle. It is trivial because D^2 is contractible, and the chosen trivialization may be taken to extend the fixed boundary frame. The relative tubular-neighborhood theorem gives

$$\Theta : D^2 \times D^2_\delta \longrightarrow \nu(\Phi(D^2)) \subset B^4, \qquad \Theta(x, 0) = \Phi(x),$$

for some $\delta > 0$ [13, 9]. For $\varepsilon < \delta$, regard the source thickening as

$$D^2 \times (-\varepsilon, \varepsilon) \times \{0\} \subset D^2 \times D^2_\delta$$

and define $L^4_\Phi = \Theta(\widehat{L} \times \{0\})$. Embeddedness and pairwise disjointness follow from those of $\widehat{L}$ and injectivity of Θ. Shrinking δ makes the support arbitrarily small. Finally, contractibility of B^4 gives $H_1(B^4; \mathbb{Z}) = 0$, so every oriented component is the boundary of an integral singular two-chain. □

Corollary 10.10 (Exact unconditional output)**.** *Every branch-free stable source decker graph without boundary vertices admits a finite resolved circuit system and a pairwise disjoint realization of that system in a tubular neighborhood of the lifted disk; its components admit singular two-chain fillings in B^4.*

Proof. Apply Theorems 10.7 to 10.9 successively. □

Remark 10.11 (Remaining endpoint-recovery problem). The realized link L^4_Φ lies in $\mathrm{Int}(B^4)$, whereas the classical Milnor invariants used below are attached to a link in S^3. Singular fillings do not imply disjoint embedded slice disks, and the tubular realization supplies neither a boundary-controlled bridge to the projected core nor an embedded endpoint with lower triple-point energy. Consequently the three structural theorems are unconditional but do not imply H_{nc}, $\mathsf{H}_{\mathrm{red}}$, or $\mathsf{H}_{\mathrm{app}}$. The remaining task is precisely a casewise projected package or a four-dimensional excursion equipped with an explicit embedded endpoint and a certified return to the required three-dimensional core.

Remark 10.12 (Exact scope of the cited sources). Carter–Saito give the branch transport and cancellation models, Roseman and Carter–Saito give the finite diagram and movie moves, and Behrens–Hayano give the framed cusp-merge model. Their relative forms are Theorems 10.2 to 10.4. None of these results proves the projected-disk package-level decrease required by Assumption 10.5.

Proposition 10.13 (Status after relative localization)**.** *The assertions*

$$\mathsf{H}^{\mathrm{rel}}_{\mathrm{bp}} \wedge \mathsf{H}^{\mathrm{rel}}_{\mathrm{Rm}} \wedge \mathsf{H}^{\mathrm{rel}}_{\mathrm{cm}}$$

are the unconditional theorems Theorems 10.2 to 10.4. The remaining non-clean condition is exactly $\mathsf{H}_{\mathrm{nc}} = \mathsf{H}^{\mathrm{rel}}_{\mathrm{ct}}$.

Definition 10.14 (Standard non-clean cleanification package)**.** A standard non-clean cleanification package at a stable quenching state $\mathcal{S}$ is a finite certificate record obtained from Theorems 10.2 and 10.3 and Assumption 10.5. It consists of:

(a) an interior support neighborhood disjoint from the fixed boundary collar;

(b) either an opposite-sign branch pair with a certified joining arc, an auxiliary cusp pair with a framed joining curve, or an innermost triple-point support with the complete transfer certificate;

(c) the boundary-fixed lifted isotopy or interpolation supplied by the three relative theorems or by Assumption 10.5;

(d) a finite Roseman/movie decomposition of the interpolation, with height-ordering and broken-sheet data recorded at every stable stage;

(e) a final state $\mathcal{S}'$ satisfying $\mathrm{E}(\mathcal{S}') < \mathrm{E}(\mathcal{S})$.

Such a package is valid precisely when the listed data are realized by the corresponding boundary-fixed ambient isotopy of the embedded disk, followed by the declared stable perturbation rel boundary collar.

Lemma 10.15 (Standard packages are certified quenching packages)**.** *Assume Assumption 10.5. Every standard non-clean cleanification package is a certified quenching package in the sense of Definitions 6.8 and 5.18. It preserves the boundary knot and the slice-disk realization class.*

Proof. The support is disjoint from the boundary collar, so the boundary knot, orientation, and boundary projection convention are fixed. The lifted isotopy or interpolation is part of the imported input, and the finite Roseman/movie decomposition turns it into a finite certificate record in the state space. The final stabilization is rel boundary collar. The last clause in Definition 10.14 is exactly the energy-legality condition $\mathrm{E}(\mathcal{S}') < \mathrm{E}(\mathcal{S})$. Therefore the package satisfies the boundary invariance, topological preservation, and discrete monotonicity requirements of Definition 5.7, and the soundness conclusion follows from the same isotopy-extension argument used in Proposition 5.19 and Theorem 6.21. □

10.3 Twist control and finite palette expansion

The branch-point cancellation input must be used with one extra bookkeeping convention. A local branch-pair type is not determined only by the abstract source graph: the associated double-curve arc may carry relative twisting in the fourth-coordinate normal direction. The following definitions make that twisting part of the finite certificate record and show that it is controlled by the existing interaction coordinate of E.

Definition 10.16 (Branch-arc twist degree)**.** Let b be an interior branch point of a stable quenching state $\mathcal{S}$, and let γ_b be the associated source double-decker arc in the branch-pair local model, with endpoints the paired branch points b and b'. Choose a collar-compatible tubular neighborhood

$$\nu(\gamma_b) \cong \gamma_b \times D^2$$

in the source disk, together with the endpoint framings supplied by the local Whitney-umbrella charts at b and b'. The *twist degree* $\tau(b) \in \mathbb{Z}$ is the relative winding number of the normal direction used by the broken-surface height convention along γ_b, measured between these endpoint framings.

Convention 10.17 (Twist-resolving normal form)**.** When a branch cancellation certificate is recorded, the neighborhood $\nu(\gamma_b)$ is first put in twist-resolving normal form. This means that the relative winding of Definition 10.16 is represented by a finite normal strip A_b running along γ_b, whose marked transverse intersections with the remaining source decker graph record the full twists. These marked points are part of the certificate record and are counted by the interaction coordinate $i(\mathcal{S})$ under Convention 6.2. The convention is the certificate-level form of the Carter–Saito local normal form: without recording these marked intersections, the abstract combinatorial graph alone would not control the branch-pair twisting.

Proposition 10.18 (Twist degree controlled by interaction energy)**.** *Let b be an interior branch point of a stable state $\mathcal{S}$, and put the associated branch arc γ_b in the twist-resolving normal form of Convention 10.17. Then*

$$|\tau(b)| \le i(\mathcal{S}).$$

Consequently, if $i(\mathcal{S}) \le N$, then the possible values of $\tau(b)$ lie in the finite set $\{-N, -N+1, \dots, N-1, N\}$, of cardinality at most $2N+1$.

Proof. In twist-resolving normal form, the relative winding number $\tau(b)$ is computed as the algebraic count of the marked transverse intersections of the strip A_b with the remaining source decker graph in $\nu(\gamma_b)$. Let m_b be the corresponding geometric count. The absolute value of an algebraic count is bounded by the geometric count, so $|\tau(b)| \le m_b$.

By Convention 10.17, all marked twist intersections contributing to m_b are recorded as recognized strip–decker interactions and hence are included in the interaction number $i(\mathcal{S})$ of Definition 6.1 and Convention 6.2. Thus $m_b \le i(\mathcal{S})$, proving $|\tau(b)| \le i(\mathcal{S})$. The finite-value statement is immediate from integrality of $\tau(b)$. □

Definition 10.19 (N-bounded non-clean configuration set)**.** Fix $N \in \mathbb{N}$. An N-bounded non-clean local configuration is the stable-equivalence class of an innermost branch-pair or triple-point support in an encoded quenching state whose relevant energy coordinates satisfy

$$b(\mathcal{S}), t(\mathcal{S}), a(\mathcal{S}), c(\mathcal{S}), i(\mathcal{S}), n(\mathcal{S}), q(\mathcal{S}) \le N,$$

with the normal cell structure, height ordering, boundary-collar position, and, for branch pairs, the twist degree τ included in the record. Denote the set of such local configuration classes by $\mathfrak{C}_N$.

Remark 10.20 (Coordinate-box nature of the bound). The bound in Definition 10.19 is a coordinate-box bound on the finite encoded data, not merely a comparison with a scalar or with a lexicographic energy vector. Lexicographic decrease alone permits later coordinates to increase after an earlier coordinate drops. The finite-palette statement below is therefore correctly read as an N-box statement, or in a reductio argument as a statement after fixing a finite coordinate bound for the encoded domain under consideration.

Lemma 10.21 (Finiteness of $\mathfrak{C}_N$)**.** *For every fixed N, the set $\mathfrak{C}_N$ is finite.*

Proof. The coordinates b, t, a, c bound the numbers of branch points, triple points, bad interval components, and circular components. The coordinates i, n, q bound marked interactions, normal elementary segments, and active certificate records. Therefore only finitely many encoded incidence, height-ordering, and local source-decker types occur under the coordinate bound.

For branch-pair configurations, Proposition 10.18 gives $|\tau| \le i(\mathcal{S}) \le N$, so the possible twist degrees form a finite set. Adding this finite twist datum to the already finite normal combinatorics still gives a finite set of local configuration classes. Thus $\mathfrak{C}_N$ is finite. □

Proposition 10.22 (Finite expanded palette for N-bounded non-clean states)**.** *Assume Assumption 10.5. Let $\mathcal{Q}$ be a finite certified palette and fix $N \in \mathbb{N}$. There exists a finite expanded palette $\mathcal{Q}_N^* \supseteq \mathcal{Q}$ with the following property: for every slice-realized stable state $\mathcal{S}$ whose seven energy coordinates are all at most N, if $\mathcal{S}$ is non-clean, then a finite boundary-fixed liftable $\mathcal{Q}_N^*$-cleanification package starts at $\mathcal{S}$, ends at a clean stable state, and satisfies $\mathrm{E}(\mathcal{S}') < \mathrm{E}(\mathcal{S})$. In particular, no such non-clean state is blocked relative to $\mathcal{Q}_N^*$.*

Proof. By Lemma 10.21, the set $\mathfrak{C}_N$ of possible N-bounded innermost non-clean local configurations is finite. The same finite-record argument gives only finitely many encoded stable state classes with all seven energy coordinates at most N: the numbers of branch points, triple points, bad arcs, circular components, marked interactions, normal elementary segments, and active certificate records are bounded, and the normal-cell and height-ordering data are finite. In the branch case, Proposition 10.18 supplies the missing bound on the twist degree, so the finite record really includes all branch-pair twisting data.

For each non-clean encoded state class in this finite N-box, choose one representative. Apply Theorem 10.2 until the branch coordinate is zero, recording twist degrees by Convention 10.17; then apply H_{nc} until the triple coordinate is zero. By Theorem 10.3, every chosen interpolation decomposes into finitely many certified moves. Since the initial state is non-clean, the concatenated endpoint has either smaller branch coordinate, or branch coordinate still zero and smaller triple coordinate; hence the completed cleanification package has lower lexicographic energy than the initial state.

Collect every completed package type and every finite Roseman/movie submove type appearing in the finitely many chosen cleanification packages, and add them to the original palette $\mathcal{Q}$. The resulting palette $\mathcal{Q}_N^*$ is finite because the N-box has finitely many encoded classes and each chosen cleanification package is finite.

Now let $\mathcal{S}$ be any slice-realized non-clean state with all seven energy coordinates at most N. Its encoded class is one of the classes used in the construction. Transport the corresponding chosen cleanification package to $\mathcal{S}$ by stable diagram equivalence and the recorded normal-cell, height-ordering, and twist data. By Lemma 10.15, the transported package is boundary-fixed, liftable, and energy-legal, ends at a clean stable state, and belongs to $\mathcal{Q}_N^*$. Therefore $\mathcal{S}$ cannot be terminal for the expanded palette.

The argument does not use, and does not assert, preservation of blockedness under palette expansion. Adding moves can only make terminality harder to satisfy. The point is instead uniformity: after all N-bounded encoded non-clean classes, including their branch twist degrees, have been enumerated, one finite palette handles all of them at once. □

10.4 Discharging the non-clean branch

Theorem 10.23 (Non-clean blocked-state cleanification). *Assume* H_{nc}. *Let* $\mathfrak{P}_{\mathrm{nc}}$ *be the set of all valid standard non-clean packages supplied by Theorems 10.2 and 10.3 and Assumption 10.5. For every slice-realized stable state* $\mathcal{S}$,

$$\neg\,\mathrm{Clean}(\mathcal{S}) \Longrightarrow \mathfrak{P}_{\mathrm{nc}}(\mathcal{S}) \neq \emptyset.$$

Consequently no non-clean slice-realized state is terminal for any package relation $\mathfrak{P} \supseteq \mathfrak{P}_{\mathrm{nc}}$.

Proof. Let $\mathcal{S}$ be slice-realized, stable, and non-clean. Then either $b(\mathcal{S}) > 0$, or $b(\mathcal{S}) = 0$ and $t(\mathcal{S}) > 0$. Boundary-collar degeneracies are excluded by the global boundary convention.

Choose N larger than every coordinate of $\mathrm{E}(\mathcal{S})$. Proposition 10.22 supplies a valid package type for the encoded class of $\mathcal{S}$; equivalently, that package belongs to $\mathfrak{P}_{\mathrm{nc}}(\mathcal{S})$.

Suppose first that $b(\mathcal{S}) > 0$. Theorem 10.2 supplies a boundary-fixed cancellation interpolation for a cancellable pair, and Theorem 10.3 converts it into a finite valid package P with

$$\mathrm{Init}(P) = \mathcal{S}, \qquad b(\mathrm{End}(P)) < b(\mathcal{S}).$$

Hence $\mathrm{E}(\mathrm{End}(P)) < \mathrm{E}(\mathcal{S})$, so $P \in \mathfrak{P}_{\mathrm{nc}}(\mathcal{S})$.

This already proves $\mathfrak{P}_{\mathrm{nc}}(\mathcal{S}) \neq \emptyset$ when $b(\mathcal{S}) > 0$; no iteration is needed for the stated existence conclusion.

Now assume $b(\mathcal{S}) = 0$ and $t(\mathcal{S}) > 0$. The remaining hypothesis H_{nc} supplies a finite valid package P such that

$$b(\mathrm{End}(P)) = 0, \qquad t(\mathrm{End}(P)) < t(\mathcal{S}).$$

Therefore $\mathrm{E}(\mathrm{End}(P)) < \mathrm{E}(\mathcal{S})$, and again $P \in \mathfrak{P}_{\mathrm{nc}}(\mathcal{S})$.

The two cases exhaust non-cleanness. If $\mathfrak{P} \supseteq \mathfrak{P}_{\mathrm{nc}}$, then $\mathfrak{P}(\mathcal{S}) \neq \emptyset$, so $\mathcal{S}$ is not $\mathfrak{P}$-terminal. □

Corollary 10.24 (Discharge of C1). $\mathsf{H}_{\mathrm{nc}} \Rightarrow \mathsf{C1}$.

Proof. Let $\mathcal{S}$ be slice-realized and non-clean. By Theorem 10.23, $\mathfrak{P}_{\mathrm{nc}}(\mathcal{S}) \neq \emptyset$. Hence alternative (a) of C1 holds. □

Remark 10.25 (Relative status of the discharge). Corollary 10.24 is conditional only on the restricted projected-disk triple package of Assumption 10.5. Branch cancellation, finite Roseman/movie decomposition, and framed cusp merge are supplied unconditionally by Theorems 10.2 to 10.4. The source-circuit and tubular results of Section 10.2 are also unconditional, but they do not supply the missing embedded endpoint. No effective search procedure for the remaining package is claimed.

10.5 The quenching reduction

Theorem 10.26 (Quenching reduction of the slice–ribbon conjecture)**.** *Assume* H_{Q}. *Then*

$$\forall K \in \mathcal{K}, \qquad \mathrm{Slice}(K) \Longrightarrow \mathrm{Ribbon}(K).$$

Proof. This is Theorem 8.25. Explicitly, let K satisfy $\mathrm{Slice}(K)$. Choose $\mathcal{S}_0$ by relative transversality, let $\mathfrak{P}$ contain all valid energy-legal packages supplied by the definitions, by H_{nc}, and by $\mathsf{H}_{\mathrm{red}}$, and choose $\sigma_{\mathfrak{P}}$. By Theorem 6.12, the sequence has a terminal value $\mathcal{S}_N$. Soundness and endpoint preservation give $\mathrm{Realizes}(\mathcal{S}_N, K)$. If $\mathcal{S}_N$ is non-clean blocked, H_{nc} gives an element of $\mathfrak{P}(\mathcal{S}_N)$, contradicting terminality. If it is clean blocked, $\mathsf{H}_{\mathrm{red}}$ and Theorem 8.13 give a certified non-zero Milnor invariant, whereas $\mathsf{H}_{\mathrm{app}}$ and Theorem 9.15 give vanishing of the same certified family. Equivalently, Theorem 9.16 gives the contradiction. Thus $\mathcal{S}_N$ is ribbon terminal, and Propositions 2.3 and 1.11 give $\mathrm{Ribbon}(K)$. □

Corollary 10.27 (Exact remaining condition)**.** *The unconditional results prove transition soundness, well-founded termination, endpoint preservation, lifted-source rigidity, and the stated homological facts. The entire remaining hypothesis is the single condition* H_{Q}*, whose clauses are:*

(a) *the remaining projected-disk endpoint-recovery package* $\mathsf{H}_{\mathrm{nc}} = \mathsf{H}_{\mathrm{ct}}^{\mathrm{rel}}$*, not supplied by the cited literature or by the three structural theorems of Section 10.2;*

(b) *the global clean Milnor-reduction input* $\mathsf{H}_{\mathrm{red}}$*;*

(c) *the projected-core Milnor-applicability input* $\mathsf{H}_{\mathrm{app}}$*.*

Consequently Theorem 10.26 is a reduction to one explicitly delimited completion condition rather than an unconditional proof of the conjecture.

Remark 10.28 (Relation with known approaches). The slice–ribbon implication is known for several restricted families, including the classes treated in [19, 8]. Our result is a separate conditional reduction and does not subsume those classification theorems.

Remark 10.29 (Possible refinements). Future versions of the quenching program may replace ordinary pairwise linking by more sensitive certified invariants. Three natural candidate directions are:

(a) Milnor-type higher linking data for clean closed core components, designed to detect Whitehead-type configurations invisible to pairwise linking;

(b) an intersection-tree or nesting invariant built from source filling disks, recording how innermost disks, annular regions, certified strips, and remaining decker arcs meet after the transverse counting conventions of Convention 6.2;

(c) algebraic-topological data tied to the four-dimensional normal bundle or to relative homology classes in the complement of the lifted disk and its boundary collar.

Quandle or biquandle cocycle data for broken surface diagrams, movie-move obstructions, and invariants of complements of projected source filling disks are also possible refinements. These candidates are not needed for the conditional proof proved here, but they may be useful for explicit examples, sharper palettes, or computational certificate checking.

Problem 10.30 (Open constructive tasks)**.** *The remaining tasks are both mathematical and constructive:*

(a) *construct explicit finite palettes containing the standard non-clean packages of Chapter 10 for concrete diagrammatic families;*

(b) *implement certificate search for the clean terminal Milnor rigidity data and standard non-clean packages in concrete diagrammatic families;*

(c) *construct refined invariants to test and visualize the clean terminal obstruction mechanism on explicit examples;*

(d) *develop formal-verification or computational-topology tools that can check certificates for a fixed finite palette or weak flow.*

No algorithm for these tasks is asserted.

10.6 High-dimensional codimension-two form

The weak-flow and quenching mechanisms admit the following general codimension-two formulation. For $n \geq 2$, let $K^{n-1} \subset S^{n+1}$ be a smooth embedded sphere. It is *slice* if it bounds a properly embedded smooth disk $D^n \subset B^{n+2}$. It is *ribbon* if it bounds such a disk for which the radial Morse function has no index-n critical point, equivalently, after projection to S^{n+1}, its singular strata have only ribbon decker type in the high-dimensional sense of Remark 2.5.

Problem 10.31 (High-dimensional slice–ribbon problem)**.** *For every $n \geq 2$, is every smooth slice $(n-1)$-knot $K^{n-1} \subset S^{n+1}$ ribbon in the sense above?*

Definition 10.32 (High-dimensional completion package)**.** Fix $n \geq 2$. A high-dimensional completion package consists of:

(a) a finite high-dimensional certified palette $\mathcal{Q}_n$ in the sense of Definitions 5.3 and 6.4;

(b) standard high-dimensional non-clean local packages eliminating or reducing the declared non-clean multiple-point and rank-drop strata by boundary-fixed liftable energy-decreasing moves;

(c) a clean terminal obstruction package specifying both applicability and a non-vanishing invariant on the same terminal sublink; non-vanishing must be proved or assumed and is not inferred from blockedness alone.

Theorem 10.33 (Conditional high-dimensional slice–ribbon template)**.** *For a fixed $n \geq 2$, assume a high-dimensional completion package in the sense of Definition 10.32. Then every slice $(n-1)$-knot $K^{n-1} \subset S^{n+1}$ is ribbon.*

Proof. Let $K^{n-1} \subset S^{n+1}$ bound a smooth disk $D^n \subset B^{n+2}$. Choose stable high-dimensional projection data and form the encoded state $\mathcal{S}_n$ of Definition 5.3. Run the selected finite quenching algorithm with the palette $\mathcal{Q}_n$. By Theorem 6.13, the history terminates. Boundary fixation and liftability preserve the boundary knot and the slice-disk realization class, while Theorem 8.7 keeps the relative homology and homotopy classes trivial throughout the weak-flow middle. If the terminal state is ribbon terminal, the radial Morse definition gives a ribbon disk. If it is non-clean blocked, the non-clean local part of the completion package gives a further energy-decreasing cleanification package, contradicting terminality. If it is clean blocked, the high-dimensional terminal invariant gives non-vanishing, while terminal rigidity and the corresponding high-dimensional Milnor-type vanishing theorem give vanishing. This contradiction excludes clean blocked terminal states. Hence only the ribbon terminal case remains. □

We do not prove Problem 10.31. We prove only the implication in Theorem 10.33. Its antecedent contains all dimension-dependent local cleanification and terminal-obstruction statements.

Conclusion

We have established two principal results. First, we constructed a general certified quenching mechanism for projected codimension-two disks. Valid transitions are boundary-fixed and liftable, the integer energy is well founded, every selected history terminates, and the resulting discrete dynamics has no non-terminal recurrence. These statements are independent of the completion condition.

Second, we reduced the slice–ribbon conjecture to the single quenching completion condition H_{Q} of Definition 0.1. Its three clauses correspond exactly to the remaining projected-disk recovery, clean-core reduction, and projected-core applicability problems. Once these clauses are available, the terminal alternative has no blocked branch: the non-clean branch admits a further certified package, while the clean branch gives simultaneous Milnor non-vanishing and vanishing. Therefore

$$\boxed{\mathsf{H}_{\mathrm{Q}} \Longrightarrow \forall K \in \mathcal{K}\,\big(\mathrm{Slice}(K) \Rightarrow \mathrm{Ribbon}(K)\big).}$$

Thus the contribution is a quenching framework together with an explicit one-condition reduction of the conjecture. Proving H_{Q}, or replacing it by weaker verifiable clauses, remains the geometric completion problem.

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